

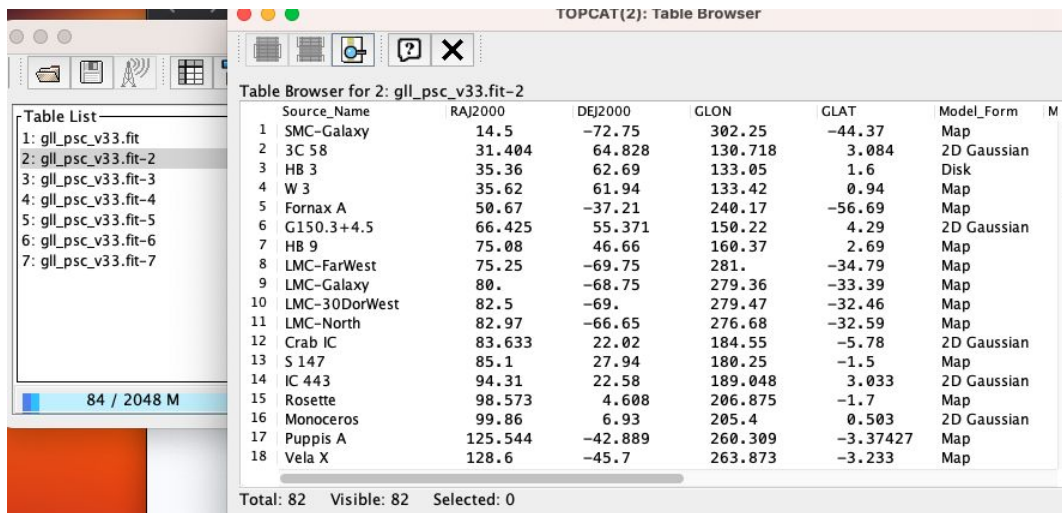
Applying ML/Deep Learning Techniques to Gamma Ray Astronomy

Krishiv Bhatia

Update: 12/08/2023

Update on Progress

- Refreshed Jerry's YouTube presentation
- Downloaded the 4FGL-DR4 FITS file from https://fermi.gsfc.nasa.gov/ssc/data/access/lat/14yr_catalog/
- Installed the TOPCAT tool on my laptop
- Viewed the 4FGL-DR4 FITS file from TOPCAT.



The screenshot shows the TOPCAT(2): Table Browser window. On the left, a 'Table List' sidebar shows several FITS files, with 'gll_psc_v33.fit-2' selected. The main window displays a table titled 'Table Browser for 2: gll_psc_v33.fit-2'. The table has columns for Source_Name, RAJ2000, DEJ2000, GLON, GLAT, Model_Form, and M. It lists 18 sources, including SMC-Galaxy, 3C 58, HB 3, W 3, Fornax A, G150.3+4.5, HB 9, LMC-FarWest, LMC-Galaxy, LMC-30DorWest, LMC-North, Crab IC, S 147, IC 443, Rosette, Monoceros, Puppis A, and Vela X. At the bottom, it shows 'Total: 82', 'Visible: 82', and 'Selected: 0'.

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Model_Form	M
1	SMC-Galaxy	14.5	-72.75	302.25	-44.37	Map	
2	3C 58	31.404	64.828	130.718	3.084	2D Gaussian	
3	HB 3	35.36	62.69	133.05	1.6	Disk	
4	W 3	35.62	61.94	133.42	0.94	Map	
5	Fornax A	50.67	-37.21	240.17	-56.69	Map	
6	G150.3+4.5	66.425	55.371	150.22	4.29	2D Gaussian	
7	HB 9	75.08	46.66	160.37	2.69	Map	
8	LMC-FarWest	75.25	-69.75	281.	-34.79	Map	
9	LMC-Galaxy	80.	-68.75	279.36	-33.39	Map	
10	LMC-30DorWest	82.5	-69.	279.47	-32.46	Map	
11	LMC-North	82.97	-66.65	276.68	-32.59	Map	
12	Crab IC	83.633	22.02	184.55	-5.78	2D Gaussian	
13	S 147	85.1	27.94	180.25	-1.5	Map	
14	IC 443	94.31	22.58	189.048	3.033	2D Gaussian	
15	Rosette	98.573	4.608	206.875	-1.7	Map	
16	Monoceros	99.86	6.93	205.4	0.503	2D Gaussian	
17	Puppis A	125.544	-42.889	260.309	-3.37427	Map	
18	Vela X	128.6	-45.7	263.873	-3.233	Map	

Update on Progress

- Downloaded ATNF Pulsar Catalogue tar gz file. Followed README to produce the executable psrcat file.
- Followed README
- Psrcat -h works but I get error in the db_file command
- What is this for? How to fix error?

```
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % ls
README          evalwrap.c      libpsrcat.a     psrcat.db
defineParams.c  evalwrap.o      makeit          psrcat.h
defineParams.o  externalCall.c  myrefs.bbl     psrcat_ref
displayOutput.c externalCall.o   parseParameters.c readCatalogue.c
displayOutput.o galcoord_ecliptic.c parseParameters.o readCatalogue.o
evaldefs.h      galcoord_ecliptic.o plotParams.c    wc_strncmp.c
evalkern.c      galcoord_ecliptic.o plotParams.o    wc_strncmp.h
evalkern.h      galcoord_ecliptic.o psrcat          wc_strncmp.o
evalkern.o      glitch.db       psrcat.c
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -h
-all          Merge with $PSRCAT_RUNDIR/obs*.db
-allref       Lists the entire bibliographic reference file
-bib           Provide bibliographic reference
-bibliography  Lists references within the catalogue
-c            List of parameters to obtain from the database
-check        Checks the catalogue for incorrect entries
-db_file      Database path and name
-descend      Sort in descending order
-e            Produce short ephemeris files suitable for TEMPO
-e2           Produce long ephemeris files
-e3           Produce ephemeris file with selected parameters
-exactMatch   Require exact match for string comparisons
-h            This help information
-l            Logical string for filtering unwanted pulsars
              e.g. p0 > 5
-listref      Provide bibliographic references at end of table
-merge        Merges multiple catalogue files. For example
              -merge "mycat*.db" will merge the public catalogue
              with all the catalogues with filenames mycat*.db in the
              local directory.
-nohead       Do not print a header at the top of the table
-nonumber     Do not print line numbers in output
-nocand       Do not include pulsar candidates
-nointerim    Do not include interim solutions
-null        Null string for displaying for unknown parameter
```

```
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -db_file ./psrcat.db -c "name p0 dm" -l "p0 > 3"
zsh: illegal hardware instruction psrcat -db_file ./psrcat.db -c "name p0 dm" -l "p0 > 3"
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar %
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar % psrcat -db_file ./psrcat.db
zsh: illegal hardware instruction psrcat -db_file ./psrcat.db
rbhatia@Rajeshs-MacBook-Pro-Display psrcat_tar %
```

Next Steps

- Could I have access to the Python scripts that you used with the 3FGL catalog
- Then, if I understand correctly, I need to reproduce your results using PyTorch?
- And then convert to using Neural Networks for Deep Learning?

Update: 01/08/2024

Progress

- 3FGLReproductionBasicLR Python code pointed to 3FGL Catalog file gll_psc_v16.fit.
- Updated it to point to 4FGL catalog file gll_psc_v33.fit.
- Tested with updated feature name
- Modified it for 1 hidden layer NN and DNN (2 hidden layers NN)

Results

Model	AGN Sensitivity	PSR Sensitivity	Accuracy of Best P Fit
Basic LR (0 hidden layer NN)	0.9724137931034482	1.0	0.9733333333333334
Neural Network (1 hidden layer)	0.9620689655172414	1.0	0.9633333333333334
Deep Neural Network (2 hidden layers)	0.9724137931034482	1.0	0.9733333333333334

Features Used

- Used the following features from columns in 4FGL catalog file gll_psc_v33.fit
 - ["PL_Index", "LP_Index", "PLEC_IndexS", "Variability_Index", "Unc_PL_Flux_Density", "Unc_LP_Flux_Density", "Unc_Energy_Flux100", "Unc_PLEC_Flux_Density", "Unc_Flux1000", "LP_beta", "Frac_Variability", "LP_SigCurv", "Signif_Avg"]
 - Found the corresponding 3FGL columns in the 4FGL list
 - Did some trial and error for other columns and checked model performance
- Column list is in pages 32-33 of Abdollahi_2020_ApJS_247_33 paper

Columns

- Printed out from code

1 DataRelease	21 PL_Flux_Density	41 Unc_PLEC_Exp_Index	61 ASSOC_4FGL
2 RAJ2000	22 Unc_PL_Flux_Density	42 PLEC_SigCurv	62 TEVCAT_FLAG
3 DEJ2000	23 PL_Index	43 PLEC_EPeak	63 ASSOC_TEV
4 GLON	24 Unc_PL_Index	44 Unc_PLEC_EPeak	64 CLASS1
5 GLAT	25 LP_Flux_Density	45 Npred	65 ASSOC1
6 Conf_68_SemiMajor	26 Unc_LP_Flux_Density	46 Flux_Band	66 ASSOC_PROB_BAY
7 Conf_68_SemiMinor	27 LP_Index	47 Unc_Flux_Band	67 ASSOC_PROB_LR
8 Conf_68_PosAng	28 Unc_LP_Index	48 nuFnu_Band	68 RA_Counterpart
9 Conf_95_SemiMajor	29 LP_beta	49 Sqrt_TS_Band	69 DEC_Counterpart
10 Conf_95_SemiMinor	30 Unc_LP_beta	50 Variability_Index	70 Unc_Counterpart
11 Conf_95_PosAng	31 LP_SigCurv	51 Frac_Variability	71 Flags
12 ROI_num	32 LP_EPeak	52 Unc_Frac_Variability	72 ASSOC_FGL
13	33 Unc_LP_EPeak	53 Signif_Peak	73 ASSOC_FHL
Extended_Source_Name	34 PLEC_Flux_Density	54 Flux_Peak	74 ASSOC_GAM1
14 Signif_Avg	35 Unc_PLEC_Flux_Density	55 Unc_Flux_Peak	75 ASSOC_GAM2
15 Pivot_Energy	36 PLEC_IndexS	56 Time_Peak	76 ASSOC_GAM3
16 Flux1000	37 Unc_PLEC_IndexS	57 Peak_Interval	77 CLASS2
17 Unc_Flux1000	38 PLEC_ExpfactorS	58 Flux_History	78 ASSOC2
18 Energy_Flux100	39 Unc_PLEC_ExpfactorS	59 Unc_Flux_History	
19 Unc_Energy_Flux100	40 PLEC_Exp_Index	60 Sqrt_TS_History	
20 SpectrumType			

Feature Transformation

- FGL4Dataset feature transformations

```
def __init__(self, wanted_type_col, fluxindices, wantedtypes, wantedfeaturenames, pointsourcecata):  
    # Create x/input data  
    xdata = []  
    ydata = []  
    # Select sources that are a part of the wanted types  
    for source in pointsourcecatalogue.data:  
        for feature_names in wantedfeaturenames:  
            print("{0} = {1}".format(*args: feature_names, source[feature_names]))  
            if source[wanted_type_col] in wantedtypes:  
                unit = []  
                # Append wanted params. Applied a log transformation to paras with highly skewed distributions.  
                unit.append(float(source[wantedfeaturenames[0]]))  
                unit.append(float(source[wantedfeaturenames[1]]))  
                unit.append(float(source[wantedfeaturenames[2]]))  
                unit.append(float(source[wantedfeaturenames[3]]))  
                unit.append(math.log(source[wantedfeaturenames[4]]))  
                if (source[wantedfeaturenames[4]] > 0) else float(source[wantedfeaturenames[4]])  
                unit.append(math.log(source[wantedfeaturenames[5]]))  
                if (source[wantedfeaturenames[5]] > 0) else float(source[wantedfeaturenames[5]])  
                unit.append(math.log(source[wantedfeaturenames[6]]))  
                if (source[wantedfeaturenames[6]] > 0) else float(source[wantedfeaturenames[6]])  
                unit.append(math.log(source[wantedfeaturenames[7]]))  
                if (source[wantedfeaturenames[7]] > 0) else float(source[wantedfeaturenames[7]])  
                unit.append(math.log(source[wantedfeaturenames[8]]))  
                if (source[wantedfeaturenames[8]] > 0) else float(source[wantedfeaturenames[8]])  
                unit.append(float(source[wantedfeaturenames[9]]))  
                unit.append(float(source[wantedfeaturenames[10]]))  
                unit.append(float(source[wantedfeaturenames[11]]))  
                unit.append(float(source[wantedfeaturenames[12]]))
```

Apply log transformations
to params with highly
skewed distributions

PL_Index = 2.5443360805511475

LP_Index = 2.5455660820007324

PLEC_IndexS = 2.539679765701294

Variability_Index = 25.699934005737305

Unc_PL_Flux_Density = 3.7675866251677195e-14

Unc_LP_Flux_Density = 4.284851688957439e-14

Unc_Energy_Flux100 = 4.238916617889388e-13

Unc_PLEC_Flux_Density = 3.952874743566419e-14

Unc_Flux1000 = 2.748007377206818e-11

LP_beta = -0.009841139428317547

Frac_Variability = 0.5357248187065125

LP_SigCurv = 0.0

Signif_Avg = 8.640633583068848

Neural Networks Configurations Used

- Used

- 0 hidden layer Neural Networks - Linear Regression
- 1 hidden layer NN
- Deep Neural Networks (2 hidden layers)

0 hidden layer NN - Linear Regression

```
Linear(in_features=13, out_features=1, bias=True)
```

NN (1 hidden layer)

```
Sequential(  
  (0): Linear(in_features=13, out_features=60, bias=True)  
  (1): ReLU()  
  (2): Linear(in_features=60, out_features=1, bias=True)  
)
```

DNN (2 hidden layers)

```
Sequential(  
  (0): Linear(in_features=13, out_features=250, bias=True)  
  (1): ReLU()  
  (2): Linear(in_features=250, out_features=40, bias=True)  
  (3): ReLU()  
  (4): Linear(in_features=40, out_features=1, bias=True)  
)
```

Other Changes & Github

Classes

- Separated the large code into classes for code modularity, improvement
- Created separate classes for:
 - **Driver:** 3FGLReproductionBasicLR_v1.py, 4FGLReproductionBasic{LR, NN, DNN}.py,
 - **Feature transformation:** FGL3Dataset.py / FGL4Dataset.py
 - **Models:** LogisticRegression.py, NeuralNetwork.py, DeepNeuralNetwork.py
 - **Utility functions:** utils.py (e.g. run_iteration)

- 3FGLReproductionBasicLR_v1.py
 - FGL3Dataset.py
 - LogisticRegression.py
 - utils.py

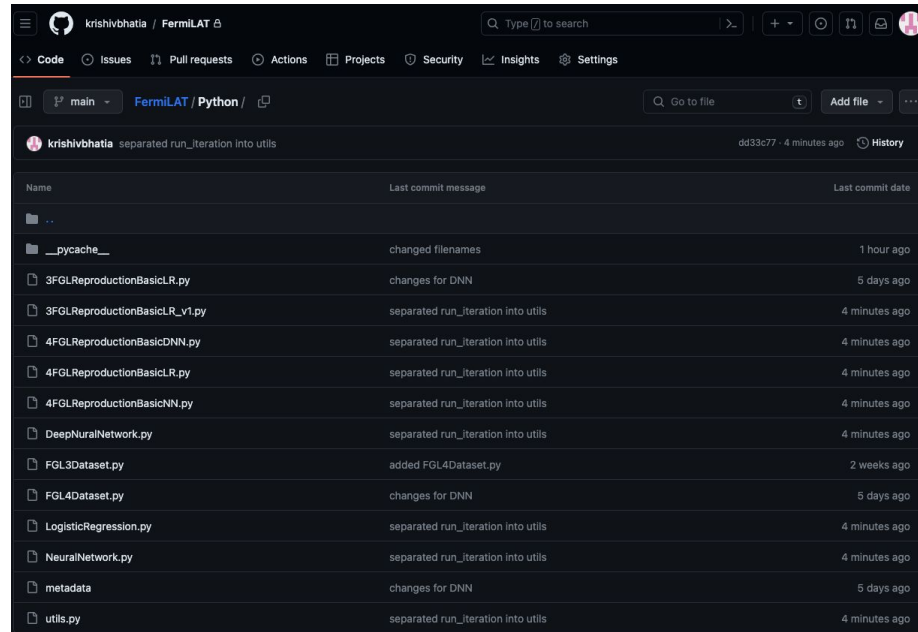
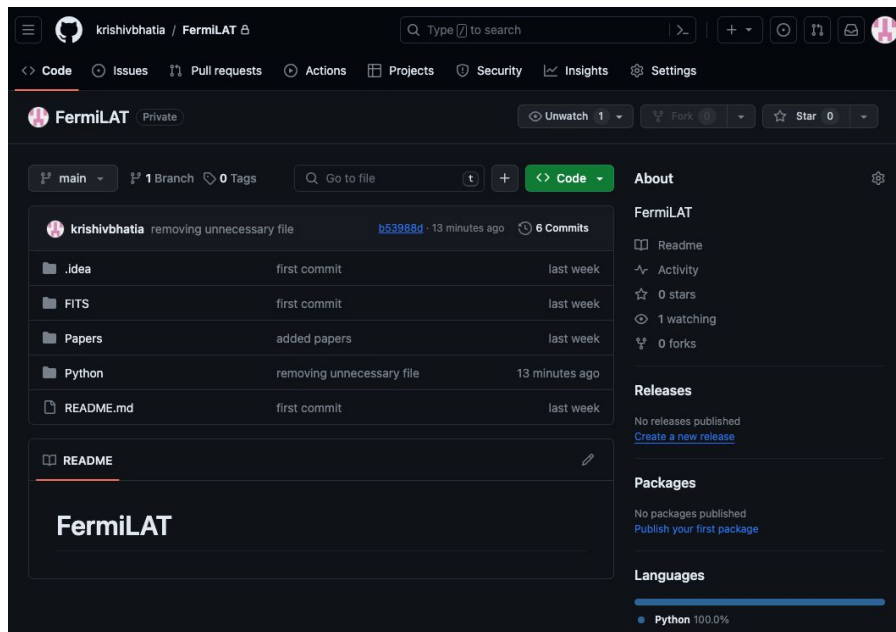
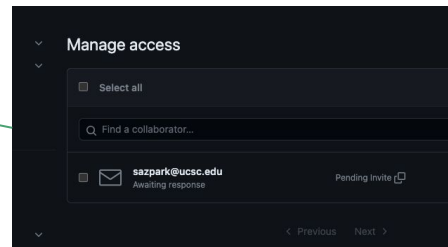
- 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py

- 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py
- 4FGLReproductionBasicDNN.py
 - FGL4Dataset.py
 - DeepNeuralNetwork.py
 - utils.py

Github Repo

- Checked into github
 - <https://github.com/krishivbhatia/FermiLAT/>

Sent you collaborator invite



Results: Linear Regression

Files for Linear Regression

- Python Scripts
 - 4FGLReproductionBasicLR.py
 - FGL4Dataset.py
 - LogisticRegression.py
 - utils.py
- Execution
 - python3 4FGLReproductionBasicLR.py

1-layer NN (No hidden layer)
Basic LR



```
main ▾ FermiLAT / Python / LogisticRegression.py   
vrbhatia changes for DNN  
1 loc) · 1.79 KB  
Blame      
  
import math  
import torch  
import torch.nn as nn  
  
class LogisticRegression(nn.Module):  
    def __init__(self, input_features):  
        print("(len(input_features[0][0]) = ", len(input_features[0][0]))  
        print("input_features[0][0] = ", input_features[0][0])  
        super(LogisticRegression, self).__init__()  
        self.model = nn.Linear(len(input_features[0][0]), 1)  
        print()  
        print(self.model)  
        print()  
  
    def forward(self, x):  
        y_predicted = torch.sigmoid(self.model(x))  
        return y_predicted
```

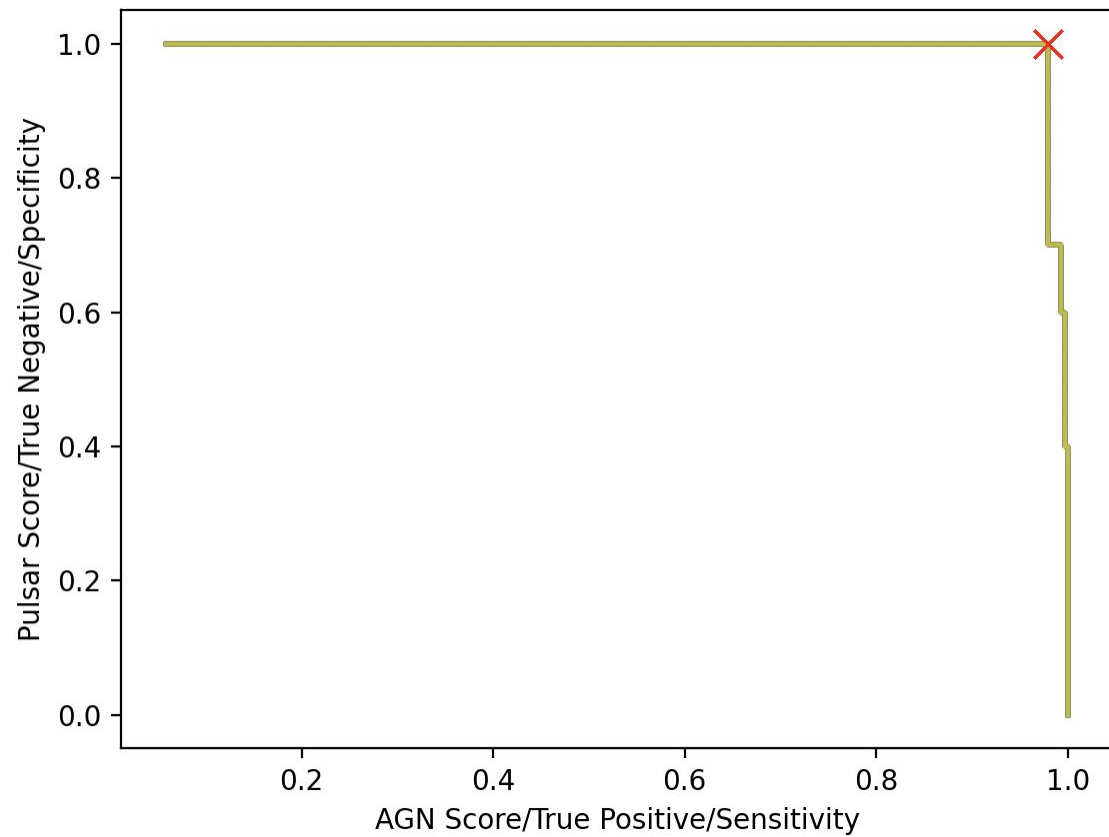
Results

- Linear(in_features=13, out_features=1, bias=True)
- **Results (200 iterations, lr = 0.00001)**
 - optimal_p = 0.9352
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Results

- Linear(in_features=13, out_features=1, bias=True)
- **Results (300 iterations, lr = 0.00001)**
 - optimal_p = 0.9352
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 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Plot



Results: NN - 1 hidden layer

Files for NN (1 hidden layer)

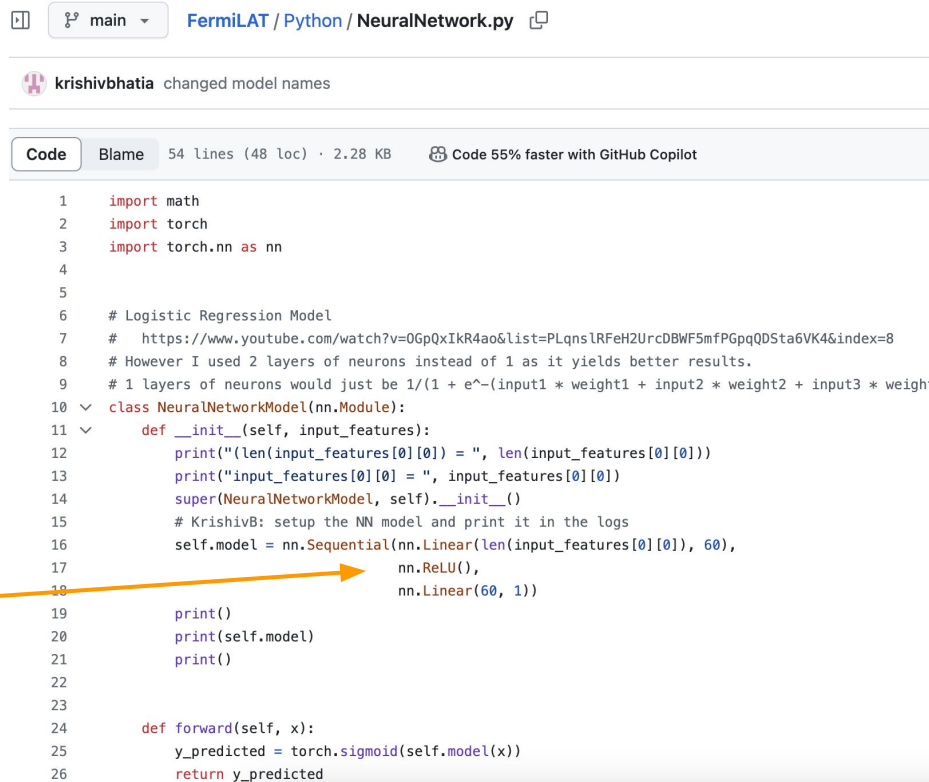
- Python Scripts

- 4FGLReproductionBasicNN.py
- FGL4Dataset.py
- NeuralNetwork.py
- utils.py

- Execution

- python3 4FGLReproductionBasicNN.py

1 hidden layers NN
ReLU activation function



```
main FermiLAT / Python / NeuralNetwork.py
krishivbhatia changed model names

Code Blame 54 lines (48 loc) · 2.28 KB Code 55% faster with GitHub Copilot

1 import math
2 import torch
3 import torch.nn as nn
4
5
6 # Logistic Regression Model
7 # https://www.youtube.com/watch?v=0GpQxIkR4ao&list=PLqns1RFeh2UrcDBWF5mfPGpqQDSta6VK4&index=8
8 # However I used 2 layers of neurons instead of 1 as it yields better results.
9 # 1 layers of neurons would just be 1/(1 + e^~(input1 * weight1 + input2 * weight2 + input3 * weight3))
10 class NeuralNetworkModel(nn.Module):
11     def __init__(self, input_features):
12         print("(len(input_features[0][0]) = ", len(input_features[0][0]))
13         print("input_features[0][0] = ", input_features[0][0])
14         super(NeuralNetworkModel, self).__init__()
15         # KrishivB: setup the NN model and print it in the logs
16         self.model = nn.Sequential(nn.Linear(len(input_features[0][0]), 60),
17                                   nn.ReLU(),
18                                   nn.Linear(60, 1))
19
20         print()
21         print(self.model)
22         print()
23
24     def forward(self, x):
25         y_predicted = torch.sigmoid(self.model(x))
26         return y_predicted
```

Results

- **Model: Sequential(**

- (0): Linear(in_features=13, out_features=60, bias=True)

- (1): ReLU() ← Activation Function

- (2): Linear(in_features=60, out_features=1, bias=True))

1 hidden layer



- **Results (300 iterations, lr = 0.0001)**

- optimal_p = 0.9525
 - Best True Positive (AGN Successfully Identified): 279 (290)
 - Best AGN/True Positive/Sensitivity : 0.9620689655172414
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9633333333333334

Results

- **Model: Sequential**
 - (0): Linear(in_features=13, out_features=60, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=60, out_features=1, bias=True))
- **Results (60 iterations, lr = 0.0001)**
 - optimal_p = 0.9525
 - Best True Positive (AGN Successfully Identified): 279 (290)
 - Best AGN/True Positive/Sensitivity : 0.9620689655172414
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9633333333333334

Results

- **Model: Sequential**
 - (0): Linear(in_features=13, out_features=40, bias=True)
 - (1): Tanh() ← tested with tanh() activation function also
 - (2): Linear(in_features=40, out_features=1, bias=True))
- **Results (300 iterations, lr = 0.0001)**
 - optimal_p = 0.9544
 - Best True Positive (AGN Successfully Identified): 278 (290)
 - Best AGN/True Positive/Sensitivity : 0.9586206896551724
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.96

Results: DNN - 2 hidden layers

Files for DNN: 2 hidden layers

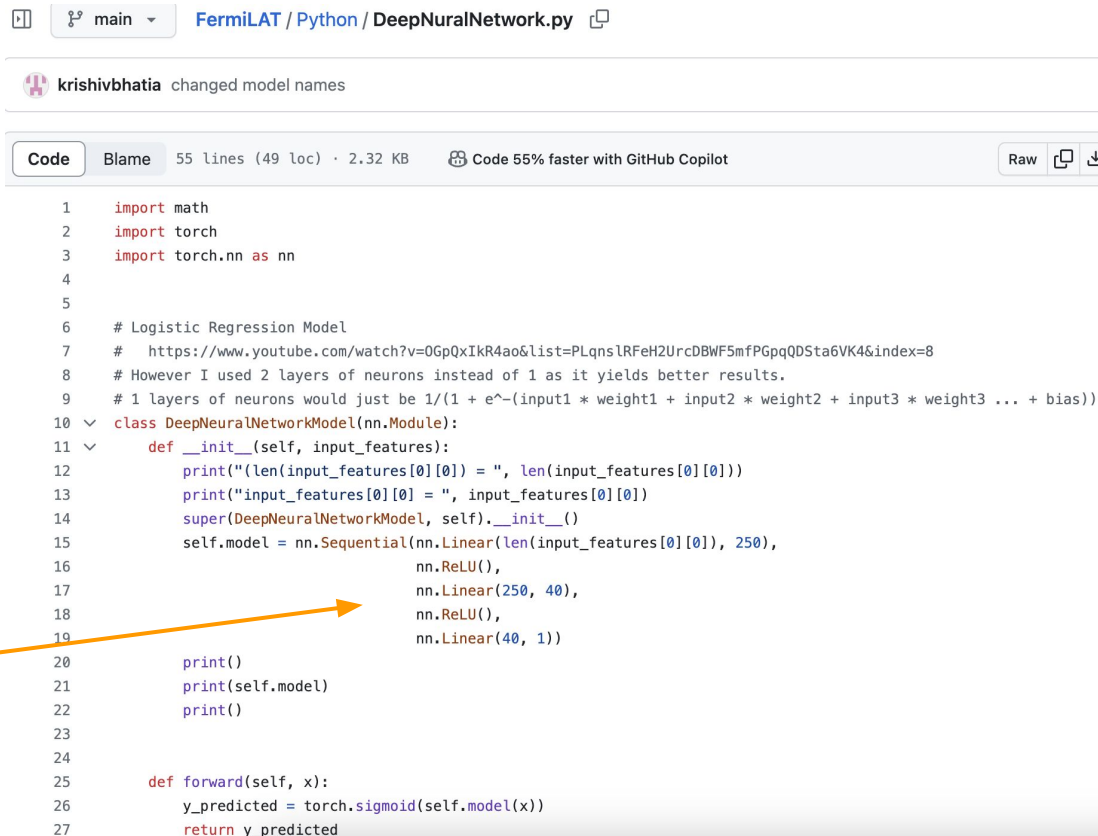
- Python Scripts

- 4FGLReproductionBasicDNN.py
- FGL4Dataset.py
- DeepNeuralNetwork.py
- utils.py

- Execution

- python3 4FGLReproductionBasicDNN.py

2 hidden layers NN
ReLU activation function



```
1 import math
2 import torch
3 import torch.nn as nn
4
5
6 # Logistic Regression Model
7 # https://www.youtube.com/watch?v=OGpQxIkR4ao&list=PLqnsIRFeH2UrcDBWF5mfPGpqQDSta6VK4&index=8
8 # However I used 2 layers of neurons instead of 1 as it yields better results.
9 # 1 layers of neurons would just be 1/(1 + e^-(input1 * weight1 + input2 * weight2 + input3 * weight3 ... + bias))
10 class DeepNeuralNetworkModel(nn.Module):
11     def __init__(self, input_features):
12         print("len(input_features[0][0]) = ", len(input_features[0][0]))
13         print("input_features[0][0] = ", input_features[0][0])
14         super(DeepNeuralNetworkModel, self).__init__()
15         self.model = nn.Sequential(nn.Linear(len(input_features[0][0]), 250),
16                                   nn.ReLU(),
17                                   nn.Linear(250, 40),
18                                   nn.ReLU(),
19                                   nn.Linear(40, 1))
20
21         print()
22         print(self.model)
23         print()
24
25     def forward(self, x):
26         y_predicted = torch.sigmoid(self.model(x))
27         return y_predicted
```

Results

- **Model: Sequential(**

- (0): Linear(in_features=13, out_features=250, bias=True)

- (1): ReLU() ← activation function

- (2): Linear(in_features=250, out_features=40, bias=True)

- (3): ReLU() ← activation function

- (4): Linear(in_features=40, out_features=1, bias=True))

2 hidden layers



- **Results (200 iterations, lr = 0.0001)**

- optimal_p = 0.9475
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Results

- **Model: Sequential**
 - (0): Linear(in_features=13, out_features=250, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=250, out_features=40, bias=True)
 - (3): ReLU()
 - (4): Linear(in_features=40, out_features=1, bias=True))
- **Results (300 iterations, lr = 0.0001)**
 - optimal_p = 0.9475
 - Best True Positive (AGN Successfully Identified): 282 (290)
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10 (10)
 - Best Pulsar/True Negative/Specificity : 1.0
 - Accuracy of Best P Fit : 0.9733333333333334

Questions

- Hardness ratios of LAT energy bands
 - 3FGL Catalog had many bands:
 - "Flux100_300", "Flux300_1000", "Flux1000_3000", "Flux3000_10000", "Flux10000_100000"
 - 4FGL Catalog has only one energy band column present:
 - Energy_Flux100
 - From Abdollahi 2020, definition: *Energy flux from 100MeV to 100GeV obtained by spectral fitting*
 - Where do we get other bands to calculate hardness ratios?

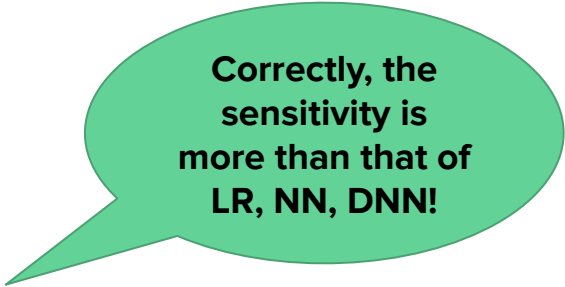
Next Steps

- Random Forest & Decision Tree implementations
 - Found Random Forest & Decision Tree implementations using PyTorch
 - <https://discuss.pytorch.org/t/how-can-i-use-knn-random-forest-models-in-pytorch/42464/2>
 - https://github.com/ValentinFigue/Sklearn_PyTorch
 - Test class shows how to use it:
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/random_forest_tests.py
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/binary_tree_tests.py
 - Use it to implement Random Forest & Decision Trees in PyTorch for 4FGL Catalog?

Update: 01/15/2024

Progress

- Completed Random Forest & Decision Tree implementations using PyTorch for 4FGL Catalog file
 - Random Forest
 - Highest sensitivity = 98.05749805749805
 - Decision Trees
 - Highest sensitivity = 98.13519813519814



Correctly, the sensitivity is more than that of LR, NN, DNN!

Background

- Referenced Random Forest & Decision Tree implementations using PyTorch
 - https://github.com/ValentinFigue/Sklearn_PyTorch
- Random Forest test class shows how to use it:
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/random_forest_tests.py
 - `random_forest = Sklearn_PyTorch.TorchRandomForestRegressor(nb_trees=10, nb_samples=2, max_depth=3)`
 - `random_forest.fit(train_features, train_labels)`
- Decision Tree test class shows how to use it:
 - https://github.com/ValentinFigue/Sklearn_PyTorch/blob/master/tests/binary_tree_tests.py
 - `binary_tree = Sklearn_PyTorch.TorchDecisionTreeClassifier(max_depth=1)`
 - `binary_tree.fit(train_features, train_labels)`

Code Updates

- Checked in code in github for:

Random Forest

- 4FGLReproduction_RF.py
 - FGL4Dataset.py
 - random_forest.py
 - utils.py

Decision Trees

- 4FGLReproduction_decision_tree.py
 - FGL4Dataset.py
 - decision_tree.py
 - decision_node.py
 - utils.py

Github Repo - Python package

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 **krishivbhatia** modifications for directories 3adc19f · 4 minutes ago [History](#)

Name	Last commit message	Last commit date
 ..		
 DecisionTrees	modifications for Random Forest and Decsion Trees	4 minutes ago
 LogisticRegression	modifications for Random Forest and Decsion Trees	4 minutes ago
 NeuralNetworks	modifications for Random Forest and Decsion Trees	4 minutes ago
 RandomForests	modifications for Random Forest and Decsion Trees	4 minutes ago
 Utils	modifications for Random Forest and Decsion Trees	4 minutes ago
 <code>__init__.py</code>	modifications for Random Forest and Decsion Trees	4 minutes ago

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 **krishivbhatia** modifications for Random Forest and Decsion Trees 5f26b63 · 5 minutes ago [History](#)

Name	Last commit message	Last commit date
 ..		
 4FGLReproduction_RF.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 __init__.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 random_forest.py	modifications for Random Forest and Decsion Trees	5 minutes ago

Github Repo - Decision Trees package

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Name	Last commit message	Last commit date
 ..		
 4FGLReproduction_DecisionTree.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 __init__.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 decision_node.py	modifications for Random Forest and Decsion Trees	5 minutes ago
 decision_tree.py	modifications for Random Forest and Decsion Trees	5 minutes ago

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 **krishivbhatia** changes in Random Forest and Decision Trees code for FermiLAT 6ebb528 · 9 minutes ago [History](#)

Name	Last commit message	Last commit date
 ..		
 FGL3Dataset.py	modifications for Random Forest and Decsion Trees	4 hours ago
 FGL4Dataset.py	modifications for Random Forest and Decsion Trees	4 hours ago
 __init__.py	modifications for Random Forest and Decsion Trees	4 hours ago
 metadata	modifications for Random Forest and Decsion Trees	4 hours ago
 utils.py	changes in Random Forest and Decision Trees code for FermiLAT	9 minutes ago

Code Details

- Method of creating classifier and invoking fit method:

4FGLReproduction_RF.py

```
# Krishiv Bhatia: Invoke TorchRandomForestClassifier
random_forest = TorchRandomForestClassifier( nb_trees: 200, nb_samples: 2500, max_depth: 15)
print("random forest = ", random_forest.nb_trees, random_forest.nb_samples, random_forest.max_depth)

# Krishiv Bhatia: split train dataset into features and labels
train_features, train_labels = dataset_to_features_labels(train_dataset)
random_forest.fit(torch.FloatTensor(train_features), torch.LongTensor(train_labels))
```

4FGLReproduction_decision_tree.py

```
# Krishiv Bhatia: Invoke TorchDecisionTreeClassifier
decision_tree = TorchDecisionTreeClassifier(20)
print("TorchDecisionTreeClassifier = ", decision_tree.max_depth)

# Krishiv Bhatia: split train dataset into features and labels
train_features, train_labels = dataset_to_features_labels(train_dataset)
decision_tree.fit(torch.FloatTensor(train_features), torch.LongTensor(train_labels))
```

Transforms dataset into
separate feature & labels
arrays (next slide)

Random Forest

Decision Trees

Transformations

- Transformed the tensor feature and labels array into regular arrays

```
main ▾ FermiLAT / Python / Utils / utils.py

Blame 150 lines (126 loc) · 4.75 KB Code 55% faster with GitHub Copilot

def mean(values):
    """
    Mean function.
    """
    m = 0.0
    for value in values:
        m = m + value/len(values)
    return m

def dataset_to_features_labels(dataset):
    # Krishiv Bhatia: train or test dataset is of the form:
    #   tensor array(train/test features), tensor array (output label)
    # or like this:
    #   tensor array(feature 0, feature 1, ... feature 12), tensor array(output label)
    # You can print it like this:
    #   for i in train_dataset:
    #       print(i)
    # An example:
    #   (tensor([ 2.1602,  1.8308,  1.6083, 13.2215, -32.9276, -32.4171, -29.4589,
    #            -32.3813, -24.3150,  0.4812,  0.0000,  2.7429,  5.4350]), tensor([1.]))
    #
    # Convert tensor feature array in dataset to numpy feature array
    train_features = ([i[i[0]].detach().numpy() for i in dataset])
    # print("train_features = ", train_features)
    # Convert tensor label array in dataset to regular array
    train_labels = ([torch.LongTensor.item(i[1]) for i in dataset])
    # print("train_labels = ", train_labels)
    return train_features, train_labels
```

References

- <https://stackoverflow.com/questions/49768306/pytorch-tensor-to-numpy-array>
- <https://stackoverflow.com/questions/54268029/how-to-convert-a-pytorch-tensor-into-a-numpy-array>
- <https://stackoverflow.com/questions/67711752/how-to-extract-tensors-to-numpy-arrays-or-lists-from-a-larger-pytorch-tensor>
- <https://stackoverflow.com/questions/47588682/how-to-cast-a-1-d-intensor-to-int-in-pytorch>
- <https://stackoverflow.com/questions/57727372/how-do-i-get-the-value-of-a-tensor-in-pytorch>

i[0] gets tensor feature array
i[1] gets tensor label array

Random Forest Results

- `random_forest = TorchRandomForestClassifier(50, 2000, 10)`
 - `correct = 1262/1287`
 - `sensitivity = 98.05749805749805`
- `random_forest = TorchRandomForestClassifier(200, 2000, 15)`
 - `correct = 1258/1287`
 - `sensitivity = 97.74669774669775`
- `random_forest = TorchRandomForestClassifier(200, 1500, 15)`
 - `correct = 1262/1287`
 - `sensitivity = 98.05749805749805`

Random Forest Results

- `random_forest = TorchRandomForestClassifier(15, 500, 10)`
 - `correct = 1261/1287`
 - `sensitivity = 97.97979797979798`
- `random_forest = TorchRandomForestClassifier(100, 500, 10)`
 - `correct = 1255/1287`
 - `sensitivity = 97.51359751359752`
- `random_forest = TorchRandomForestClassifier(50, 1000, 15)`
 - `correct = 1258/1287`
 - `sensitivity = 97.74669774669775`
- `random_forest = TorchRandomForestClassifier(50, 1500, 10)`
 - `correct = 1259`
 - `sensitivity = 97.82439782439782`

Decision Tree Results

- `decision_tree = TorchDecisionTreeClassifier(10)`
 - `correct = 1263`
 - `sensitivity = 98.13519813519814`
- `decision_tree = TorchDecisionTreeClassifier(20)`
 - `correct =`
 - `sensitivity =`
- `decision_tree = TorchDecisionTreeClassifier(30)`
 - `correct = 1263`
 - `sensitivity = 98.13519813519814`
- `decision_tree = TorchDecisionTreeClassifier(80)`
 - `correct = 1263`
 - `sensitivity = 98.13519813519814`

Update: 01/22/2024

Progress

- Completed the problem of Energy Band Level Hardness ratios
- Column: Flux_Band
- Printed it out. Numpy array with 8 numbers
 - `source[flux_col] = [1.318705452994784e-12, 3.918336766162156e-10, 3.155801175935835e-10, 1.0454052207231612e-10, 2.0413597895396762e-11, 6.700519479541089e-12, 1.909341825270805e-12, 2.2682772056883754e-16]`

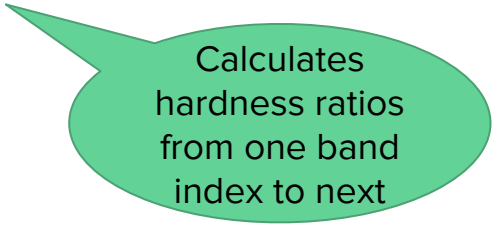
Transformation

- In FGL4Dataset.py
- Converted numpy array to list

- `flux_list = res = source[flux_col].tolist()`

- Calculated hardness ratios

```
# Calculate hardness ratios (also considered in the paper)
for index in range(1, len(flux_list)):
    unit.append((flux_list[index] - flux_list[index-1] + 0.0) /
                (flux_list[index] + flux_list[index-1] + 0.0))
xdata.append(unit)
```



Calculates
hardness ratios
from one band
index to next

Logistic Regression

- Model:
 - **Linear(in_features=20, out_features=1, bias=True)**
- Iterations = 30/100
 - Best P Threshold Value : 0.972
 - Score : 1.0586206896551724
 - Best Score : 1.9655172413793105
 - Best True Positive (AGN Successfully Identified): 280
 - AGN count : 290
 - Best AGN/True Positive/Sensitivity : 0.9655172413793104
 - Best True Negative (PSR Successfully Identified): 10
 - PSR count : 10
 - Best Pulsar/True Negative/Specificity : 1.0
 - **Accuracy of Best P Fit : 0.9666666666666667**
 - Best correct : 290
 - No of samples : 300

Neural Networks

- Model:
 - Sequential(
 - (0): Linear(in_features=20, out_features=60, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=60, out_features=1, bias=True))
- Iterations = 30
 - Best P Threshold Value : 0.976
 - Score : 1.0551724137931036
 - Best Score : 1.9586206896551723
 - Best True Positive (AGN Successfully Identified): 278
 - AGN count : 290
 - Best AGN/True Positive/Sensitivity : 0.9586206896551724
 - Best True Negative (PSR Successfully Identified): 10
 - PSR count : 10
 - Best Pulsar/True Negative/Specificity : 1.0
 - **Accuracy of Best P Fit : 0.96**
 - Best correct : 288
 - No of samples : 300

Deep Neural Networks

- Model:
 - **Sequential**
 - (0): **Linear(in_features=20, out_features=250, bias=True)**
 - (1): **ReLU()**
 - (2): **Linear(in_features=250, out_features=50, bias=True)**
 - (3): **ReLU()**
 - (4): **Linear(in_features=50, out_features=1, bias=True)**
 -)
- Results
 - Best P Threshold Value : 0.971
 - Score : 1.0586206896551724
 - Best Score : 1.9724137931034482
 - Best True Positive (AGN Successfully Identified): 282
 - AGN count : 290
 - Best AGN/True Positive/Sensitivity : 0.9724137931034482
 - Best True Negative (PSR Successfully Identified): 10
 - PSR count : 10
 - Best Pulsar/True Negative/Specificity : 1.0
 - **Accuracy of Best P Fit : 0.9733333333333334**
 - Best correct : 292
 - No of samples : 300

Deep Neural Networks

- DNN
- Model:
 - Sequential(
 - (0): Linear(in_features=20, out_features=250, bias=True)
 - (1): ReLU()
 - (2): Linear(in_features=250, out_features=40, bias=True)
 - (3): ReLU()
 - (4): Linear(in_features=40, out_features=1, bias=True))
 - Best P Threshold Value : 0.967
 - Score : 1.0517241379310345
 - Best Score : 1.9689655172413794
 - Best True Positive (AGN Successfully Identified): 281
 - AGN count : 290
 - Best AGN/True Positive/Sensitivity : 0.9689655172413794
 - Best True Negative (PSR Successfully Identified): 10
 - PSR count : 10
 - Best Pulsar/True Negative/Specificity : 1.0
 - **Accuracy of Best P Fit : 0.97**
 - Best correct : 291
 - No of samples : 300

Random Forests

- Model:
 - **# Trees: 200**
 - **# Samples: 2500**
 - **Depth: 15**
- Results
 - Correct = 1258
 - Total = 1286
 - Sensitivity = 97.82439782439782

Random Forests

- Model:
 - **# Trees: 200**
 - **# Samples: 1500**
 - **Depth: 15**
- Results
 - Correct = 1258
 - Total = 1286
 - Sensitivity = 97.74669774669775

Random Forests

- Model:
 - **# Trees: 200**
 - **# Samples: 500**
 - **Depth: 15**
- Results
 - Correct = 1254
 - Total = 1286
 - Sensitivity = 97.43589743589743

Decision Trees

- Decision Trees
- Model:
 - **Depth: 20**
 - Correct = 1259
 - Total = 1286
 - Sensitivity = 97.82439782439782

Next Steps

- Could you kindly clarify what needs to be done with the following tasks
 - associated/unassociated sources
 - AGN types

Update: 02/09/2024

Progress

- Simplified Logistic Regression & Neural Networks Python code to train and then test on the entire 30% test dataset
- This resolves the earlier problem where the test was showing only PSR count = 10 & AGN count = 290
- New scripts:
 - LR: 4FGLReproductionBasicLRnew.py
 - NN: 4FGLNNnew.py
 - DNN: 4FGLDNNnew.py
 - Only difference in the above scripts is the model trained and tested

Training

```
torch.set_default_tensor_type(torch.DoubleTensor)
torch.manual_seed(82)
LogRegModel = LogisticRegression(train_dataset)
# Stochastic Gradient Descent. I could use Adams but Adams is worse than regular SGD unless you
optimizer = torch.optim.SGD(LogRegModel.parameters(), lr=0.0001)
print("optimizer = ", optimizer.__class__)
# Binary Cross Entropy Loss which is the loss method that would most likely be used in this scenario
criterion = nn.BCELoss()
tot_iter = 100
print("total iterations = ", tot_iter)
for iteration in range(0, tot_iter):
    dev_inputs = dev_labels = []
    print("Iteration # ", iteration)
    total_samples = len(train_dataset)
    # KrishivB: Put run_iteration code in utils.py, imported it, and call here
    for epoch in range(30):
        (dev_inputs, dev_labels) = run_iteration(LogRegModel, iteration, train_dataset, total_samples, criterion,
                                                optimizer, dev_inputs, dev_labels, append_dev_input_label: True)
```

Define model once

Iterations for loop

Train model in iterations

Testing

Enumerate over
test dataset

Predict

Get actual
value

Check if predicted
matches with actual

```
correct = correct_agm = correct_psr = total_agm = total_psr = 0
for i, (inputs, labels) in enumerate(test_dataset):
    y_predicted = LogRegModel(inputs) # Insert/fit Model for prediction here
    predicted_item = y_predicted.data.item()
    print(" i, y_predicted, y_predicted.data, predicted_item = ", i, y_predicted, y_predicted.data, predicted_item)
    predicted_val = 0 if (math.isnan(predicted_item)) else round(predicted_item)
    label_val = labels.item()
    total_agm += 1 if label_val == 1 else 0
    total_psr += 1 if label_val == 0 else 0
    match = (predicted_val == label_val)
    if match:
        correct += 1
        correct_agm += 1 if label_val == 1 else 0
        correct_psr += 1 if label_val == 0 else 0
    print("i, predicted_item, predicted_val, label_val, correct, match = ",
          i, predicted_item, predicted_val, label_val, correct, match)
```

Results

Logistic Regression

- # iterations = 50
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1241 20
 - total, total_agn, total_psr = 1287 1244 43
 - **total sensitivity = 0.9797979797979798**
 - agn sensitivity = 0.997588424437299
 - psr sensitivity = 0.46511627906976744

- # iterations = 750
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1239 22
 - total, total_agn, total_psr = 1287 1244 43
 - **total sensitivity = 0.9797979797979798**
 - agn sensitivity = 0.9959807073954984
 - psr sensitivity = 0.5116279069767442

- # iterations = 100, 150, 200
- Results
 - Correct = 1262
 - correct_agn, correct_psr = 1240 22
 - total, total_agn, total_psr = 1287 1244 43
 - **total sensitivity = 0.9805749805749806**
 - agn sensitivity = 0.9967845659163987
 - psr sensitivity = 0.5116279069767442

- # iterations = 1000, 1250
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1238 23
 - total, total_agn, total_psr = 1287 1244 43
 - **total sensitivity = 0.9797979797979798**
 - agn sensitivity = 0.9951768488745981
 - psr sensitivity = 0.5348837209302325

Neural Network

- # iterations = 10
- Results
 - Correct = 1261
 - correct_agm, correct_psr = 1241 20
 - total, total_agm, total_psr = 1287 1244 43
 - total sensitivity = 0.9797979797979798
 - agm sensitivity = 0.997588424437299
 - psr sensitivity = 0.46511627906976744

- # iterations = 50
- Results
 - Correct = 1265
 - correct_agm, correct_psr = 1240 25
 - total, total_agm, total_psr = 1287 1244 43
 - total sensitivity = 0.9829059829059829
 - agm sensitivity = 0.9967845659163987
 - psr sensitivity = 0.5813953488372093

- # iterations = 100
- Results
 - Correct = 1264
 - correct_agm, correct_psr = 1239 25
 - total, total_agm, total_psr = 1287 1244 43
 - total sensitivity = 0.9821289821289821
 - agm sensitivity = 0.9959807073954984
 - psr sensitivity = 0.5813953488372093

- # iterations = 150
- Results
 - Correct = 1265
 - correct_agm, correct_psr = 1237 28
 - total, total_agm, total_psr = 1287 1244 43
 - total sensitivity = 0.9829059829059829
 - agm sensitivity = 0.9943729903536977
 - psr sensitivity = 0.6511627906976745

Deep Neural Network

- # iterations = 10
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1241 19
 - total, total_agn, total_psr = 1287 1244 43
 - total sensitivity = 0.9790209790209791
 - agn sensitivity = 0.997588424437299
 - psr sensitivity = 0.4418604651162791

- # iterations = 25
- Results
 - Correct = 1266
 - correct_agn, correct_psr = 1240 26
 - total, total_agn, total_psr = 1287 1244 43
 - total sensitivity = 0.9836829836829837
 - agn sensitivity = 0.9967845659163987
 - psr sensitivity = 0.6046511627906976

- # iterations = 100
- Results
 - Correct = 1267
 - correct_agn, correct_psr = 1239 28
 - total, total_agn, total_psr = 1287 1244 43
 - total sensitivity = 0.9844599844599845
 - agn sensitivity = 0.9959807073954984
 - psr sensitivity = 0.6511627906976745

- # iterations = 150
- Results
 - Correct = 1267
 - correct_agn, correct_psr = 1236 29
 - total, total_agn, total_psr = 1287 1244 43
 - total sensitivity = 0.9829059829059829
 - agn sensitivity = 0.9935691318327974
 - psr sensitivity = 0.6744186046511628

Observations

- I found overall AGN sensitivity to be > 0.995 throughout but PSR sensitivity to be around 0.5.
- With the NN & DNN model, PSR sensitivity started going up with more training (as # iterations increased).
- In the case of NN with 150 iterations (each iteration having 30 epochs), PSR sensitivity increased to 0.65.
- In the case of DNN, with 25 iterations, PSR sensitivity increased to 0.60 and with 100 iterations, increased to 0.65, and increased to 0.67 with 150 iterations.
- So we may need to find what is the problem with PSR?
 - Is it related to the choice of model, features, or training?

Update: 02/19/2024

Progress

- Fix the problem of low Pulsar count
- Added “msp” in wantedtypes list



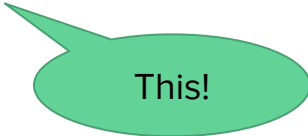
This!

```
|# KrishivB: Added AGN in caps  
|wantedtypes = ["PSR", "psr", "YNG", "yng", "MSP", "msp", "FSRQ", "fsrq", "BLL", "bll", "BCU", "bcu", "RDG", "rdg",  
|               "NLSY1", "nlsy1", "AGN", "agn", "ssrq", "sey"]
```

Progress

- Added “MSP”, “msp” to the **source[wanted_type_col]** check in FGL4Dataset.py
- It was missing. Only “PSR”, “psr” were present.

```
# Calculate hardness ratios (also considered in the paper)
for index in range(1, len(flux_list)):
    unit.append((flux_list[index] - flux_list[index-1] + 0.0) /
                (flux_list[index] + flux_list[index-1] + 0.0))
xdata.append(unit)
if source[wanted_type_col] in ("PSR", "psr", "MSP", "msp"):
    ydata.append([0]) # 0 is for pulsar
else:
    ydata.append([1]) # 1 is for AGN
```



This!

Logistic Regression

- # iterations = 25
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1172 88
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9699769053117783**
 - agn sensitivity = 0.9890295358649789
 - psr sensitivity = 0.7719298245614035

- # iterations = 200, 500
- Results
 - Correct = 1264
 - correct_agn, correct_psr = 1177 87
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9730561970746728**
 - agn sensitivity = 0.9932489451476794
 - psr sensitivity = 0.7631578947368421

- # iterations = 50, 100, 150
- Results
 - Correct = 1264
 - correct_agn, correct_psr = 1177 87
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9730561970746728**
 - agn sensitivity = 0.9932489451476794
 - psr sensitivity = 0.7631578947368421

- # iterations = 300
- Results
 - Correct = 1265
 - correct_agn, correct_psr = 1178 87
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9738260200153964**
 - agn sensitivity = 0.9940928270042194
 - psr sensitivity = 0.7631578947368421

Neural Network

- # iterations = 25
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1173 88
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.970746728252502**
 - agn sensitivity = 0.9898734177215189
 - psr sensitivity = 0.7719298245614035

- # iterations = 200
- Results
 - Correct = 1261
 - correct_agn, correct_psr = 1178 83
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.970746728252502**
 - agn sensitivity = 0.9940928270042194
 - psr sensitivity = 0.7280701754385965

- # iterations = 50, 100
- Results
 - Correct = 1257
 - correct_agn, correct_psr = 1176 81
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9676674364896074**
 - agn sensitivity = 0.9924050632911392
 - psr sensitivity = 0.7105263157894737

- # iterations = 300
- Results
 - Correct = 1262
 - correct_agn, correct_psr = 1180 82
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9715165511932256**
 - agn sensitivity = 0.9957805907172996
 - psr sensitivity = 0.7192982456140351

Deep Neural Network

- # iterations = 25
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1172 88
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9699769053117783**
 - agn sensitivity = 0.9890295358649789
 - psr sensitivity = 0.7719298245614035

- # iterations = 200
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1175 85
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9699769053117783**
 - agn sensitivity = 0.9915611814345991
 - psr sensitivity = 0.7456140350877193

- # iterations = 100
- Results
 - Correct = 1258
 - correct_agn, correct_psr = 1176 82
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.968437259430331**
 - agn sensitivity = 0.9924050632911392
 - psr sensitivity = 0.7192982456140351

- # iterations = 300
- Results
 - Correct = 1260
 - correct_agn, correct_psr = 1173 87
 - total, total_agn, total_psr = 1299 1185 114
 - **total sensitivity = 0.9699769053117783**
 - agn sensitivity = 0.9898734177215189
 - psr sensitivity = 0.7631578947368421

Random Forest

- **Model**

- # Trees: 200
- # Samples: 500
- Depth: 15

- **Results**

- Correct = 1245
- sensitivity = 95.84295612009238

- **Model**

- # Trees: 300
- # Samples: 1500
- Depth: 15

- **Results**

- Correct = 1251
- sensitivity = 96.30484988452656

- **Model**

- # Trees: 200
- # Samples: 1000
- Depth: 15

- **Results**

- Correct = 1246
- sensitivity = 95.91993841416475

Update: 02/26/2024

Progress

- Classified unassociated sources as PSR or AGN
 - Initially, trained and tested models with regular dataset
 - Then, read dataset with field Class1 == null
 - Used trained model to predict whether the unassociated sources are PSRs or AGNs
- Also, plotted measure of outlyingness of the unassociated sources from the class.

Outlyingness Measure

- Predictions ranged from [0.0, 1.0]
- AGNs
 - Predictions were in range [0.5, 1.0]
 - The more near it was to 1.0, the closer it was to the class
 - The farther away it was from 1.0 (or closer to 0.5), the more outlying it was from the class
- PSRs
 - Predictions were in range [0.0, 0.5]
 - The more near it was 0.0, the closer it was to the class
 - The farther away it was from 0.0 (or closer to 0.5), the more outlying it was from the class

Code change

- New files

- Logistic Regression
 - 4FGL_LR_cls1_null.py
- Neural Networks
 - 4FGL_NN_cls1_null.py
 - 4FGL_DNN_cls1_null.py
- Random Forests
 - 4FGL_RF_cls1_null.py
- Utils
 - FGL4Dataset_cls1_null.py



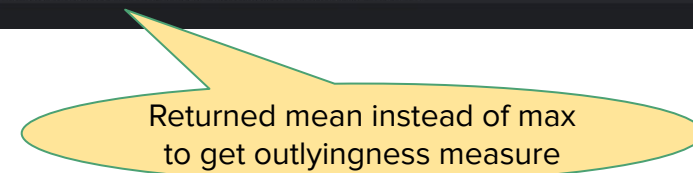
Read dataset with field Class1 == null

- Updated files

- Random Forests
 - random_forest.py



```
11 11 11
predictions = []
for tree, index_features in zip(self.trees, self.trees_features):
    sampled_vector = torch.index_select(vector, dim=0, index=index_features)
    predictions.append(tree.predict(sampled_vector))
print("predictions=", predictions)
print("sum(predictions)={}, len(self.trees)={}, sum(predictions)/len(self.trees)={}",
      sum(predictions), len(self.trees), sum(predictions)/len(self.trees))
return sum(predictions)/len(self.trees)
# return max(set(predictions), key=predictions.count)
```



Returned mean instead of max
to get outlyingness measure

Code change

- FGL4Dataset_cls1_null.py

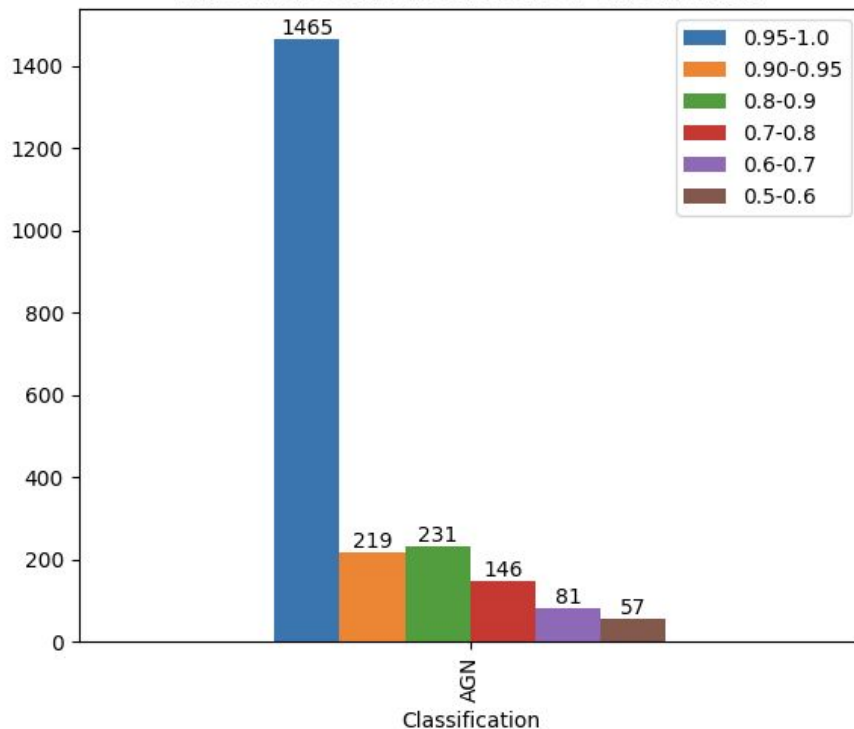
```
class FGL4Dataset_Cls1_Null(Dataset):
    # Krishiv Bhatia
    def __init__(self, wanted_type_col, flux_col, wanted_types, wanted_feature_names, poi
        # Create x/input data
        xdata = []
        ydata = []
        # Select sources that are a part of the wanted types
        for source in point_source_catalogue.data:
            # for feature_names in wanted_feature_names:
            #     print("{0} = {1}".format(feature_names, source[feature_names]))
            # print("(type(source[flux_col]) = ", type(source[flux_col]))
            flux_list = res = source[flux_col].tolist()
            # print("source[flux_col] = {0}".format(flux_list))
            print("source[wanted_type_col] = ", source[wanted_type_col])
            if source[wanted_type_col] == '':
                unit = []
                # Append wanted params. Applied a log transformation to paras with highly
                unit.append(float(source[wanted_feature_names[0]]))
                unit.append(float(source[wanted_feature_names[1]]))
                unit.append(float(source[wanted_feature_names[2]]))
                unit.append(float(source[wanted_feature_names[3]]))
```

Read dataset with field Class1 == null

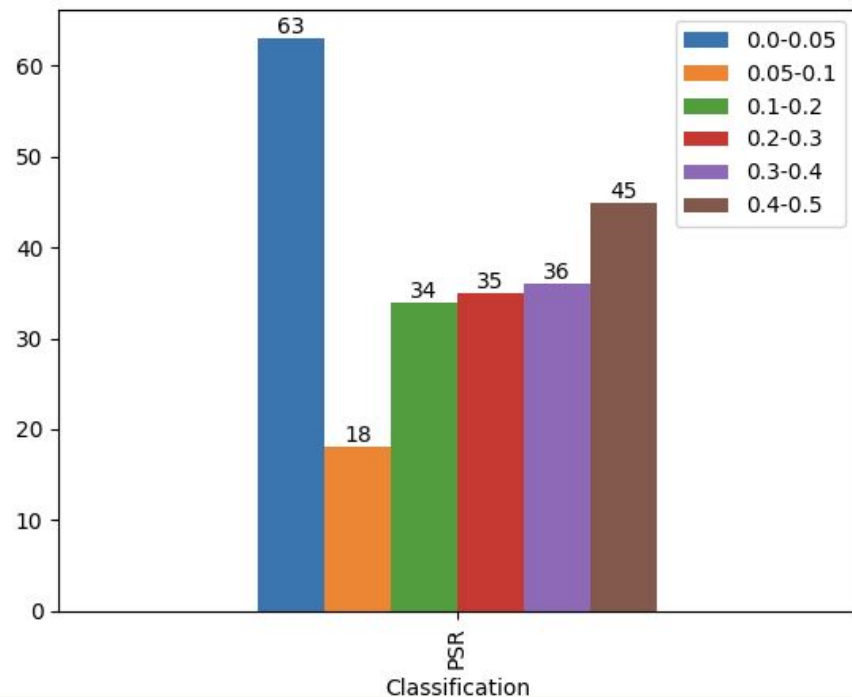
Logistic Regression

- # iterations = 50 Total: 2430 unassociated sources

2199 unassociated sources AGN classification



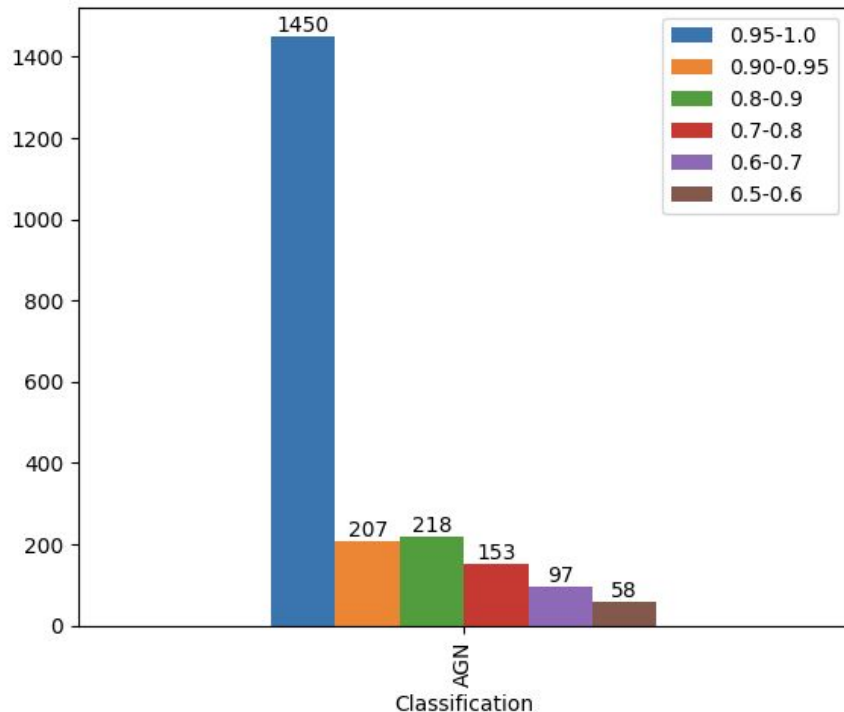
231 unassociated sources PSR classification



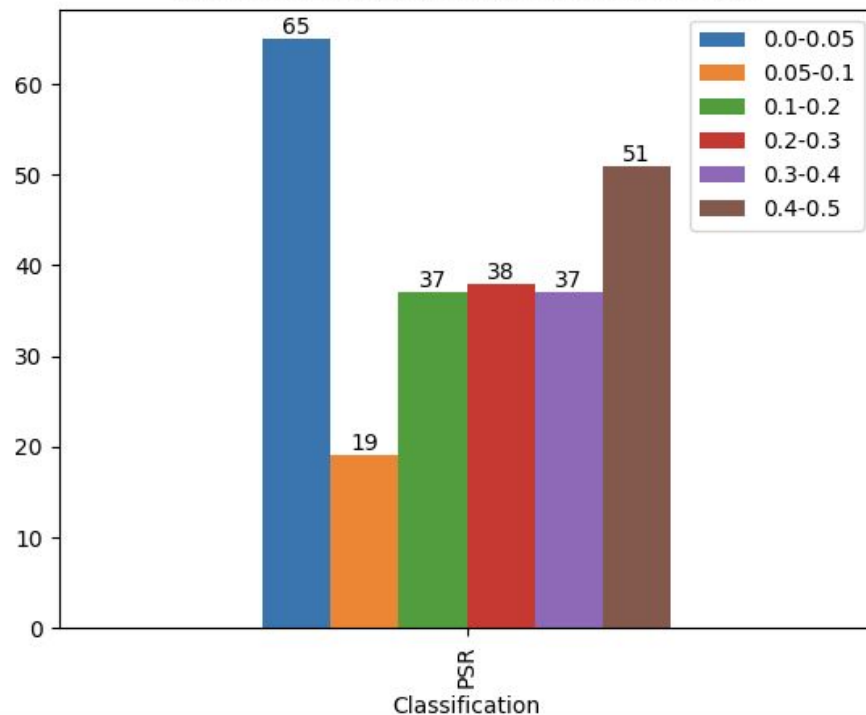
Logistic Regression

- # iterations = 100 Total: 2430 unassociated sources

2183 unassociated sources AGN classification



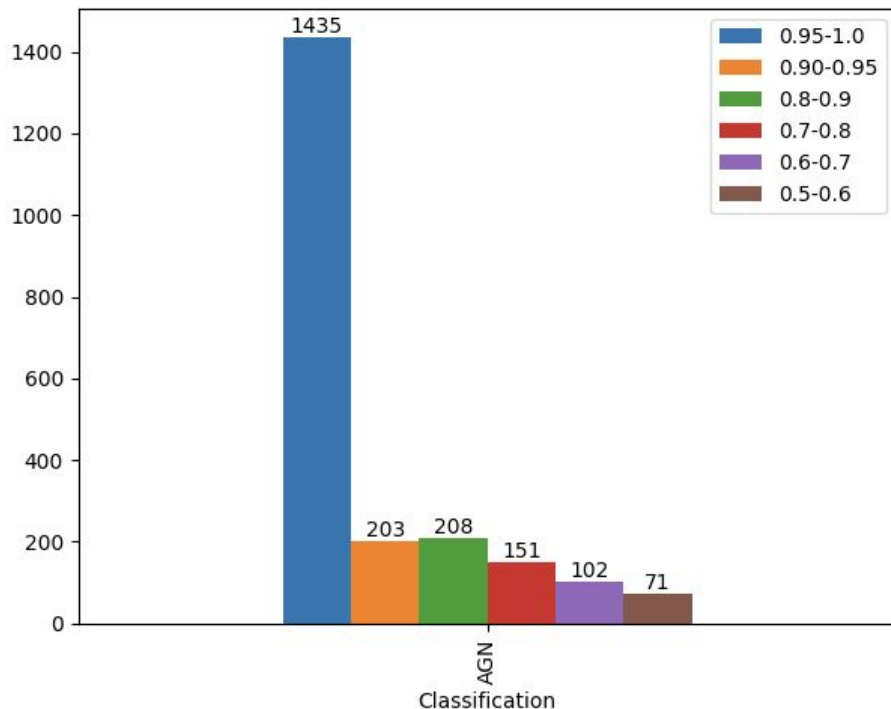
247 unassociated sources PSR classification



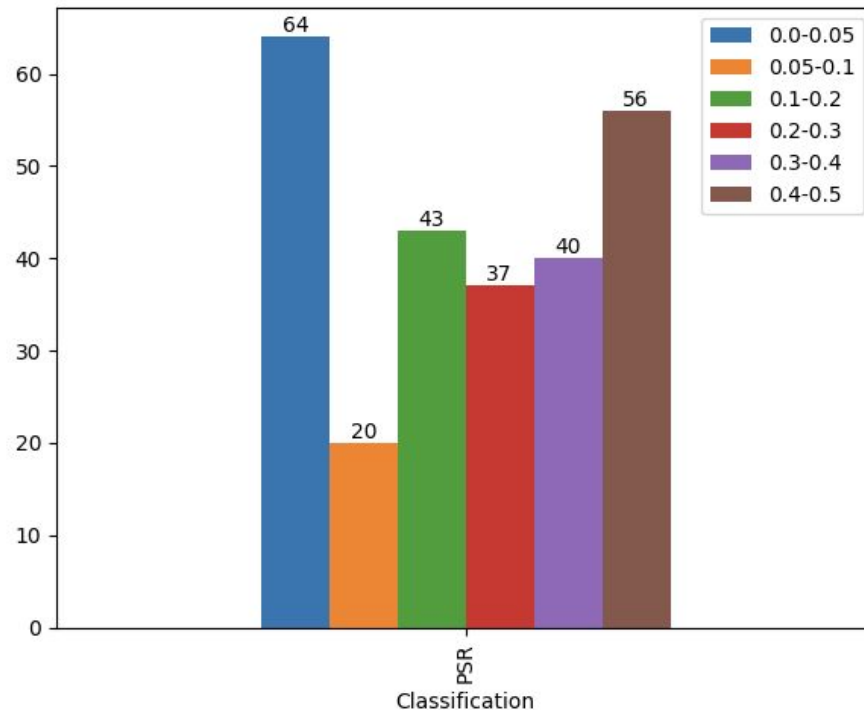
Logistic Regression

- # iterations = 200 Total: 2430 unassociated sources

2170 unassociated sources AGN classification



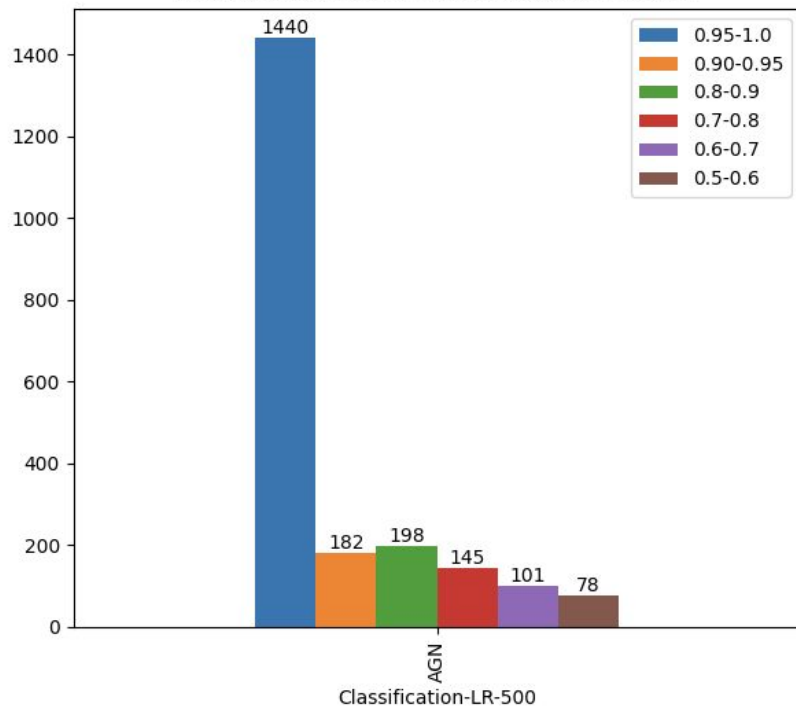
260 unassociated sources PSR classification



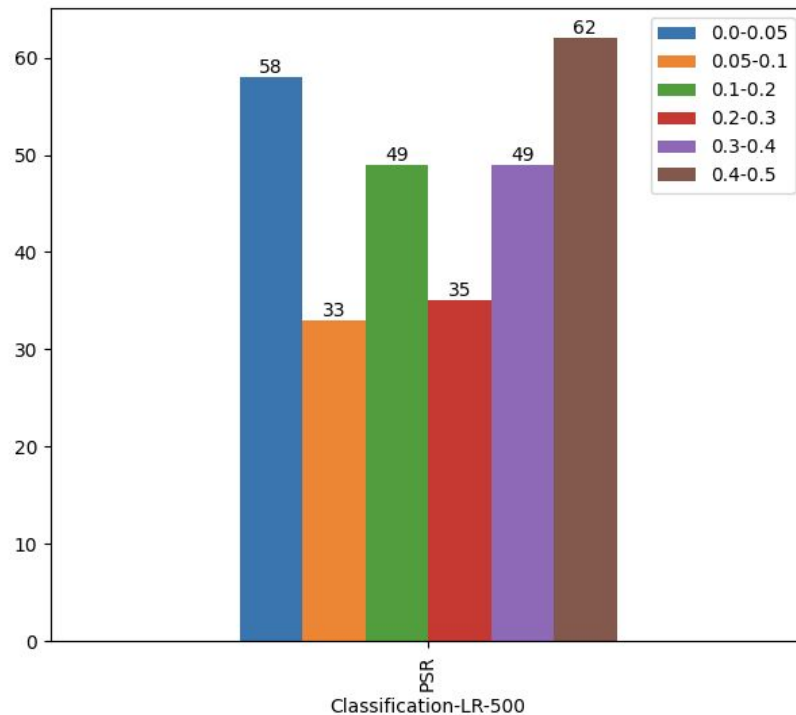
Logistic Regression

- # iterations = 500 Total: 2430 unassociated sources

2144 unassociated sources AGN classification

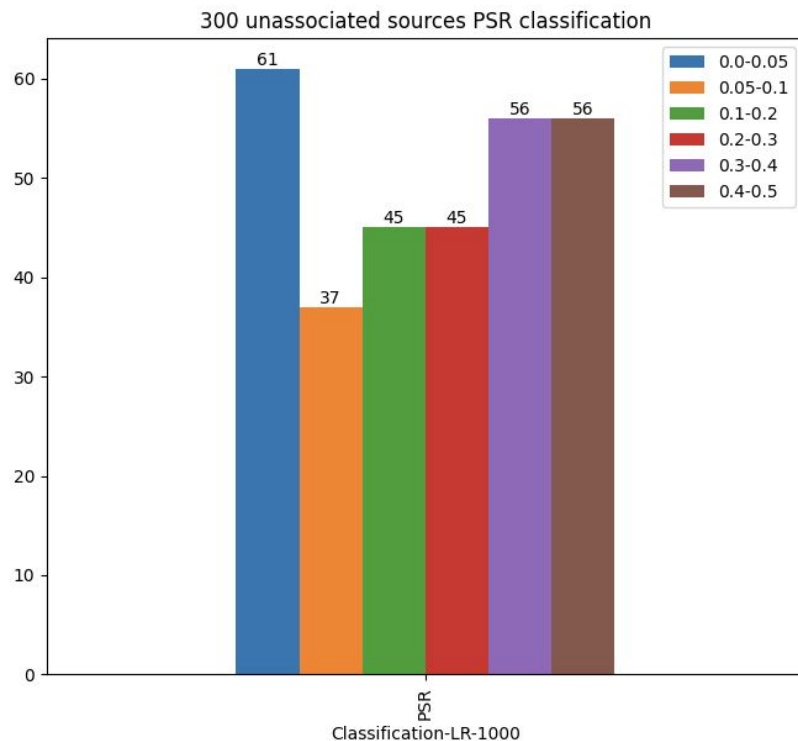
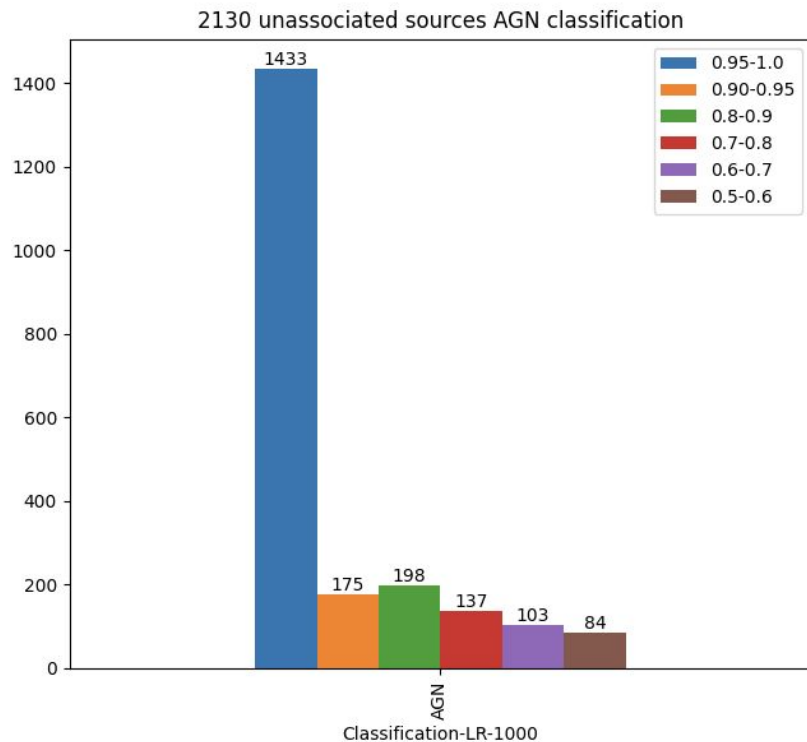


286 unassociated sources PSR classification



Logistic Regression

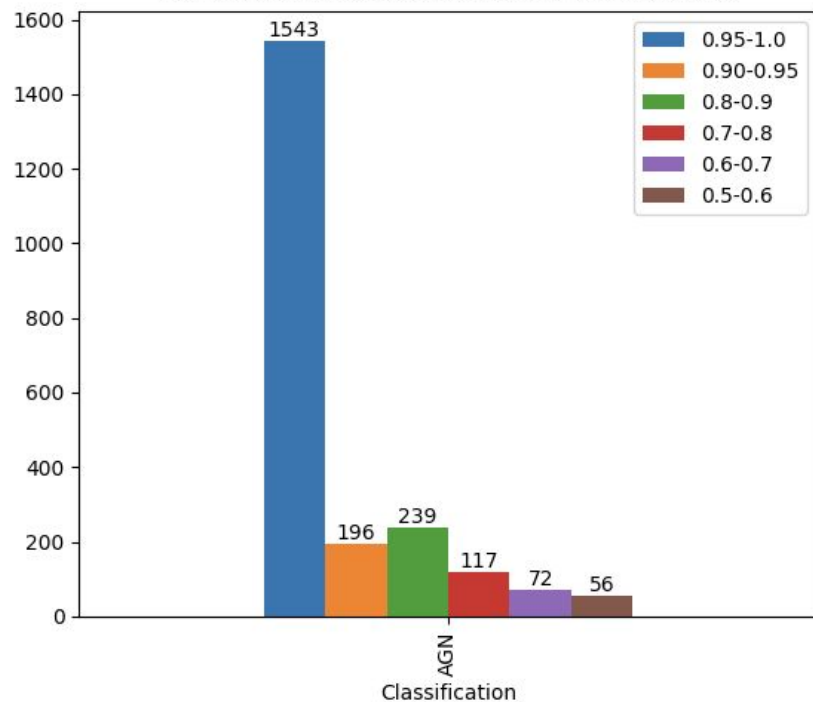
- # iterations = 1000 Total: 2430 unassociated sources



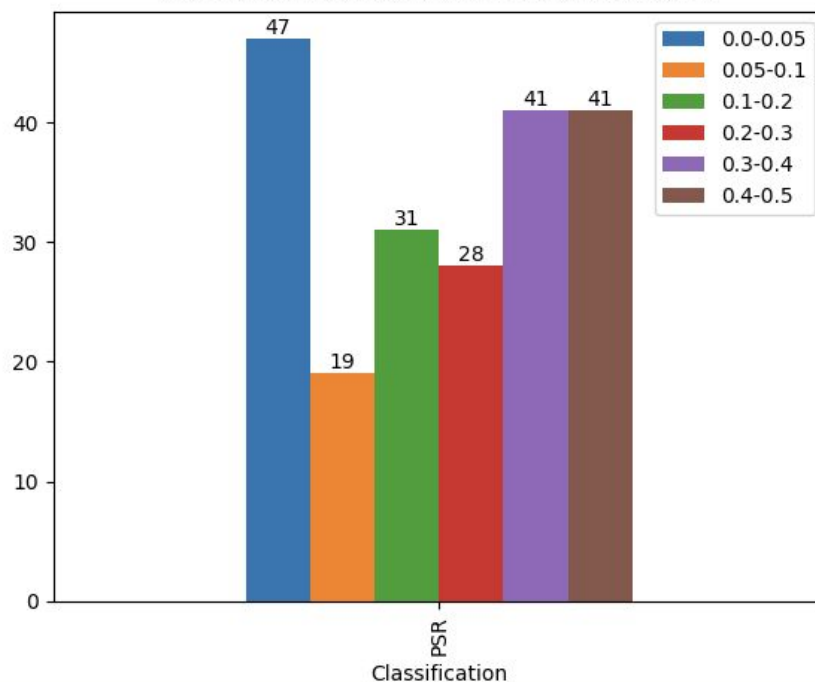
Neural Networks

- # iterations = 50 Total: 2430 unassociated sources

2223 unassociated sources AGN classification



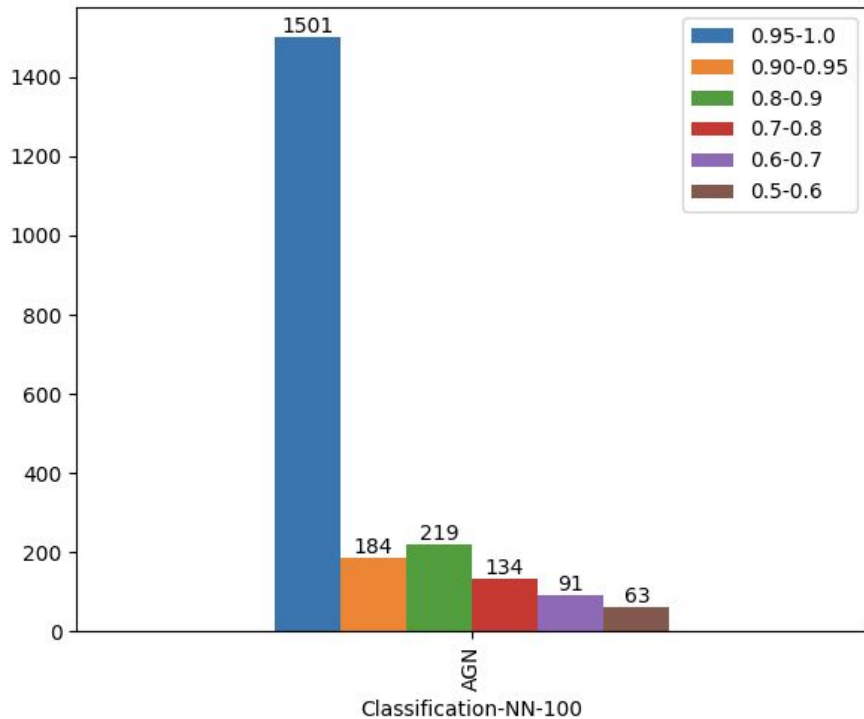
207 unassociated sources PSR classification



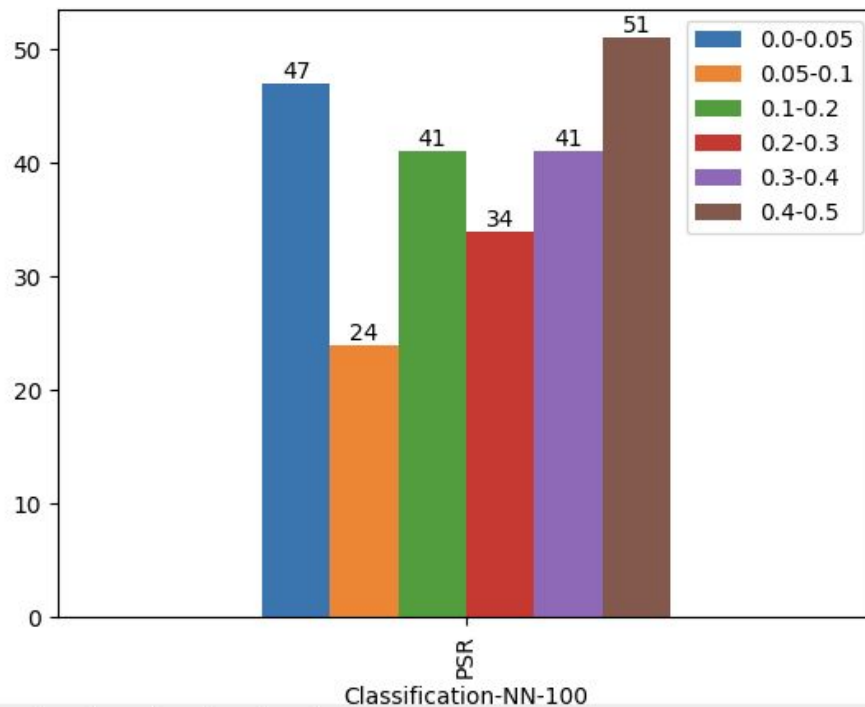
Neural Networks

- # iterations = 100 Total: 2430 unassociated sources

2192 unassociated sources AGN classification



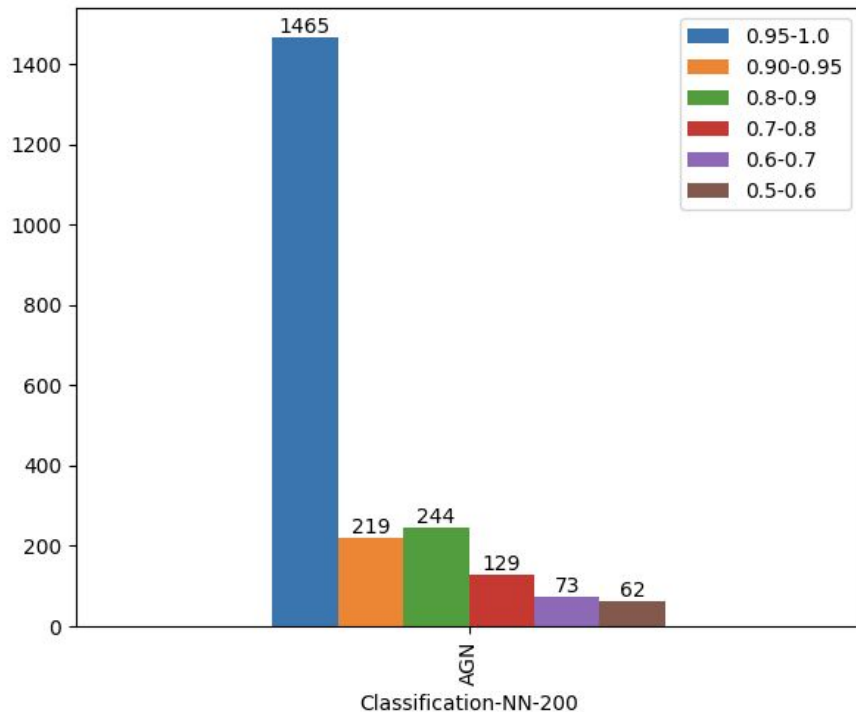
238 unassociated sources PSR classification



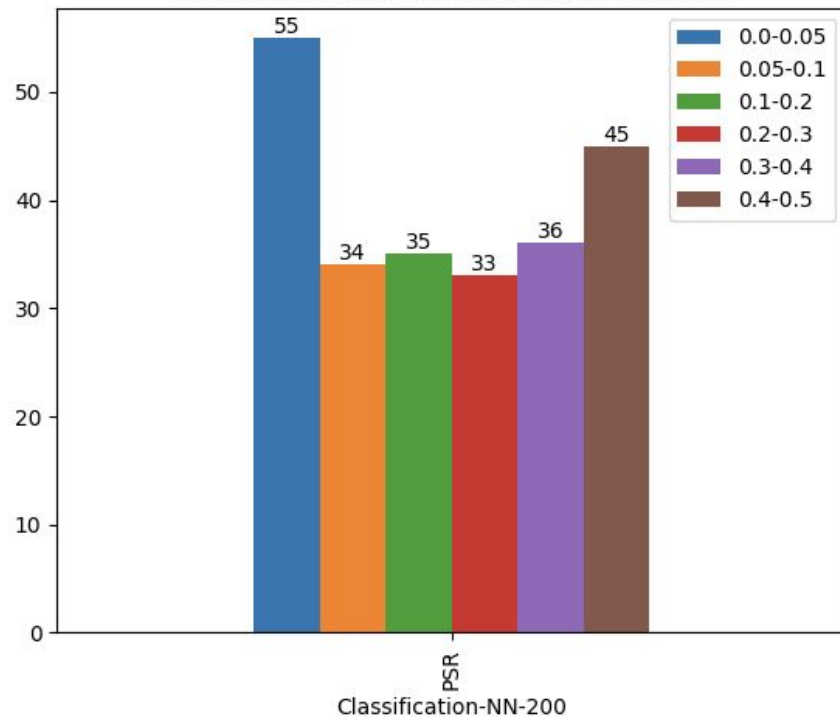
Neural Networks

- # iterations = 200 Total: 2430 unassociated sources

2192 unassociated sources AGN classification



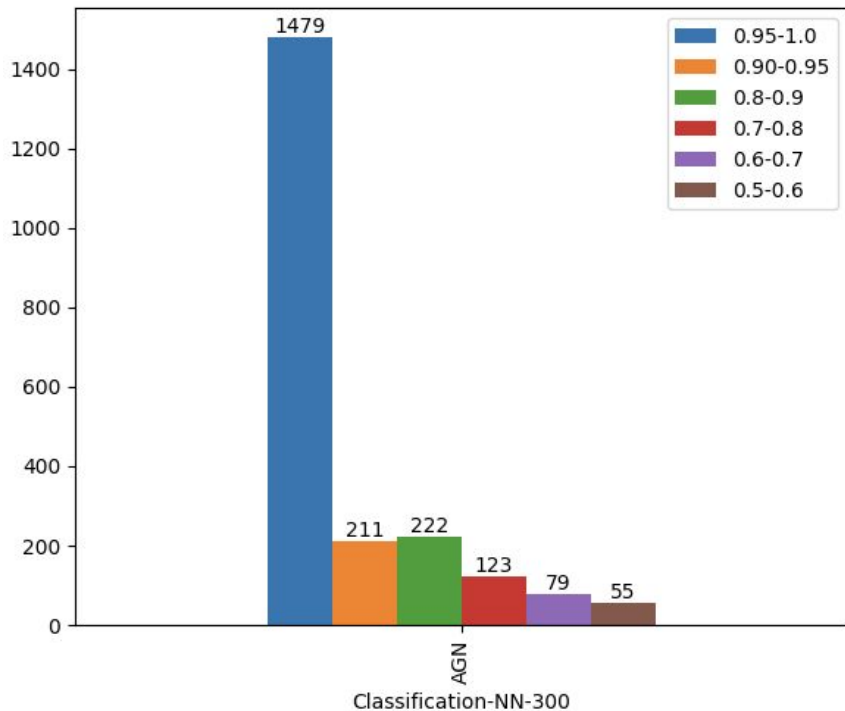
238 unassociated sources PSR classification



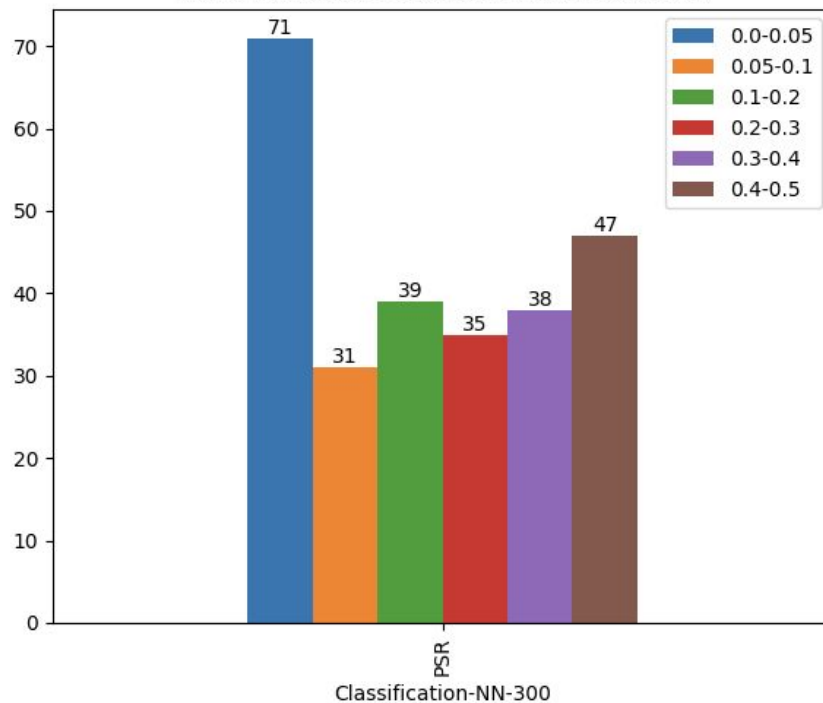
Neural Networks

- # iterations = 300 Total: 2430 unassociated sources

2169 unassociated sources AGN classification



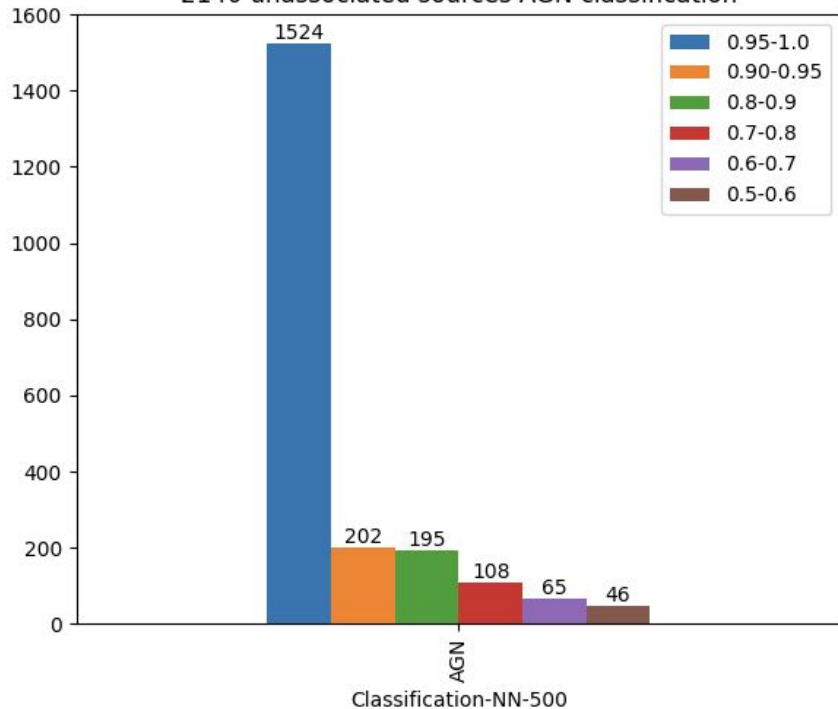
261 unassociated sources PSR classification



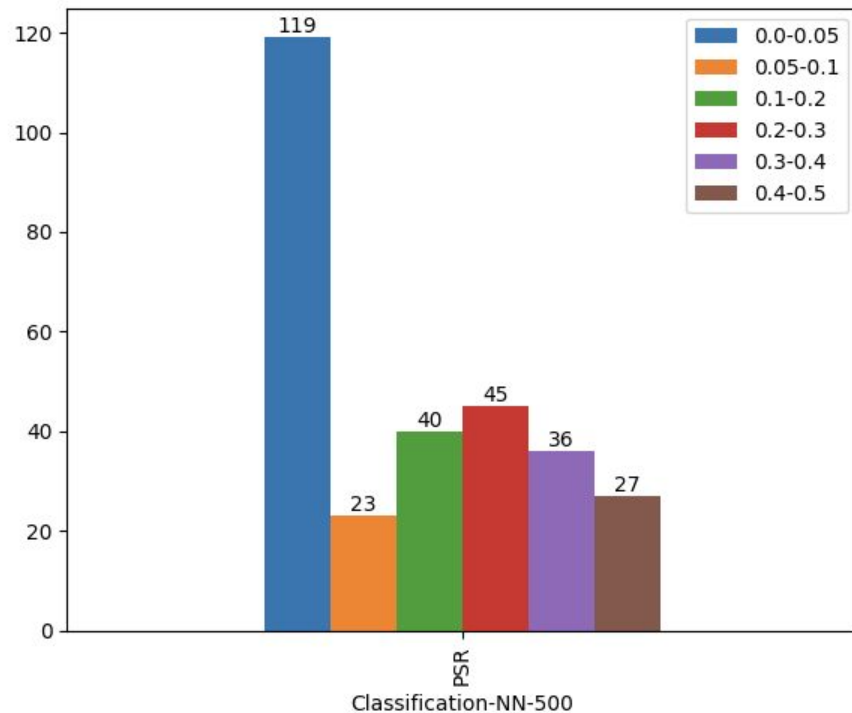
Neural Networks

- # iterations = 500 Total: 2430 unassociated sources

2140 unassociated sources AGN classification



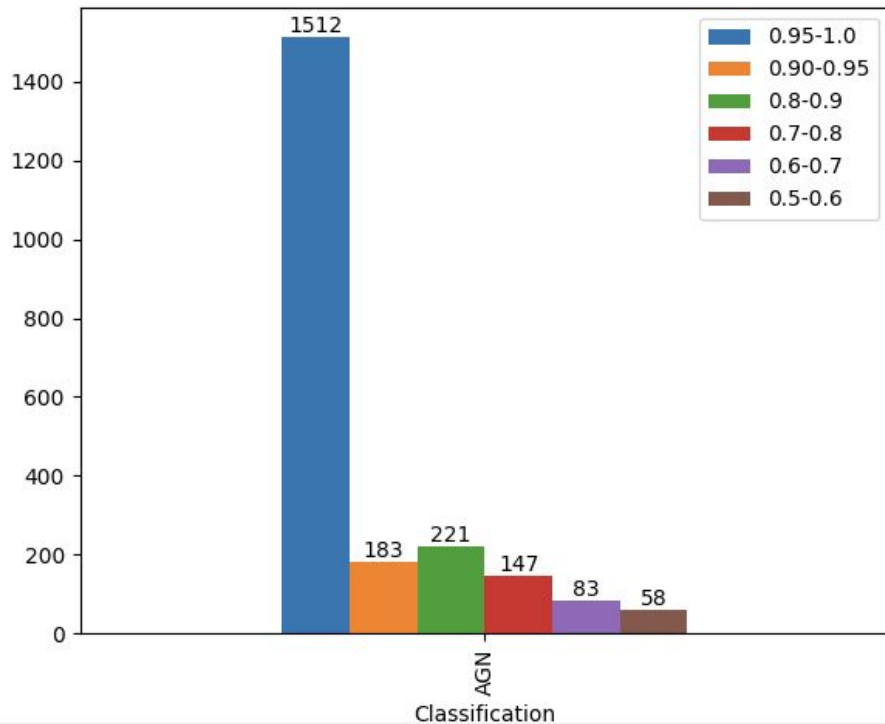
290 unassociated sources PSR classification



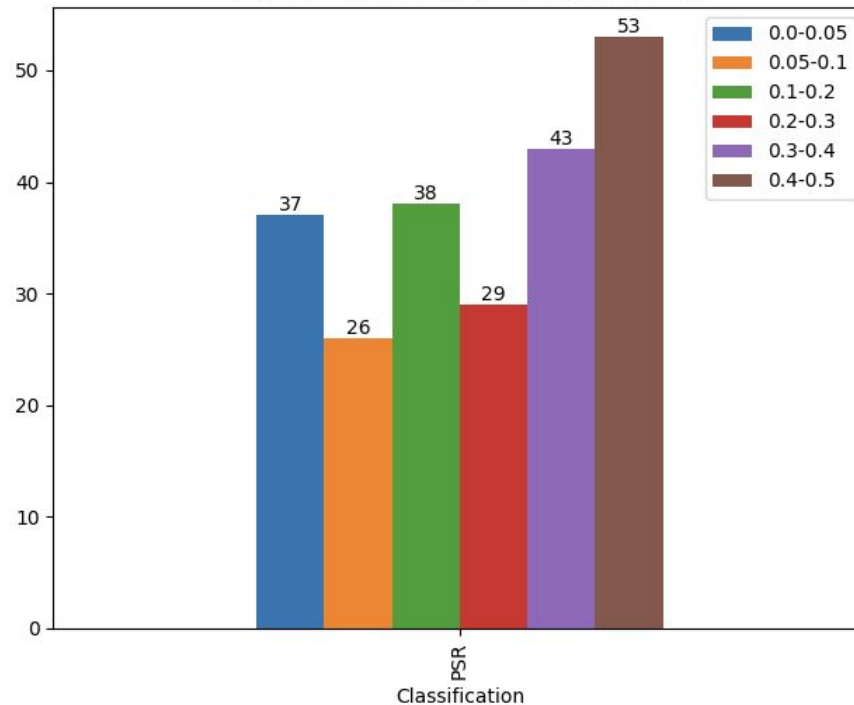
Deep Neural Networks

- # iterations = 50 Total: 2430 unassociated sources

2204 unassociated sources AGN classification



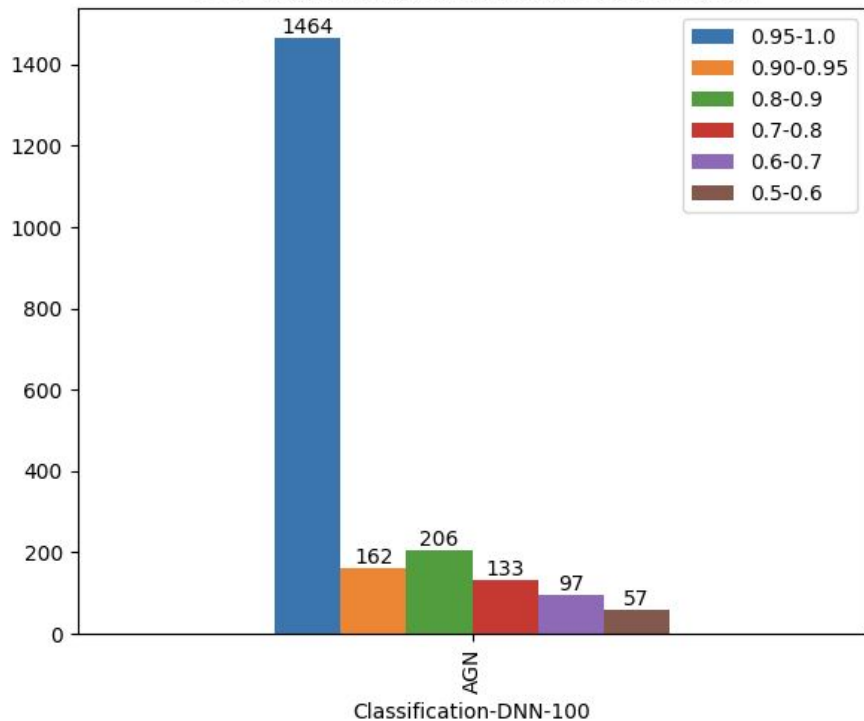
226 unassociated sources PSR classification



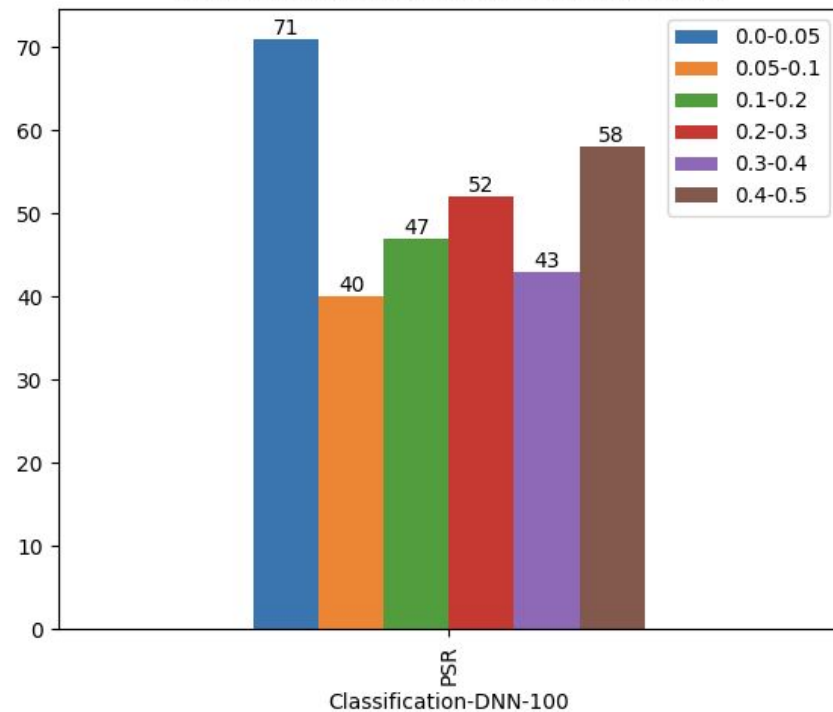
Deep Neural Networks

- # iterations = 100 Total: 2430 unassociated sources

2119 unassociated sources AGN classification



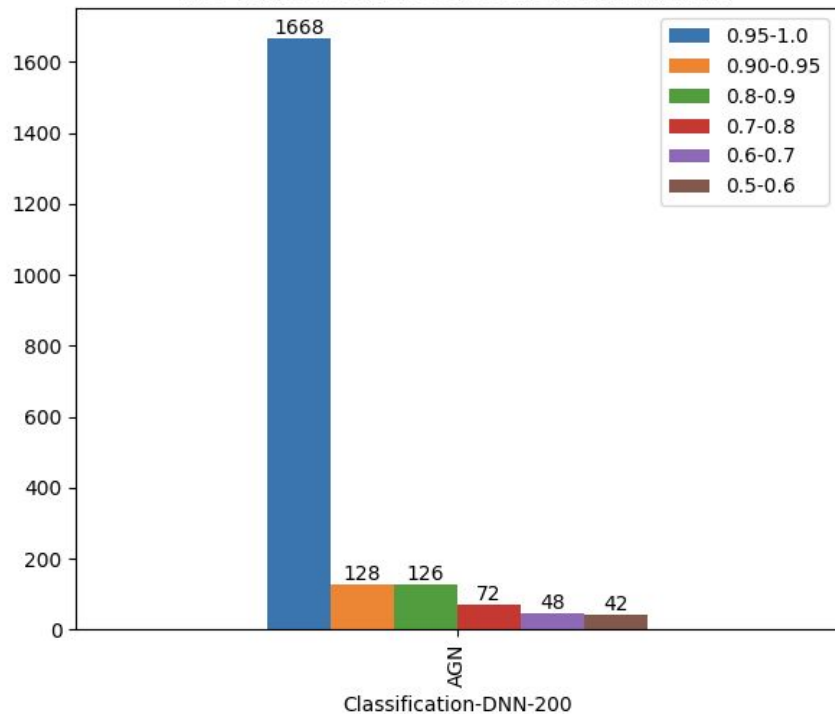
311 unassociated sources PSR classification



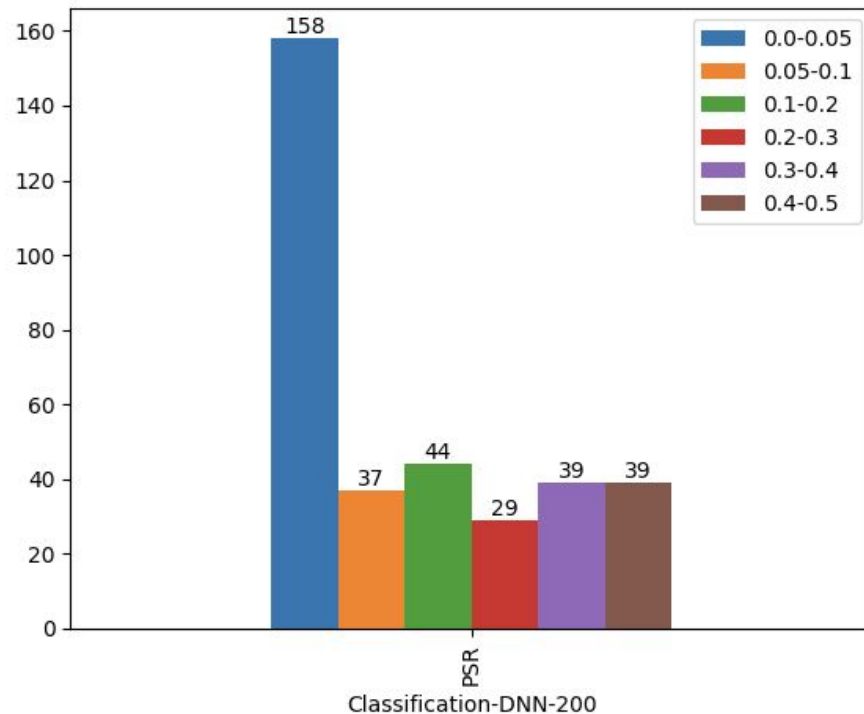
Deep Neural Networks

- # iterations = 200 Total: 2430 unassociated sources

2084 unassociated sources AGN classification

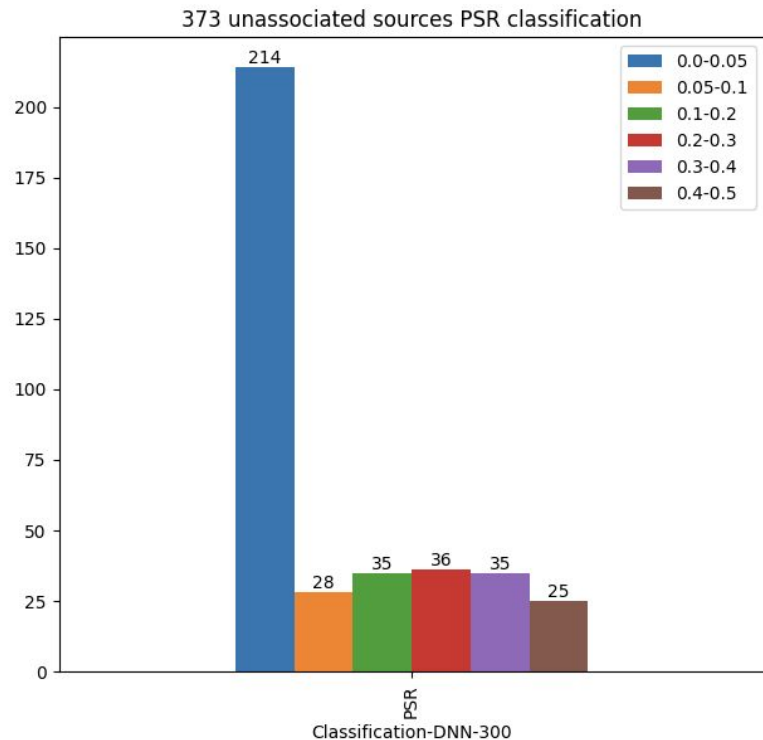
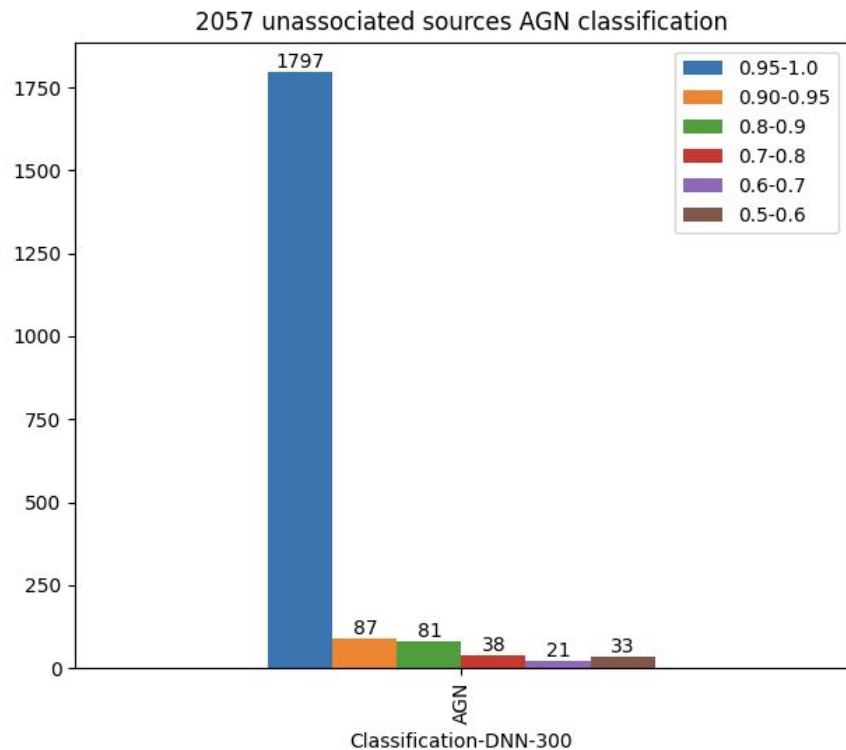


346 unassociated sources PSR classification



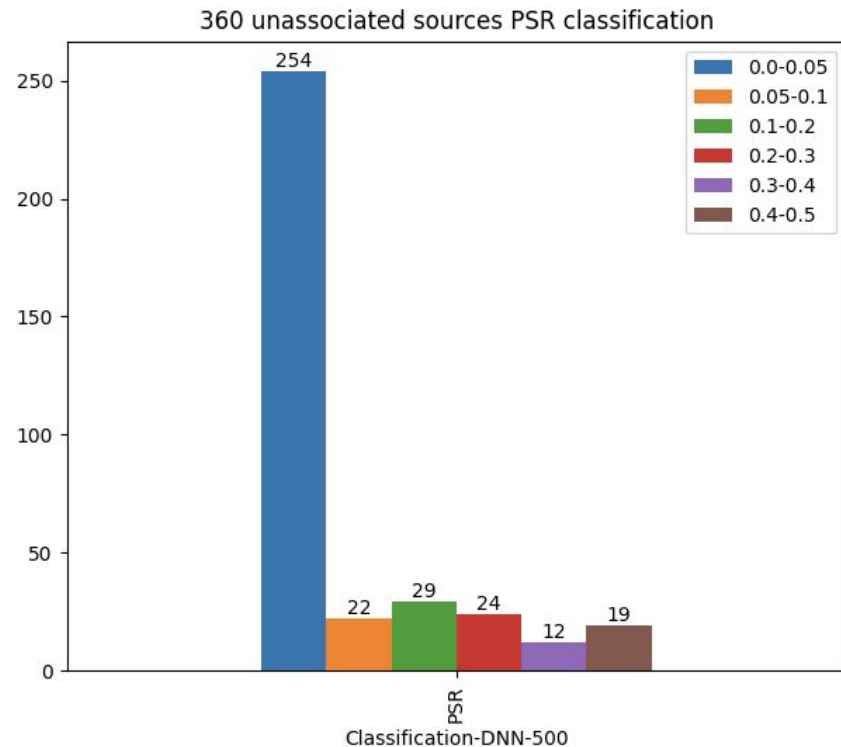
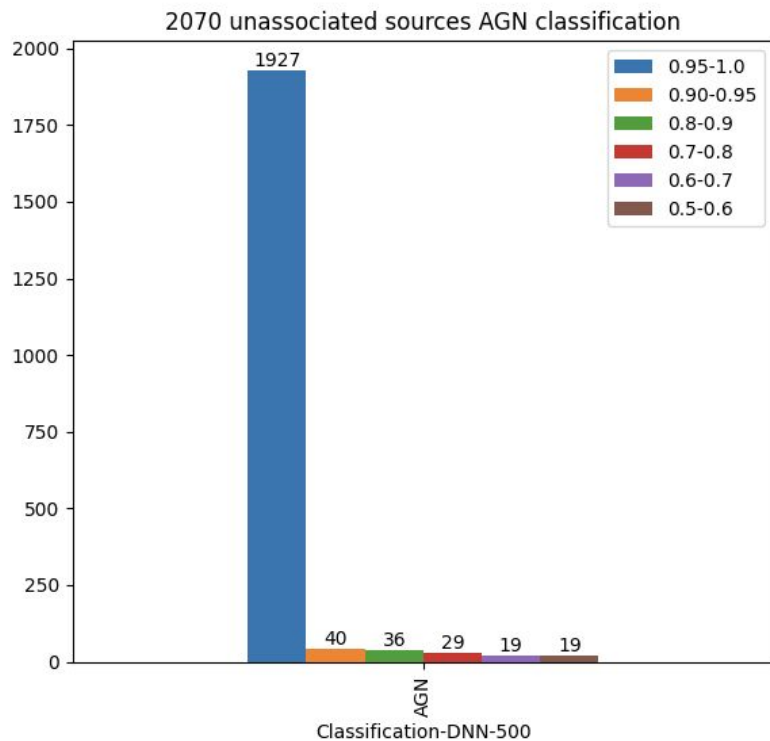
Deep Neural Networks

- # iterations = 300 Total: 2430 unassociated sources



Deep Neural Networks

- # iterations = 500 Total: 2430 unassociated sources

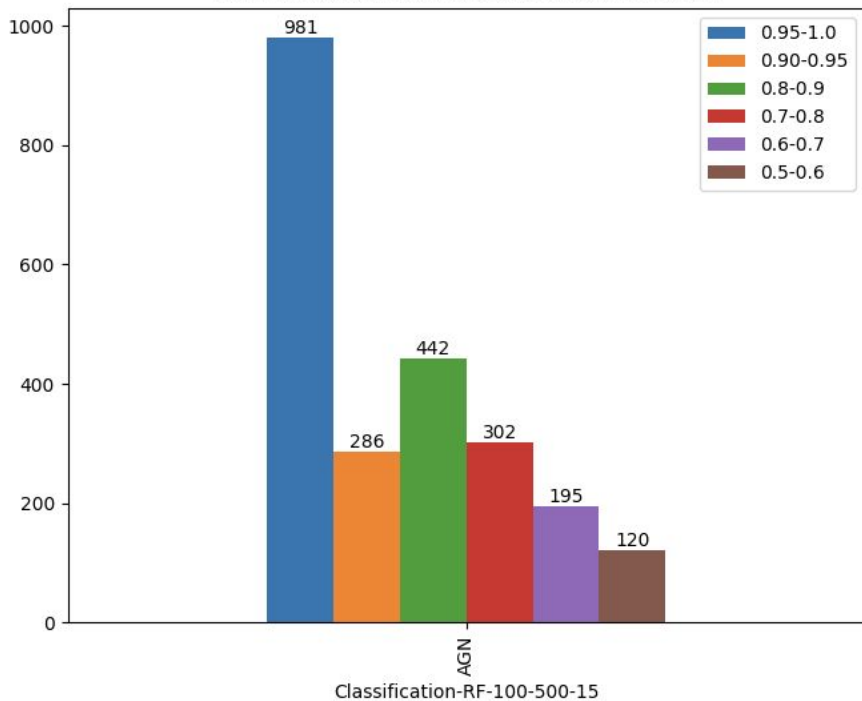


Random Forests

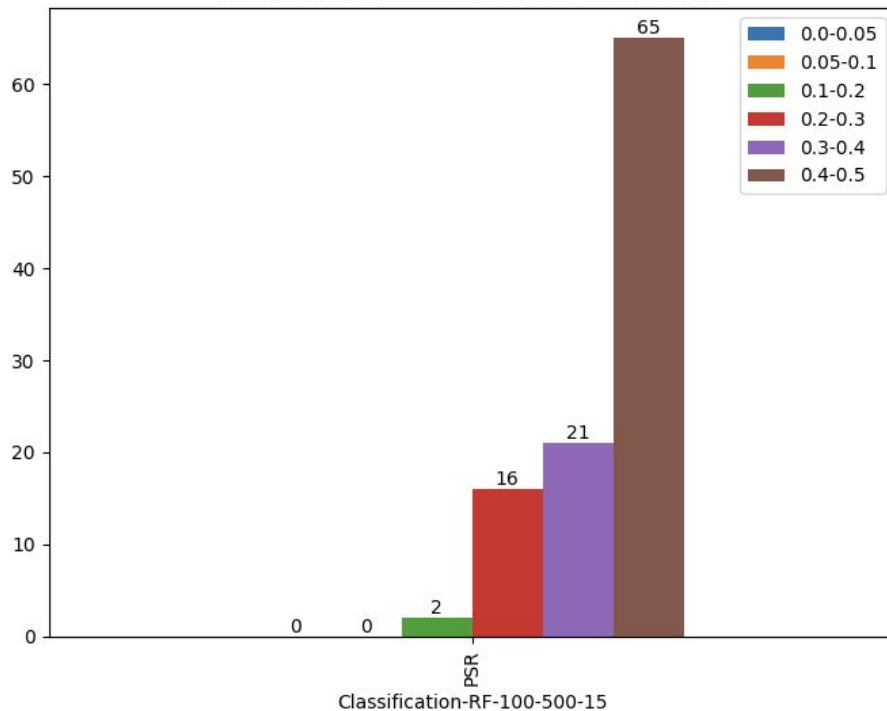
- #trees: 100, # samples: 500, depth: 15

Total: 2430 unassociated sources

2326 unassociated sources AGN classification



104 unassociated sources PSR classification



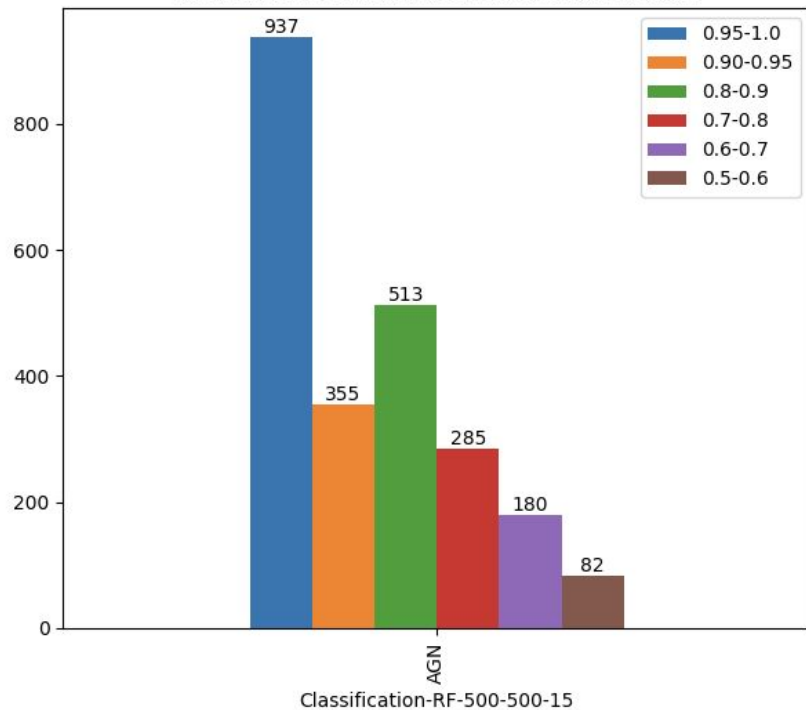
[illegible]

Random Forests

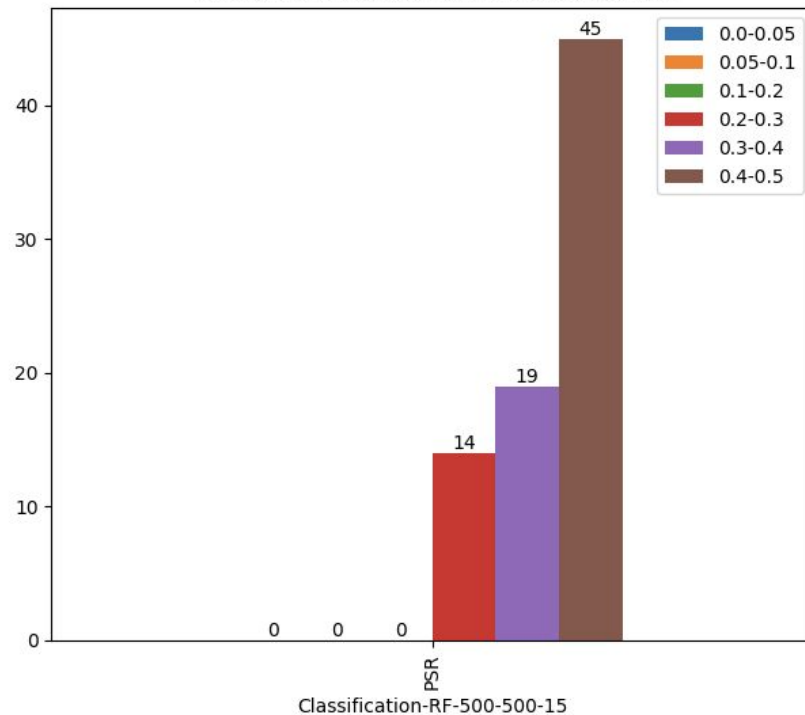
- #trees: 500, # samples: 500, depth: 15

Total: 2430 unassociated sources

2352 unassociated sources AGN classification



78 unassociated sources PSR classification

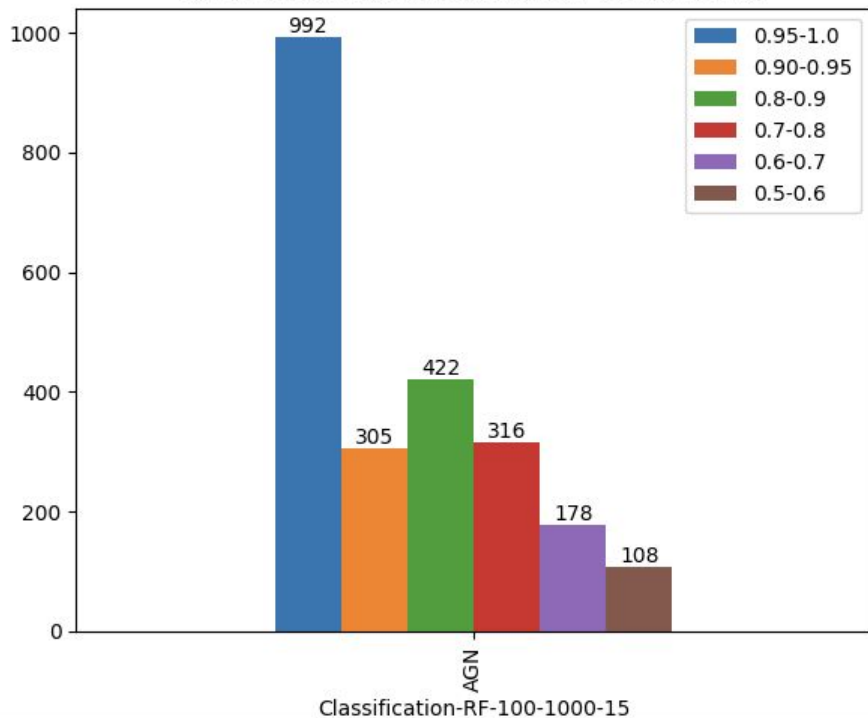


Random Forests

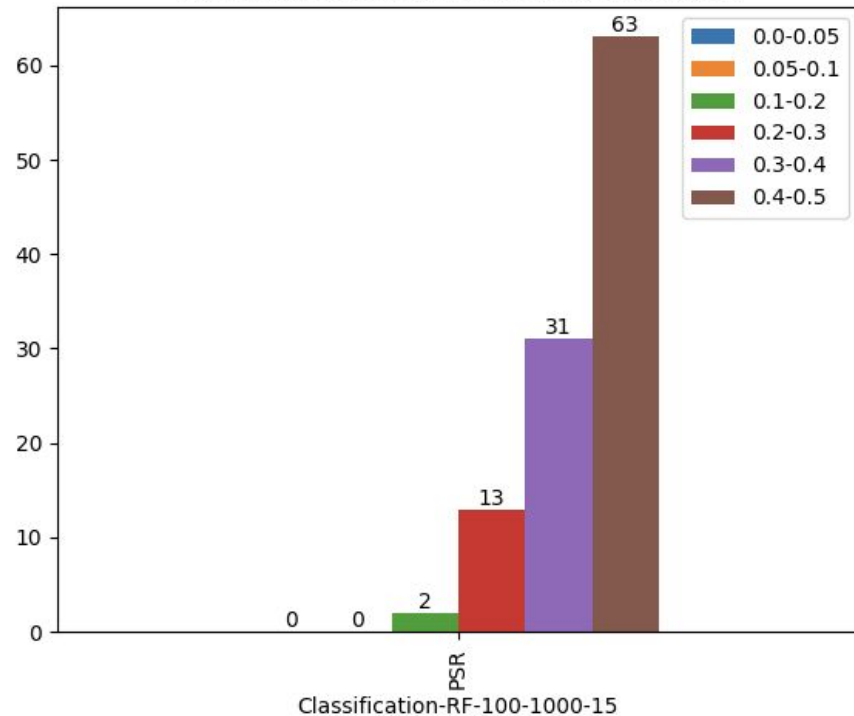
- #trees: 100, # samples: 1000, depth: 15

Total: 2430 unassociated sources

2321 unassociated sources AGN classification



109 unassociated sources PSR classification

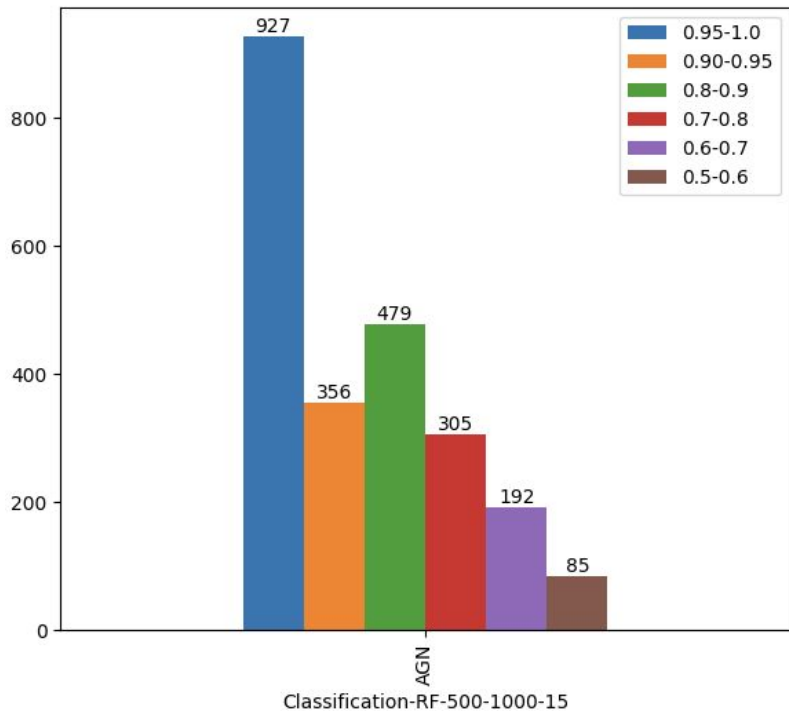


Random Forests

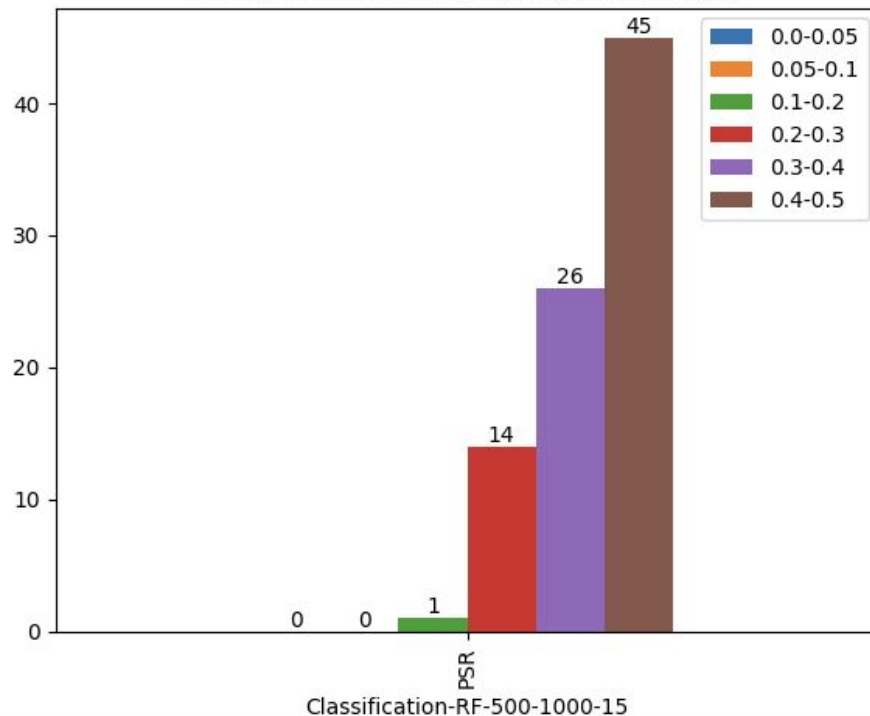
- #trees: 500, # samples: 1000, depth: 15

Total: 2430 unassociated sources

2344 unassociated sources AGN classification



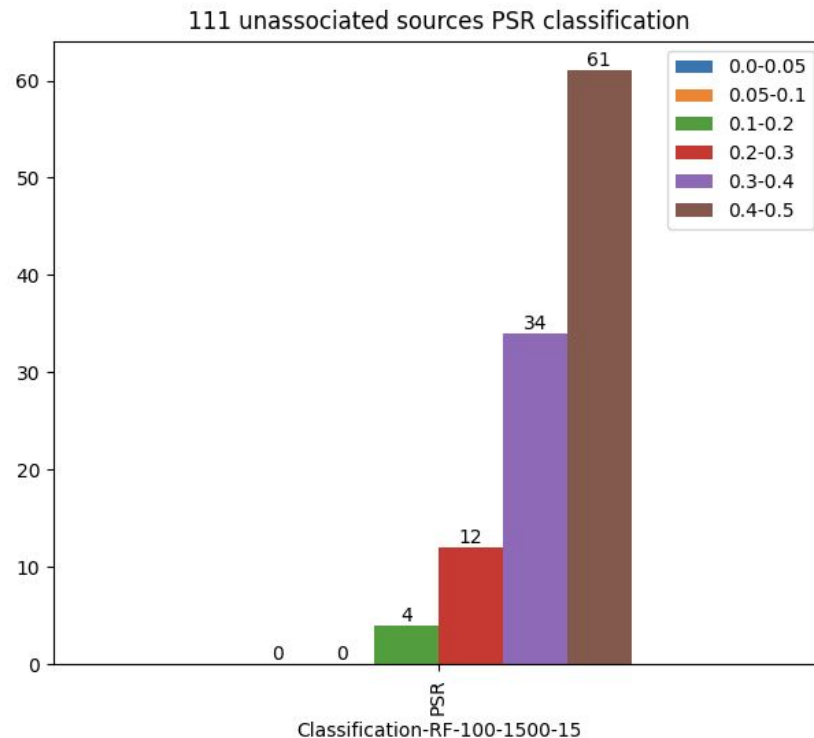
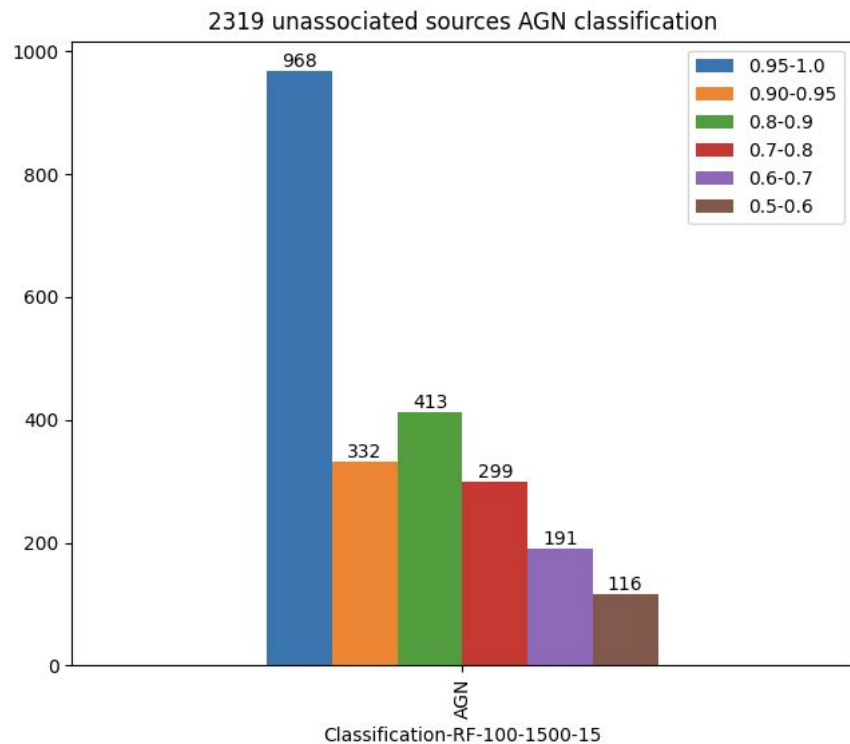
86 unassociated sources PSR classification



Random Forests

- #trees: 100, # samples: 1500, depth: 15

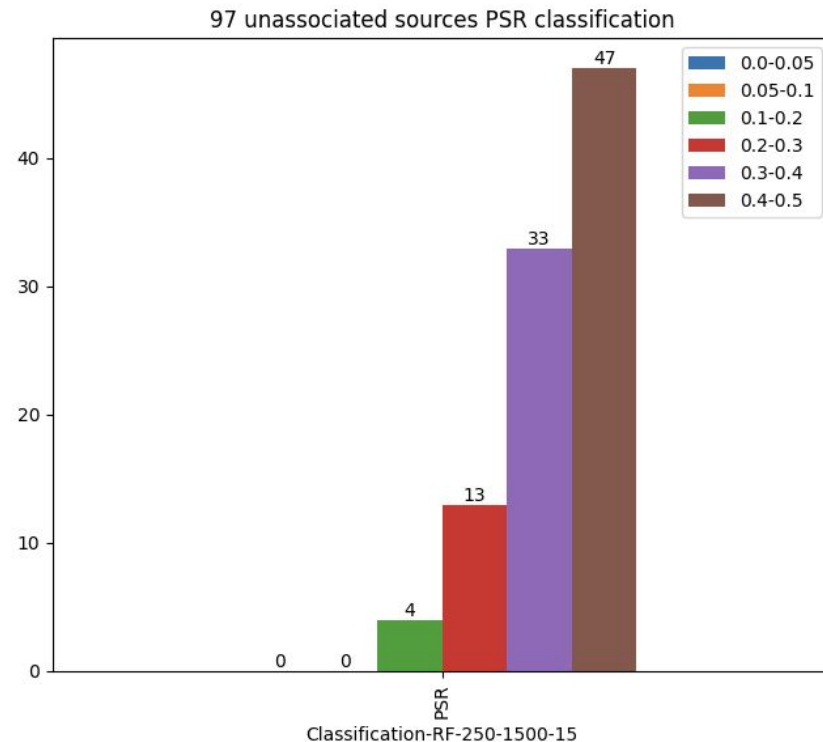
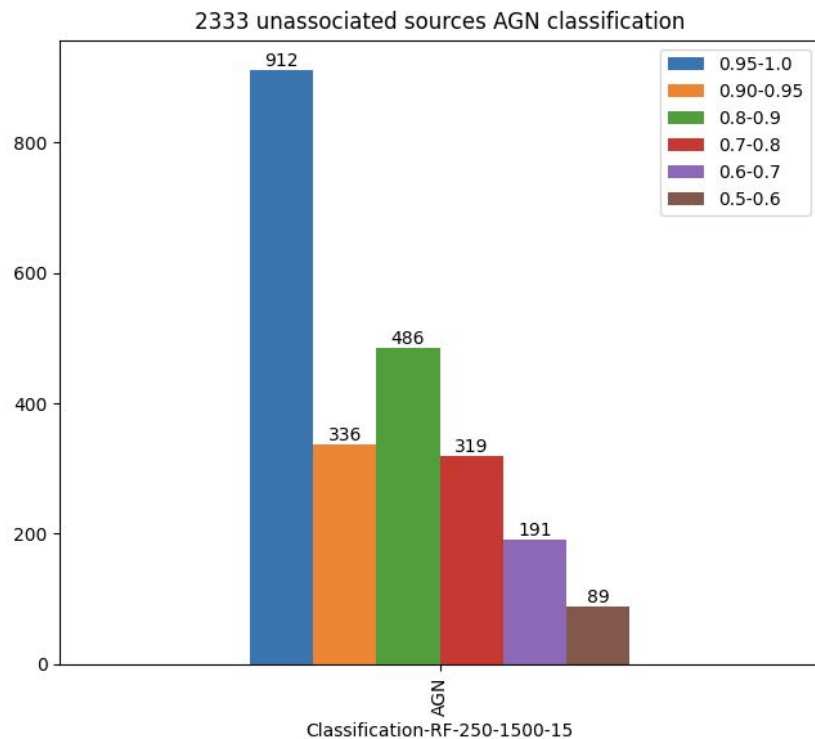
Total: 2430 unassociated sources



Random Forests

- #trees: 250, # samples: 1500, depth: 15

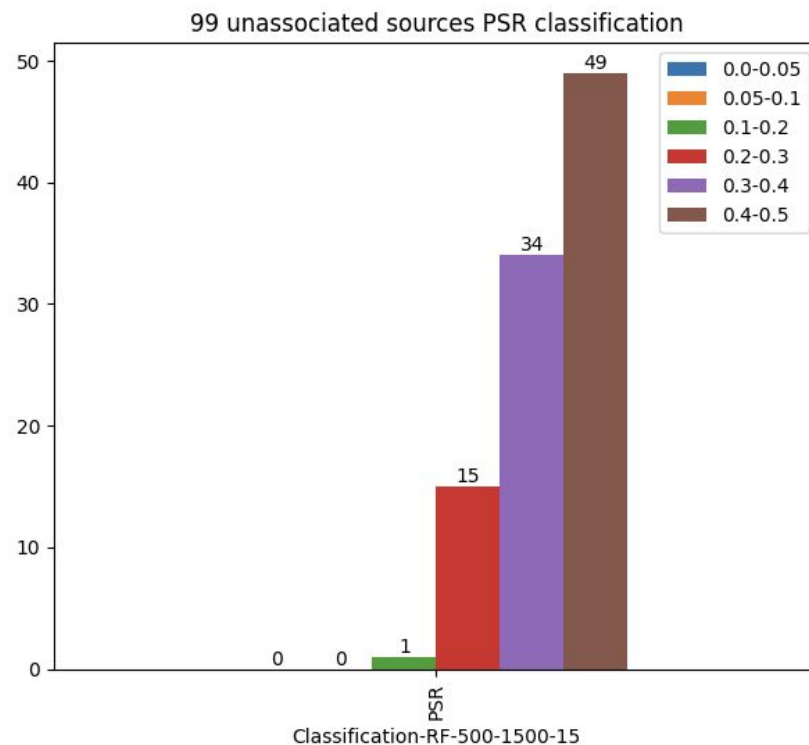
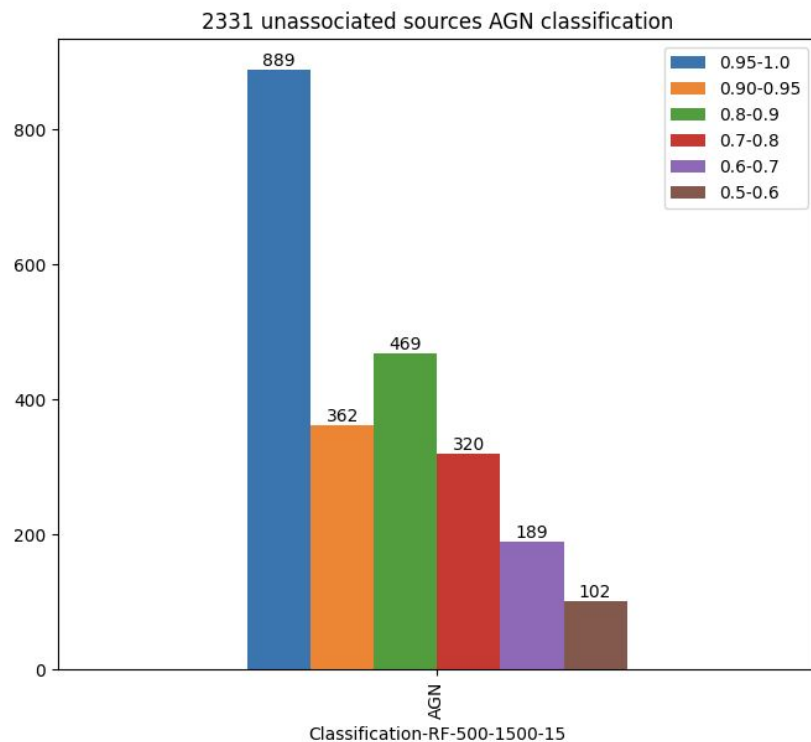
Total: 2430 unassociated sources



Random Forests

- #trees: 500, # samples: 1500, depth: 15

Total: 2430 unassociated sources

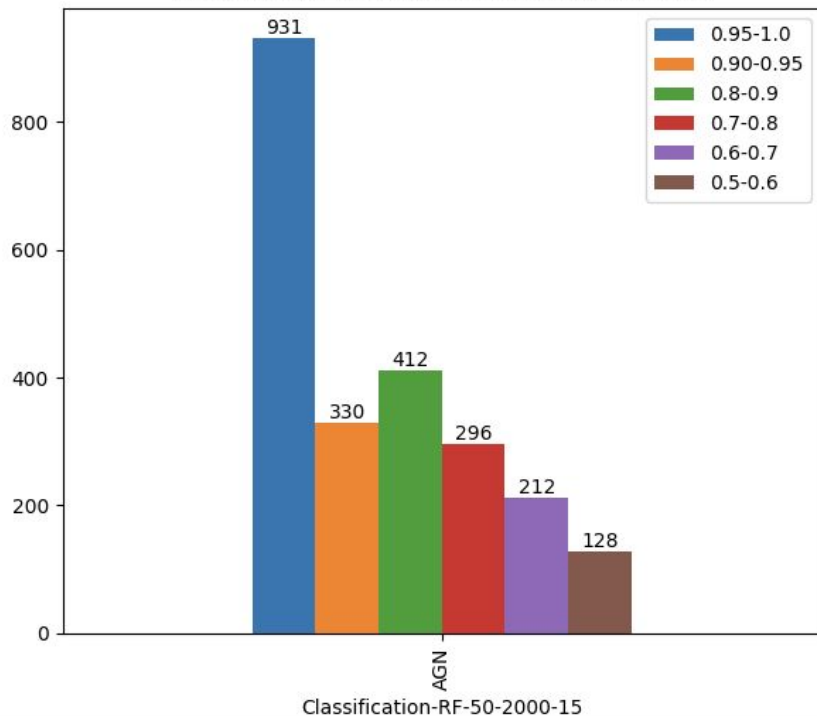


Random Forests

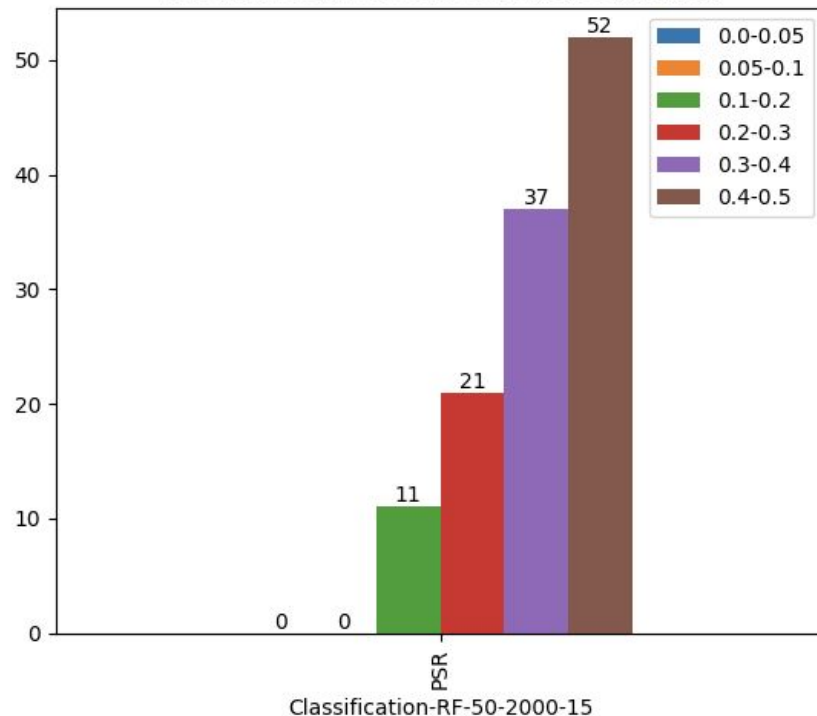
- #trees: 50, # samples: 2000, depth: 15

Total: 2430 unassociated sources

2309 unassociated sources AGN classification



121 unassociated sources PSR classification

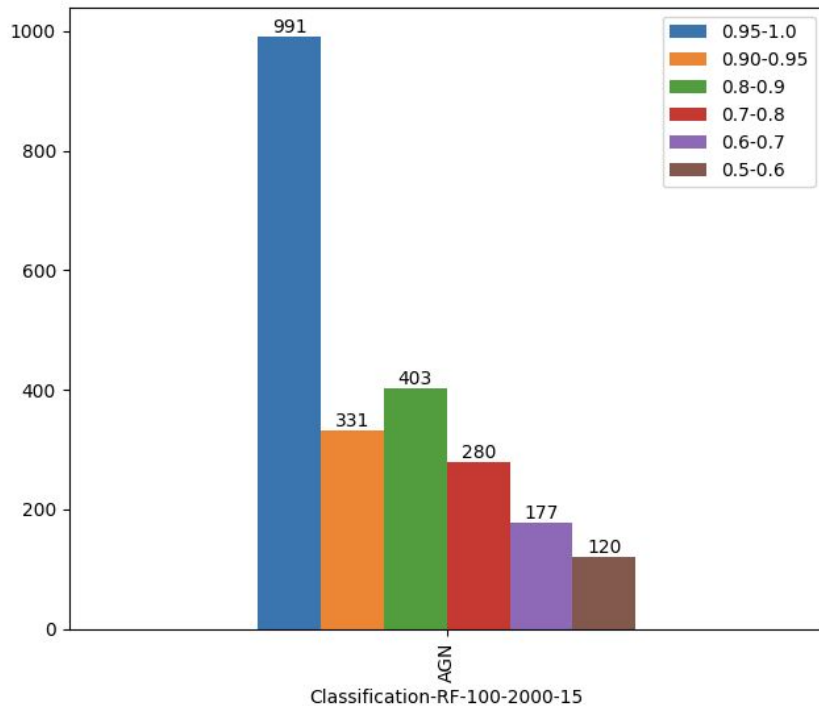


Random Forests

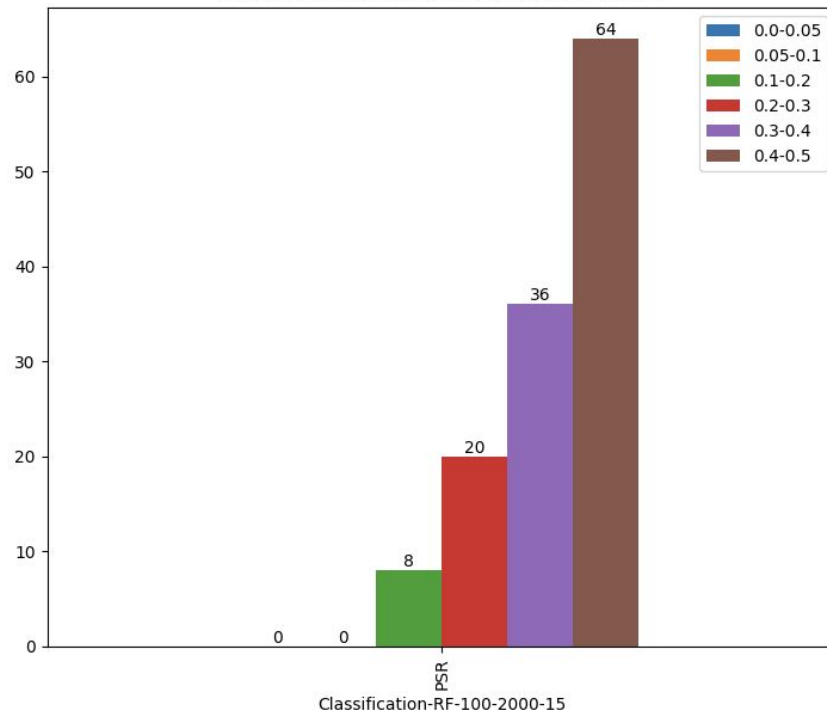
- #trees: 100, # samples: 2000, depth: 15

Total: 2430 unassociated sources

2302 unassociated sources AGN classification



128 unassociated sources PSR classification

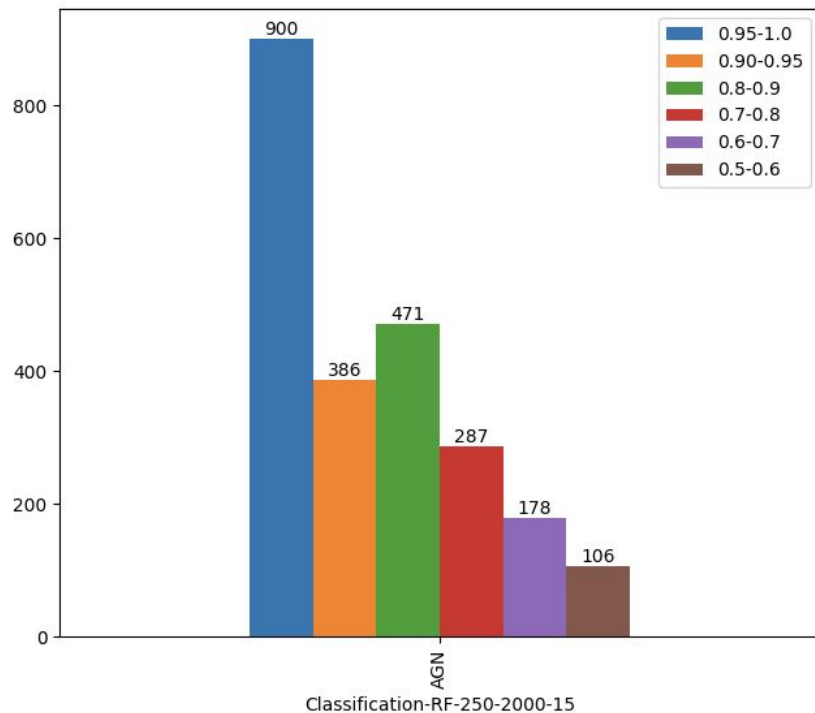


Random Forests

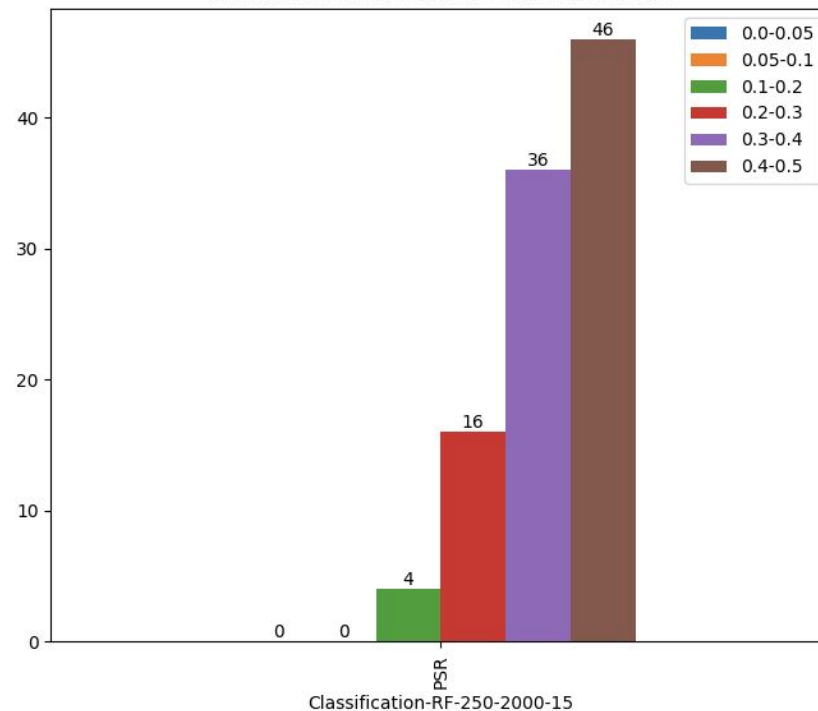
- #trees: 250, # samples: 2000, depth: 15

Total: 2430 unassociated sources

2328 unassociated sources AGN classification



102 unassociated sources PSR classification

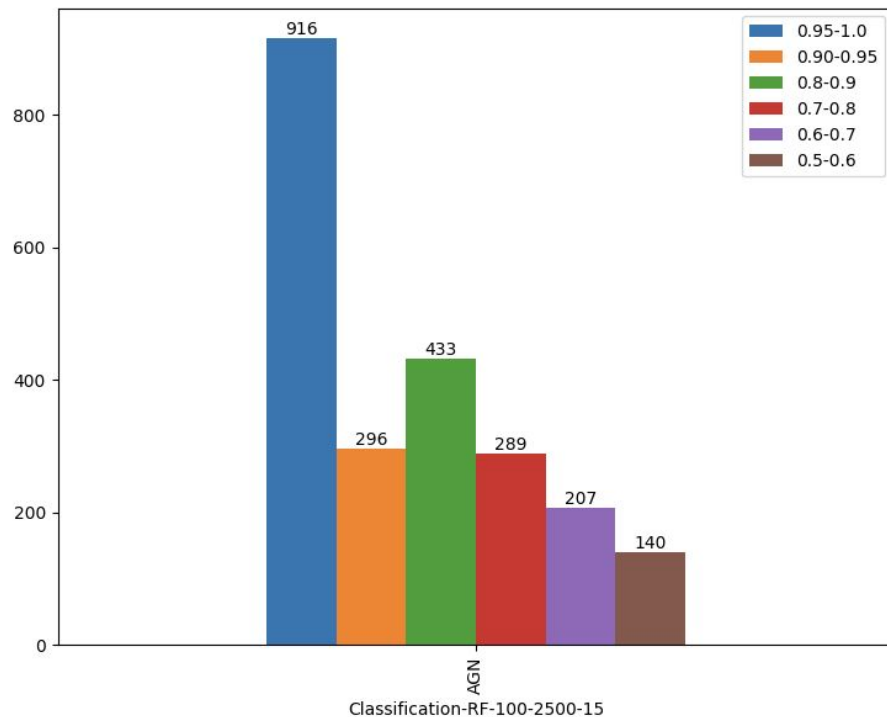


Random Forests

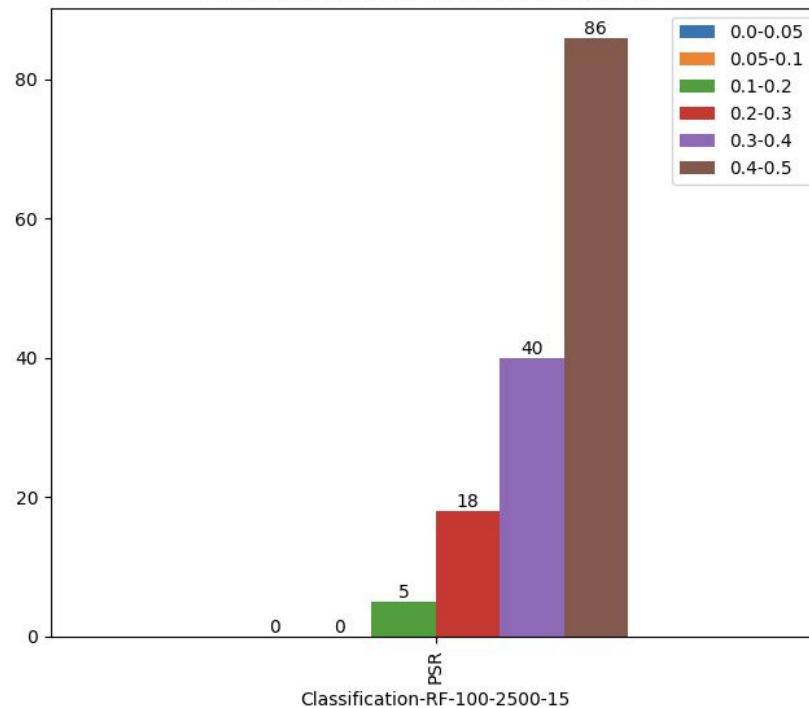
- #trees: 100, # samples: 2500, depth: 15

Total: 2430 unassociated sources

2281 unassociated sources AGN classification



149 unassociated sources PSR classification

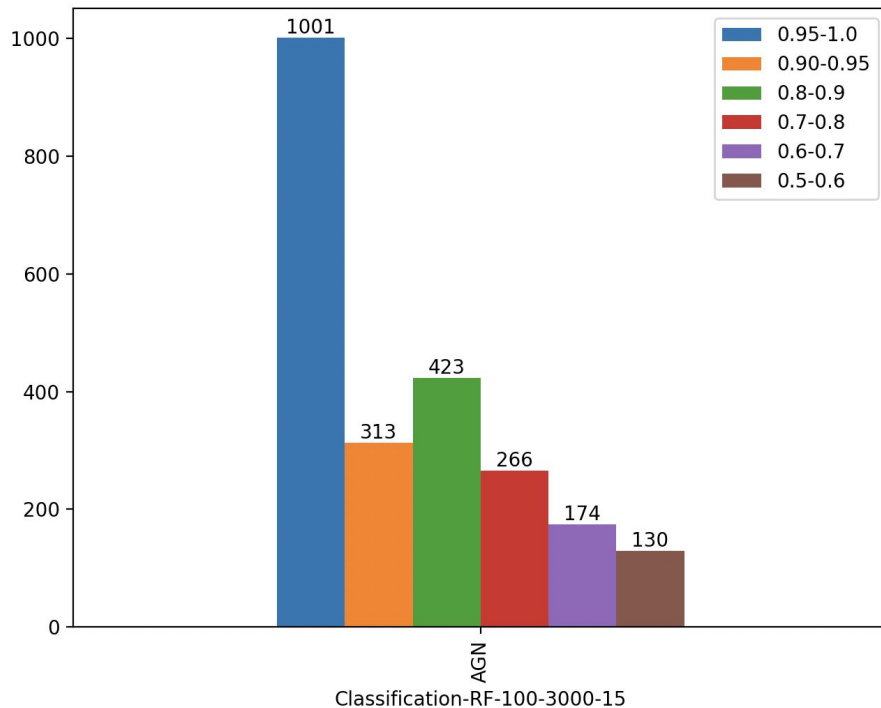


Random Forests

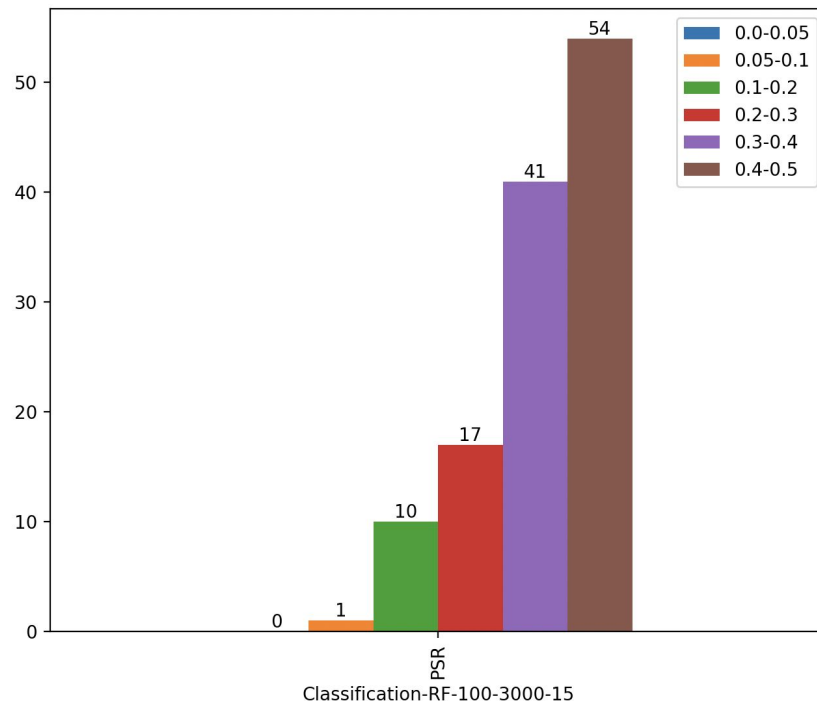
- #trees: 100, # samples: 3000, depth: 15

Total: 2430 unassociated sources

2307 unassociated sources AGN classification



123 unassociated sources PSR classification



Results

- LR:
 - Firmly predicted 1433-1465 AGNs in [0.95-1.0] range
 - Firmly predicted 58-65 PSRs in [0.0-0.05] range until 200 iterations.
- Neural Networks
 - Firmly predicted 1465-1543 AGNs [0.95-1.0] range
 - Firmly predicted 47-119 PSRs [0.0-0.05] range
- Deep Neural Networks
 - Firmly predicted 1464-1927 AGNs [0.95-1.0] range
 - Firmly predicted 37-254 PSRs [0.0-0.05] range
 - More iterations predicted more AGNs in [0.95-1.0] range and more PSRs in [0.0-0.05] range
- Random Forests
 - Firmly predicted 927-992 AGNs [0.95-1.0] range
 - Firmly predicted 0 PSRs in the [0.0-0.05] range
 - Predicted 0-4 PSRs in the [0.5-0.10] range
 - Weakly predicted 45-65 PSRs in [0.45-0.5] range

Update: 03/22/2024

Progress

- Solved the **unbalanced dataset problem** by multiplying total count of Pulsar rows
- So to multiply by 10, created 9 more rows for each Pular row

Code Changes

```
unit.append(float(source[wanted_feature_names[9]]))
unit.append(float(source[wanted_feature_names[10]]))
unit.append(float(source[wanted_feature_names[11]]))
unit.append(float(source[wanted_feature_names[12]]))
# Calculate hardness ratios (also considered in the paper)
for index in range(1, len(flux_list)):
    unit.append((flux_list[index] - flux_list[index-1] + 0.0) /
                (flux_list[index] + flux_list[index-1] + 0.0))
xdata.append(unit)
if source[wanted_type_col] in ("PSR", "psr", "MSP", "msp"):
    ydata.append([0]) # 0 is for pulsar
else:
    ydata.append([1]) # 1 is for AGN
# Multiply Pulsar rows by 10. Create 9 more rows for each Pulsar row
if source[wanted_type_col] in ("PSR", "psr", "MSP", "msp"):
    for i in range(9):
        xdata.append(unit)
        ydata.append([0]) # 0 is for pulsar
self.x = torch.from_numpy(np.array(xdata, float))
self.y = torch.from_numpy(np.array(ydata, float))
self.size = len(xdata)
```

Multiply Pulsar rows
by creating 9 more
rows for each Pulsar
row to get 10x rows

Code Changes

- Create **3 more PSR rows** for each PSR row

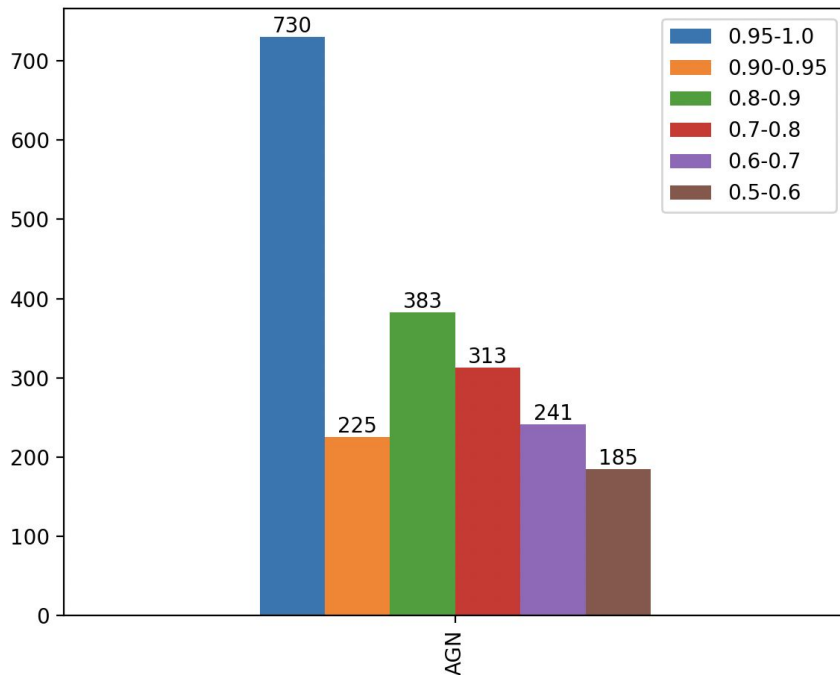
```
correct = 1527
test_size = 1587
correct_agm = 1170
total_agm = 1187 ←
correct_psr = 357
total_psr = 400 ←
sensitivity = 96.21928166351607
agm_sensitivity = 98.56781802864364
psr_sensitivity = 89.25
```

Code Changes

- #trees: 20, # samples: 500, depth: 15

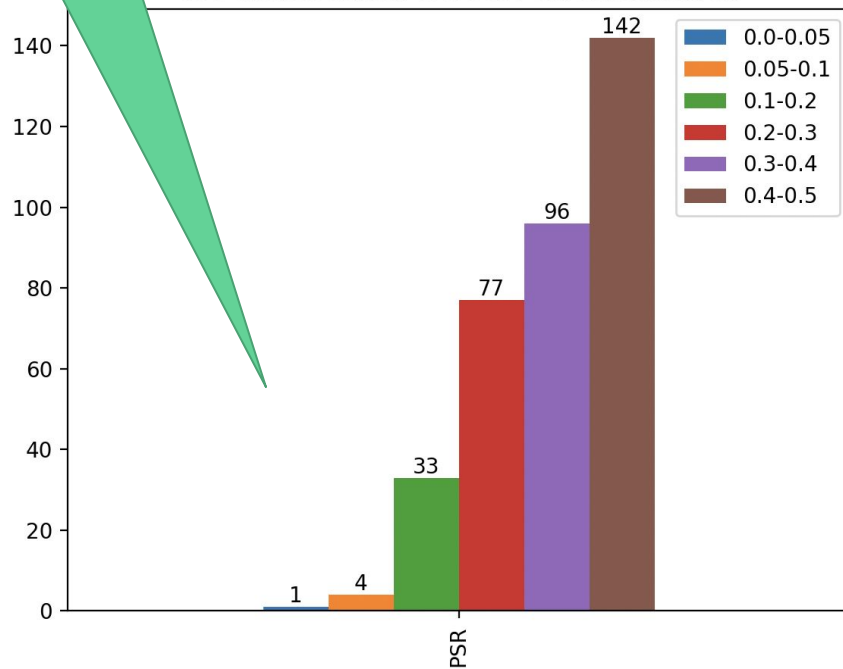
Now PSRs have
been strongly
identified

2077 unassociated sources AGN classification



Classification-RF-20-500-15

353 unassociated sources PSR classification



Classification-RF-20-500-15

Code Changes

- Create **4 more PSR rows** for each PSR row

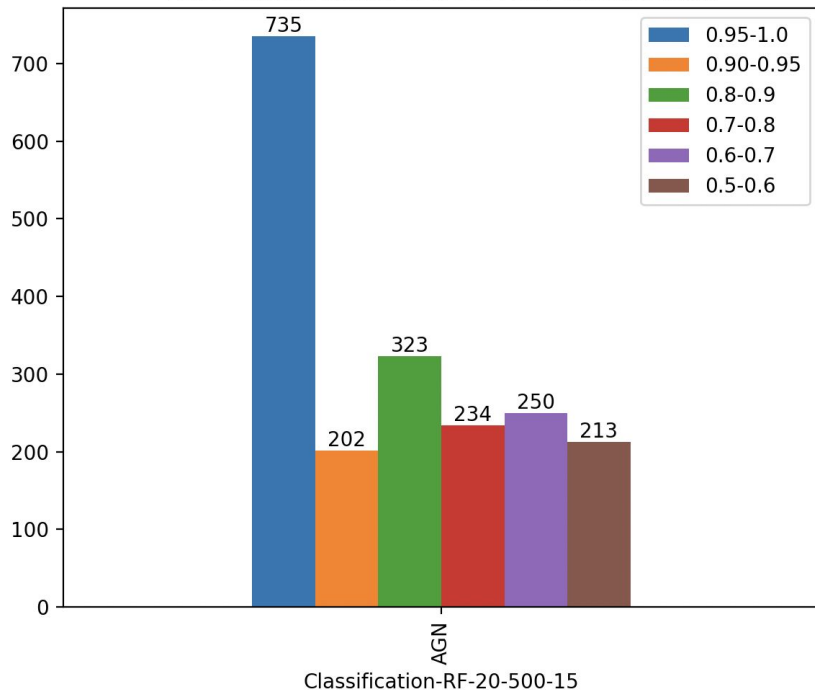
```
correct = 1649
test_size = 1683
correct_agm = 1171
total_agm = 1190 ←
correct_psr = 478
total_psr = 493 ←
sensitivity = 97.97979797979798
agm_sensitivity = 98.40336134453781
psr_sensitivity = 96.95740365111561
```

Code Changes

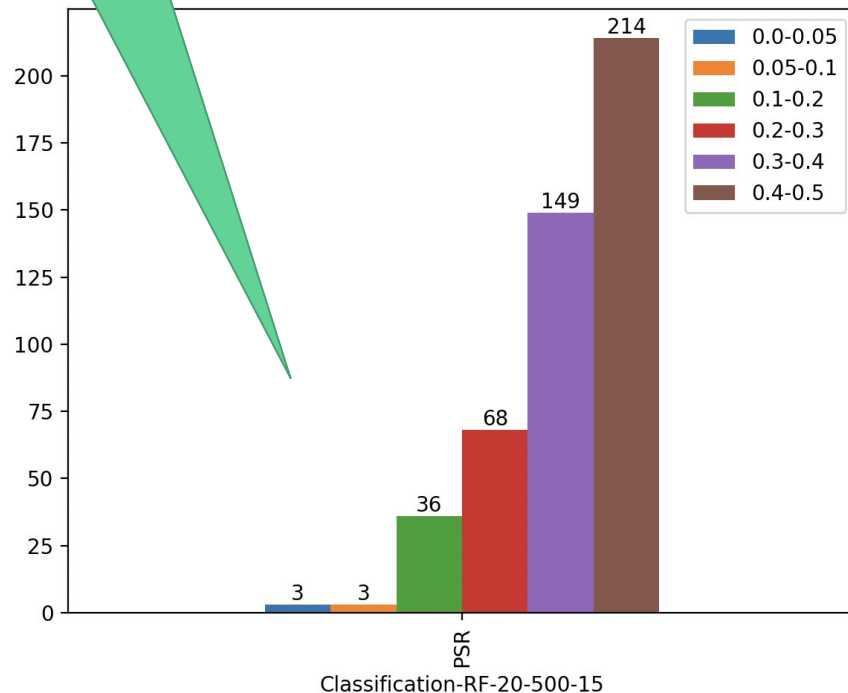
- #trees: 20, # samples: 500, depth: 15

Now PSRs have
been strongly
identified

1957 unassociated sources AGN classification



473 unassociated sources PSR classification



Code Changes

- Create **9 more PSR rows** for each PSR row

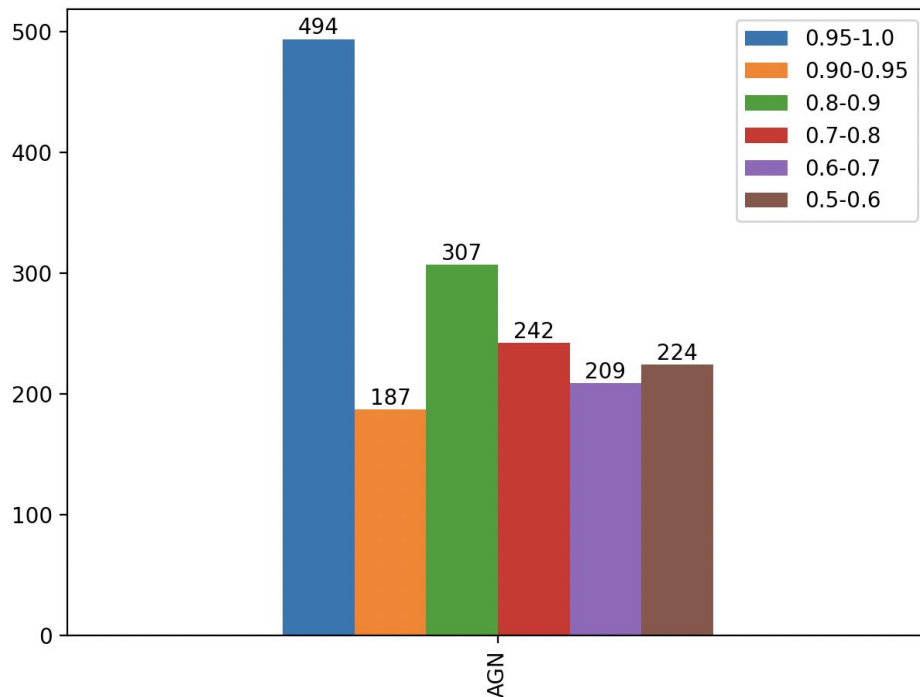
```
correct = 2112
test_size = 2163
correct_agm = 1169
total_agm = 1196 ←
correct_psr = 943
total_psr = 967 ←
sensitivity = 97.64216366158114
agm_sensitivity = 97.74247491638796
psr_sensitivity = 97.51809720785936
```

Code Changes

- #trees: 20, # samples: 500, depth: 15

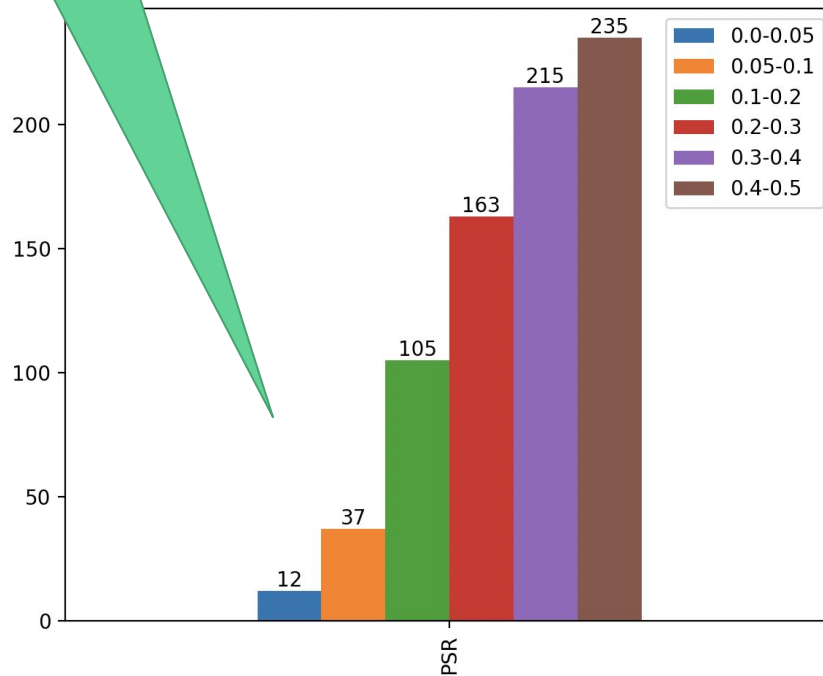
Now PSRs have
been strongly
identified

1663 unassociated sources AGN classification



Classification-RF-20-500-15

767 unassociated sources PSR classification



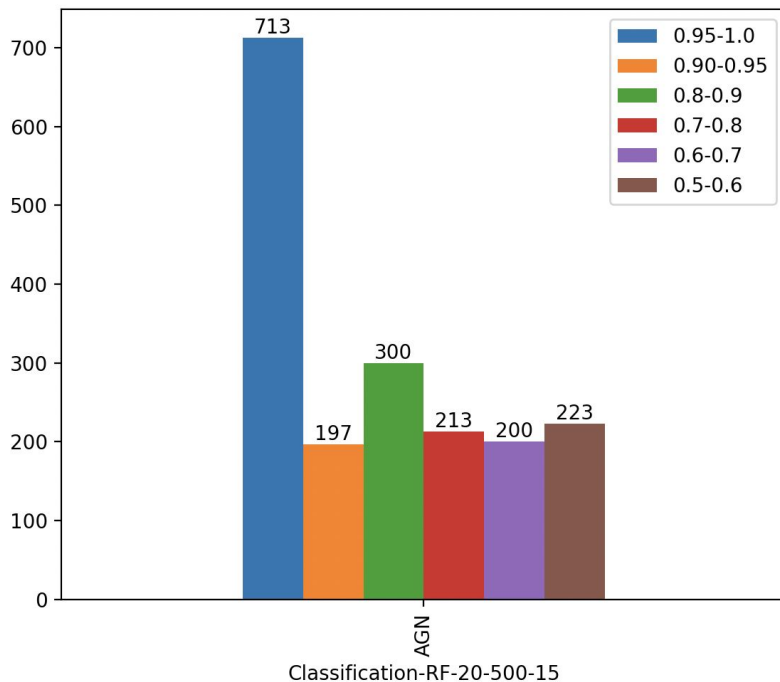
Classification-RF-20-500-15

Code Changes

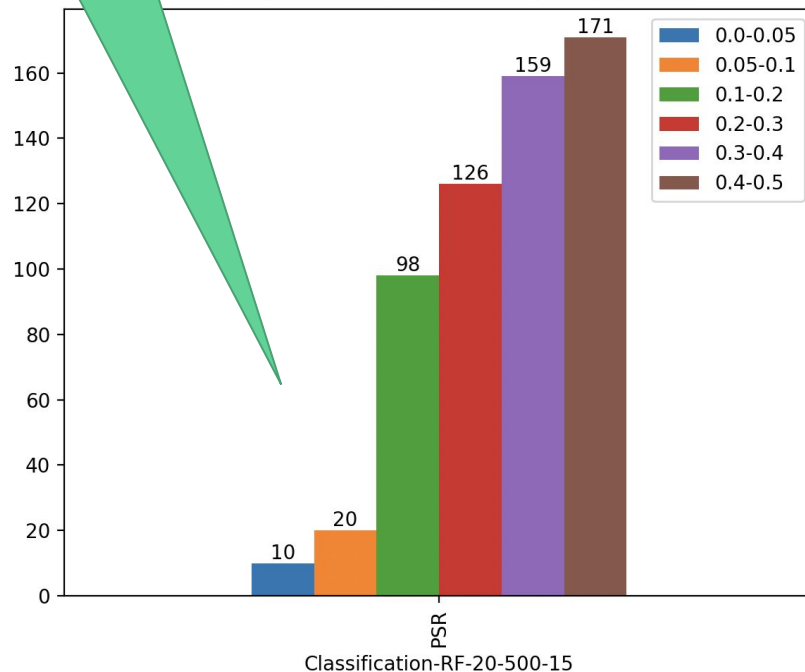
- #trees: 20, # samples: 500, depth: 15

Now PSRs have
been strongly
identified

1846 unassociated sources AGN classification



584 unassociated sources PSR classification



Code Changes

- Create **11 more PSR rows** for each PSR row

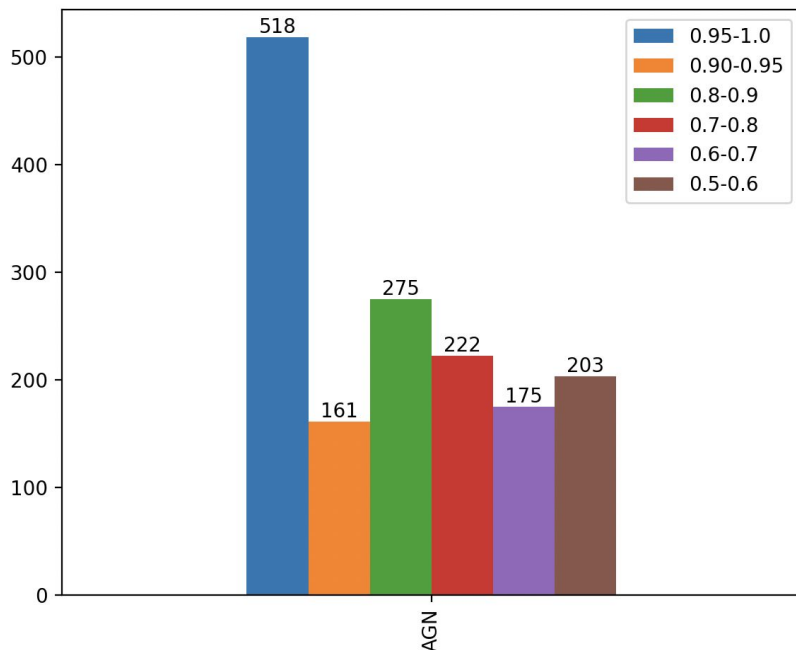
```
correct = 2298
test_size = 2355
correct_agm = 1180
total_agm = 1223 ←
correct_psr = 1118
total_psr = 1132 ←
sensitivity = 97.57961783439491
agm_sensitivity = 96.4840556009812
psr_sensitivity = 98.76325088339223
```

Code Changes

- #trees: 20, # samples: 500, depth: 15

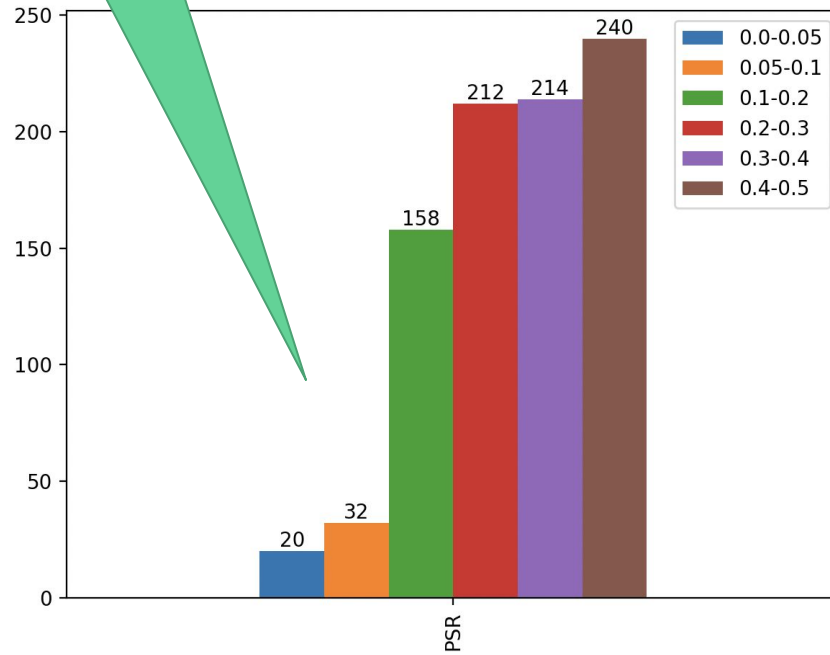
Now PSRs have
been strongly
identified

1554 unassociated sources AGN classification



Classification-RF-20-500-15

876 unassociated sources PSR classification



Classification-RF-20-500-15

Results

- As more rows were duplicated for Pulsars bringing them closer to the AGN rows, the PSRs were more strongly identified for unassociated sources.
- Discuss measure of closeness
 - Sum of predictions of individual trees / number of trees

Update: 04/10/2024

Sort Fields

Catalog	Sort Fields
BSL	sqrt_TS sqrt_TS23 sqrt_TS35
1FGL	Signif_avg Signif_peak sqrt_TS30_100 sqrt_TS100_300 sqrt_TS300_1000 sqrt_TS1000_3000 sqrt_TS3000_10000 sqrt_TS10000_100000

Catalog	Sort Fields
2FGL	Signif_avg Signif_peak Signif_curve sqrt_TS30_100 sqrt_TS100_300 sqrt_TS300_1000 sqrt_TS1000_3000 sqrt_TS3000_10000 sqrt_TS10000_100000
3FGL	Signif_avg Signif_peak Signif_curve sqrt_TS30_100 sqrt_TS100_300 sqrt_TS300_1000 sqrt_TS1000_3000 sqrt_TS3000_10000 sqrt_TS10000_100000

Catalog	Sort Fields
4FGL-DR1	Signif_avg Signif_peak Signif2_peak
4FGL-DR1	Signif_avg Signif_peak
4FGL-DR2	Signif_avg Signif_peak
4FGL-DR3	Signif_avg Signif_peak
4FGL-DR4	Signif_avg Signif_peak

Process

- Opened the FITS files in TopCat
- Found unassociated sources (class1 is null) in catalogs like BSL, 1/2/3FGL, 4FGL DR1/2/3/4
- If exists in 3FGL or 4FGL, looked prior associations in columns shown below

Table Browser for 9: gll_psc_v16_3FGL.fit

	Flux_History	Unc_Flux_History	OFGL_Name	1FGL_Name	2FGL_Name
2740	o (2.6007621E-8, 2.6720793E-8, 2.856525E-8,...	((-2.0991994E-8, 2.2942334E-8), (-2.364542...			
2742	o (2.2480405E-8, 1.0676823E-8, 1.2815452E-...	((-5.183114E-9, 5.8511316E-9), (-3.7521533...		1FGL J2112.5-3044	2FGL J2112.5-3042
2744	o (1.5292892E-8, 3.6051482E-8, 2.5758847E-...	((-1.2139872E-8, 1.3199144E-8), (-1.382965...			

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

	Unc_Flux_History	Sqrt_TS_History	ASSOC_FGL
1806	E-... ((-2.2582012E-9, 2.3462454E-9), (-2.349689...	(13.781338, 22.222631, 22.628527, 24.8065...	3FGL J0850.0+4855
251	E-... ((-1.2570351E-9, 1.3408505E-9), (-1.214068...	(14.391105, 17.910778, 18.003508, 18.3883...	3FGL J0106.5+4855
2302	2.... ((-3.7645798E-9, 3.7645798E-9), (-2.491857...	(26.477457, 13.359099, 16.868233, 15.9703...	3FGL J1103.9-5357
5258	7.7... ((-8.396821E-10, 8.396821E-10), (-8.894406...	(16.858236, 19.474714, 16.138826, 15.2607...	3FGL J2112.5-3044
4825	F-... ((-1.4758287E-9, 1.5485185E-9), (-1.319461...	(15.454493, 13.886233, 15.747384, 18.8581...	3FGL J1918.2-4110

Prior
Association

Table Browser for 41: gll_psc_v33_4FGL_DR4.fit

	Sqrt_TS_History	ASSOC_4FGL	ASSOC_FGL
3499	1.9291846E-9), (-4.10009E-9), (12.577743, 85.35981, 17.117466, 5.910452...	4FGL J1312.6+4828	3FGL J1312.7+4828
2942	3.350813E-9), (-1.4934102E-9), (17.42409, 16.94912, 17.928568, 18.0316, 1...	4FGL J1120.0-2204	3FGL J1119.9-2204
4384	4.4978483E-9), (-4.02377E-9), (27.16054, 28.981466, 14.323773, 2.578804,...	4FGL J1603.8-4903	3FGL J1603.9-4903
6518	8.4200646E-10), (-8.9265... (17.27783, 19.530153, 16.588514, 15.34258...	4FGL J2112.5-3043	3FGL J2112.5-3044

Vela, Crab, Greminga

Table Browser for 1: gll_psc3month_BSL_v2.fit

	Source_Name	RA	DEC	GLON	GLAT	Sqrt_TS	CLASS1	ASSOC1	ASSOC2	Conf_95_Radius	Flux100_1000	Unc_Flux100_1000	Flux1000_1
179	0FGL J2021.5+4026	305.398	40.439	78.23	2.07	69.6	PSR	LAT PSR J2021+4026		0.053	1.23600E-6	5.10000E-8	1.0600
24	0FGL J0238.6+1636	39.663	16.613	156.775	-39.112	85.7	bzb	CGRaBS J0238+1636	AO 0235+164	0.052	6.07000E-7	2.10000E-8	6.8100
136	0FGL J1709.7+4428	257.427	-44.475	343.106	-2.679	85.8	PSR	PSR J1709-4429		0.048	1.15300E-6	5.00000E-8	1.5820
114	0FGL J1504.4+1030	226.115	10.505	11.409	54.577	88.2	bzq	CGRaBS J1504+1029	PKS 1502+106	0.054	6.34000E-7	2.10000E-8	5.8600
158	0FGL J1836.2+5924	279.056	59.406	88.855	24.996	98.	PSR	LAT PSR J1836+5925		0.053	4.44000E-7	1.70000E-8	8.3600
41	0FGL J0534.6+2201	83.653	22.022	184.562	-5.764	139.2	PSR	PSR J0534+2200	Crab	0.046	2.04000E-6	9.20000E-8	1.5400
200	0FGL J2254.0+1609	343.502	16.151	86.125	-38.187	149.1	bzq	CGRaBS J2253+1608	3C 454.3	0.051	2.11700E-6	4.30000E-8	9.8300
49	0FGL J0634.0+1745	98.503	17.76	195.155	4.285	283.2	PSR	PSR J0633+1746	Geminga	0.043	2.86200E-6	3.80000E-8	6.1610
63	0FGL J0835.4-4510	128.865	-45.17	263.56	-2.767	374.2	PSR	PSR J0835-4510	Vela	0.042	8.03100E-6	5.70000E-8	1.1200

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Flux_Density	Spectral_Index	Conf_68_SemiMajor	Conf_68_SemiMinor
862	1FGL J1512.8-0906	228.207	-9.1025	351.284	40.1392	BZQ		PKS 1510-08		101.448	61.8159	411.604	5.93235E-10	2.40756	0.009578	
257	1FGL J0457.0-2325	74.2595	-23.4228	223.719	-34.9024	bzq		PKS 0454-234		115.436		404	1.64470E-10	2.20649	0.010875	
10	1FGL J0007.0+7303	1.7565	73.0523	119.66	10.4634	PSR		LAT PSR J0007+7...		119.66			7.02733E-11	1.96724		
712	1FGL J1256.2-0547	194.053	-5.7911	305.117	57.0603	BZQ		3C 279		127.848			2.28447E-10	2.32061	0.01013	
1395	1FGL J2253.9+1608	343.495	16.1495	86.1169	-38.1851	BZQ		3C 454.3		154.277			7.31831E-10	2.46619	0.009924	
991	1FGL J1709.7+4429	257.428	-44.4856	343.098	-2.68594	PSR		PSR J1709-4429		168.265			3.18954E-10	2.09819		
848	1FGL J1504.4+1029	226.102	10.4927	11.3786	54.581	BZQ		PKS 1502+106		188.71		488.027	3.98856E-10	2.22063	0.007615	
1129	1FGL J1836.2+5925	279.058	59.4251	88.8758	24.999	PSR		LAT PSR J1836+5...		191.12			1.83301E-10	1.98779		
298	1FGL J0534.5+2200	83.6332	22.0145	184.558	-5.78403	PSR	PWN	PSR J0534+2200	Crab	277.644	174.338	494.555	1.00005E-9	2.24752		
357	1FGL J0633.9+1746	98.4756	17.7703	195.134	4.26605	PSR		PSR J0633+1746	Geminga	549.801	352.127	663.604	1.25520E-9	1.92058		
460	1FGL J0835.3-4510	128.836	-45.1763	263.552	-2.78715	PSR	PWN	PSR J0835-4510	Vela	640.132	390.091	582.982	3.66952E-9	2.0244		

Look here!

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Pivot_Energy	Sqrt_TS300_1000	Spectral_Index	ASSOC_TEV
776	2FGL J1104.4+3812	166.12	38.2134	179.818	65.0348	bzb		Mkn 421		131.491	1162.36	58.993	1.77058	Markarian 421
422	2FGL J0617.2+2234e	94.31	22.58	189.048	3.0337	SNR		SNR G189.1-03...	IC 443	132.926	640.492	75.9253	1.7912	IC443
762	2FGL J1057.9-5226	164.496	-52.4485	285.985	6.64999	PSR		PSR J1057-5226		133.667	652.629	81.5958	1.01356	
181	2FGL J0240.5+6113	40.1298	61.2249	135.677	1.08177	HMB		LS I+61 303		133.955	505.083	98.3982	2.18321	LSI +61 303
886	2FGL J1229.1+0202	187.277	2.04167	289.954	64.3492	BZQ		3C 273		140.633	231.151	90.3276	2.4524	
917	2FGL J1256.1-0547	194.042	-5.79369	305.097	57.0581	BZQ		3C 279		147.848	294.806	95.6947	2.22124	3C279
381	2FGL J0538.8-4405	84.706	-44.0838	250.08	-31.0919	BZB		PKS 0537-441		164.992	371.966	98.0431	2.01159	
878	2FGL J1224.9+2122	186.226	21.38	255.07	81.6598	BZQ		4C +21.35		182.403	317.119	115.726	2.12155	4C+21.35
7	2FGL J0007.0+7303	1.77352	73.0545	119.665	10.4647	PSR		LAT PSR J0007+...		189.52	727.398	95.97	1.44987	
1107	2FGL J1512.8-0906	228.207	-9.10336	351.283	40.1383	BZQ		PKS 1510-08		202.414	259.642	128.682	2.28839	PKS 1510-089
1276	2FGL J1709.7+4429	257.433	-44.4852	343.101	-2.68855	PSR		PSR J1709-4429		256.76	607.611	147.288	1.62488	HESS J1708-443
1470	2FGL J1836.2+5926	279.065	59.4339	88.8862	24.9969	PSR		LAT PSR J1836+...		303.698	550.467	190.746	1.23136	
1800	2FGL J2253.9+1609	343.497	16.1526	86.1217	-38.1839	BZQ		3C 454.3		339.435	297.596	255.301	2.22639	
373	2FGL J0534.5+2201	83.6315	22.0196	184.553	-5.78261	PSR		PSR J0534+2200	Crab	376.634	485.021	240.037	2.14454	Crab
439	2FGL J0633.9+1746	98.4785	17.7739	195.132	4.27019	PSR		PSR J0633+1746	Geminga	791.02	510.79	486.28	1.22089	Geminga
570	2FGL J0835.3-4510	128.839	-45.1792	263.556	-2.78707	PSR		PSR J0835-4510	Vela	1027.8	433.286	654.836	1.52334	Vela X

Vela, Crab, Greminga

Table Browser for 9: gll_psc_v16_3FGL.fit

	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name
2013	78.2317	2.08315	655.304	3.80554E-10	237.049	134.378	58.7847	1.56906	PSR	LAI PSR J2021+4026			
1483	255.059	81.6615	306.157	5.16984E-10	219.674	137.609	4.99403	2.18542	FSRQ	4C +21.35		4C +21.35	
20	119.662	10.4606	913.709	7.55546E-11	288.746	145.293	22.6506	1.20806	PSR	LAT PSR J0007+7303		CTA 1	
622	184.552	-5.77575	635.591	5.54130E-10	30.6699	177.273	15.8003	2.04501	PSR	PSR J0534+2200	Crab	Crab Pulsar	
2114	343.098	-2.68919	849.288	3.10332E-10	360.818	188.988	28.4751	1.03076	PSR	PSR J1709-4429		HESS J1708-443	
2391	88.8821	24.9983	747.86	2.01251E-10	438.122	260.318	71.7768	0.886055	PSR	LAT PSR J1836+5925			
2913	86.1214	-38.1918	412.747	1.33123E-9	480.741	308.06	18.2082	1.63463	FSRQ	3C 454.3			
961	263.55463	-2.78727	832.557	2.33308E-9	1048.96	528.538	53.9787	1.00295	PSR	PSR J0835-4510	Vela	Vela Pulsar	
726	195.133	4.27009	927.902	9.37508E-10	1012.14	568.663	84.973	0.916183	PSR	PSR J0633+1746	Geminga	Geminga	
620	184.553	-5.78121	1000.	1.83367E-11	—∞		0.	1.64	PWN	Crab		Crab	
956	263.333	-3.10593	1000.	2.59146E-11	—∞		0.	2.41006	PWN	Vela X		Vela X	Vela X
1848	320.27	-1.27147	1000.	1.78847E-12	—∞		0.	1.57134	PWN	MSH 15-52		MSH 15-52	MSH 15-52
2613	78.2413	2.19658	1000.	4.00366E-13	—∞		0.	1.53	SNR	Gamma Cygni		VER J2019+407	gamma Cygni
2639	79.6012	1.39613	1000.	6.84591E-11	—∞		0.	2.175	SFR	Cygnus Cocoon		MGRO J2031+41	Cygnus Cocoon

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name	Pivot_Energy	Signif_Peak	LP_SigCurv
2792	4FGL J1427.9-4206	216.987	-42.106	321.449	17.2695	258.031	FSRQ		PKS 1424-41				677.451	216.453	17.1327
1309	4FGL J0721.9+7120	110.488	71.3485	143.985	28.0224	260.283	BLL		S5 0716+71		S5 0716+714		717.339	141.324	9.47317
2048	4FGL J1104.4+3812	166.119	38.207	179.833	65.0357	294.278	BLL		Mkn 421		Markarian 421		1286.47	146.43	3.21282
25	4FGL J0007.0+7303	1.768	73.0512	119.663	10.4617	329.094	PSR		PSR J0007+7303		CTA 1		1452.3		62.4081
3441	4FGL J1709.7-4429	257.43	-44.4854	343.099	-2.68665	374.423	PSR		PSR J1709-4429				1411.45		74.0677
4881	4FGL J2253.9+1609	343.496	16.1506	86.1185	-38.1849	443.845	FSRQ		3C 454.3				517.462	355.842	38.1457
4002	4FGL J1836.2+5925	279.056	59.4279	88.8783	25.0006	466.08	PSR		PSR J1836+5925				1361.54		111.226
1560	4FGL J0835.3-4510	128.837	-45.1781	263.554	-2.78773	496.139	PSR		PSR J0835-4510	Vela X field	Vela Pulsar		1746.46		196.328
1178	4FGL J0633.9+1746	98.483	17.7723	195.135	4.27298	792.691	PSR		PSR J0633+1746		Geminga		1585.48		207.659
994	4FGL J0534.5+2201i	83.633	22.02	184.553	-5.78148		PWN		Crab Nebula		Crab	Crab IC	10000.		0.

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

	Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name
698	4FGL J0319.8+4130	1	49.9575	41.5121	150.58	-13.2581	235.649	952.578	RDG		NGC 1275		NGC 1275	
569	4FGL J0240.5+6113	1	40.1426	61.2287	135.68	1.08765	236.903	1185.92	HMB		LSI +61 303		LSI +61 303	
3321	4FGL J1512.8-0906	1	228.215	-9.1064	351.288	40.1312	238.707	723.638	FSRQ		PKS 1510-089		PKS 1510-0...	
904	4FGL J0428.6-3756	1	67.173	-37.9403	240.71	-43.6147	249.909	808.988	bll		PKS 0426-380			
2273	4FGL J1057.9-5227	1	164.498	-52.4528	285.987	6.64635	269.544	1030.07	PSR		PSR J1057-52...			
2756	4FGL J1256.1-0547	1	194.842	-5.7887	305.095	57.0632	291.633	466.046	FSRQ		3C 279		3C 279	
3144	4FGL J1427.9-4206	1	216.987	-42.106	321.449	17.2695	293.386	645.788	FSRQ		PKS 1424-41			
1463	4FGL J0721.9+7120	1	110.488	71.3485	143.985	28.0224	298.165	734.586	BLL		S5 0716+71		S5 0716+714	
2306	4FGL J1104.4+3812	1	166.119	38.207	179.833	65.0357	325.523	1263.75	BLL		Mkn 421		Markarian 4...	
5504	4FGL J2232.6+1143	1	338.152	11.7306	77.4385	-38.5832	349.938	531.25	FSRQ		CTA 102			
30	4FGL J0007.0+7303	1	1.768	73.0512	119.663	10.4617	351.673	1472.99	PSR		PSR J0007+7...		CTA 1	
3893	4FGL J1709.7-4429	1	257.43	-44.4854	343.099	-2.68665	408.691	1462.71	PSR		PSR J1709-44...			
5577	4FGL J2253.9+1609	1	343.496	16.1506	86.1185	-38.1849	465.154	232.352	FSRQ		3C 454.3			
4559	4FGL J1836.2+5925	1	279.056	59.4279	88.8783	25.0006	523.634	1024.48	PSR		PSR J1836+5...			
1750	4FGL J0835.3-4510	1	128.837	-45.1781	263.554	-2.78773	548.109	1760.38	PSR		PSR J0835-45...	Vela X field	Vela Pulsar	
1323	4FGL J0633.9+1746	1	98.483	17.7723	195.135	4.27298	840.475	1328.4	PSR		PSR J0633+1...		Geminga	
1118	4FGL J0534.5+2201i	1	83.633	22.02	184.553	-5.78148		10000.	PWN		Crab Nebula		Crab	Crab IC

Vela, Crab, Gremlinga

Table Browser for 34: gll_psc_v31_4FGL_DR3.fit

Source Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name
653	4FGLJ0240.5+6113	1	40.1426	61.2287	135.68	1.08765	1260.67	249.812	HMB	LSI +61 303		LSI +61 3...	
1036	4FGLJ0428.6-3756	1	67.173	-37.9403	240.71	-43.6147	850.508	256.684	bll	PKS 0426-380			
2616	4FGLJ1057.9-5227	1	164.498	-52.4528	285.987	6.64635	1408.27	277.55	PSR	PSR J1057-5226			
3598	4FGLJ1427.9-4206	1	216.987	-42.106	321.449	17.2695	1367.49	297.359	FSRQ	PKS 1424-41			
3153	4FGLJ1256.1-0547	1	194.042	-5.7887	305.095	57.0632	954.15	302.454	FSRQ	3C 279		3C 279	
1694	4FGLJ0721.9+7120	1	110.488	71.3405	143.985	28.0224	1510.63	321.002	BLL	SS 0716+71		SS 0716+...	
2650	4FGLJ1104.4+3812	1	166.119	38.207	179.833	65.0357	1258.26	343.769	BLL	Mkn 421		Markarian ...	
6329	4FGLJ2232.6+1143	1	338.152	11.7306	77.4385	-38.5832	1082.42	345.948	FSRQ	CTA 102			
37	4FGLJ0007.0+7303	1	1.768	73.0512	119.663	10.4617	2213.57	370.992	PSR	PSR J0007+7303		CTA 1	
4471	4FGLJ1709.7-4429	1	257.43	-44.4854	343.099	-2.68665	2078.25	413.705	PSR	PSR J1709-4429		HESS PSR ...	
6412	4FGLJ2253.9+1609	1	343.496	16.1506	86.1185	-38.1849	891.907	454.905	FSRQ	3C 454.3			
5226	4FGLJ1836.2+5925	1	279.056	59.4279	88.8783	25.0006	1428.31	555.913	PSR	PSR J1836+5925			
2014	4FGLJ0835.3-4510	1	128.837	-45.1781	263.554	-2.78773	1914.57	665.559	PSR	PSR J0835-4510	Vela X fi...	Vela Pulsar	
1534	4FGLJ0633.9+1746	1	98.483	17.7723	195.135	4.27298	1670.18	884.643	PSR	PSR J0633+1746		Geminga P...	
1294	4FGLJ0534.5+2201i	1	83.633	22.02	184.553	-5.78148	10000.		PWN	Crab Nebula		Crab	Crab IC

Table Browser for 41: gll_psc_v33_4FGL_DR4.fit

Source Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Extended_Source_Name	Signif_Avg	Signif_Peak	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV
1216	4FGLJ0457.0-2324	1	74.2608	-23.4149	223.71	-34.8991	247.863	106.397	692.189	FSRQ		PKS 0454-234		
6706	4FGLJ2158.8-3013	1	329.714	-30.2251	17.7308	-52.2434	248.351	100.358	1168.97	bll		PKS 2155-304		PKS 2155-304
1411	4FGLJ0538.8-4405	1	84.7089	-44.0862	250.083	-31.0904	249.871	153.837	653.401	BLL		PKS 0537-441		
850	4FGLJ0319.8+4130	1	49.9575	41.5121	150.58	-13.2581	251.542	138.776	974.9	RDG		NGC 1275		NGC 1275
1110	4FGLJ0428.6-3756	1	67.173	-37.9403	240.71	-43.6147	258.51	112.784	873.599	bll		PKS 0426-380		
697	4FGLJ0240.5+6113	1	40.1426	61.2287	135.68	1.08765	261.254	109.965	1517.29	HMB		LSI +61 303		LSI +61 303
2828	4FGLJ1057.9-5227	1	164.498	-52.4528	285.987	6.64635	297.348	1445.88	1445.88	PSR		PSR J1057-52...		
3891	4FGLJ1427.9-4206	1	216.987	-42.106	321.449	17.2695	305.414	216.333	1550.06	FSRQ		PKS 1424-41		
1840	4FGLJ0721.9+7120	1	110.488	71.3405	143.985	28.0224	318.785	130.895	1606.17	BLL		SS 0716+71		SS 0716+714
3412	4FGLJ1256.1-0547	1	194.042	-5.7887	305.095	57.0632	319.282	303.431	980.925	FSRQ		3C 279		3C 279
6837	4FGLJ2232.6+1143	1	338.152	11.7306	77.4385	-38.5832	340.599	400.519	1098.69	FSRQ		CTA 102		
6722	4FGLJ2202.7+4216	1	330.695	42.2821	92.6007	-10.4441	343.185	280.56	992.378	BLL		BL Lac		BL Lacertae
2866	4FGLJ1104.4+3812	1	166.119	38.207	179.833	65.0357	363.7	147.932	1328.17	BLL		Mkn 421		Markarian 421
40	4FGLJ0007.0+7303	1	1.768	73.0512	119.663	10.4617	401.705	147.932	2247.63	PSR		PSR J0007+73...		CTA 1
4830	4FGLJ1709.7-4429	1	257.43	-44.4854	343.099	-2.68665	431.37	147.932	2136.25	PSR		PSR J1709-44...		HESS PSR B17...
6924	4FGLJ2253.9+1609	1	343.496	16.1506	86.1185	-38.1849	448.024	365.922	908.003	FSRQ		3C 454.3		
5643	4FGLJ1836.2+5925	1	279.056	59.4279	88.8783	25.0006	587.57	1496.23	1496.23	PSR		PSR J1836+59...		
2175	4FGLJ0835.3-4510	1	128.837	-45.1781	263.554	-2.78773	689.218	1976.37	1976.37	PSR		PSR J0835-45...	Vela X field	Vela Pulsar
1660	4FGLJ0633.9+1746	1	98.483	17.7723	195.135	4.27298	897.975	1705.33	1705.33	PSR		PSR J0633+17...		Geminga Pulsar
1390	4FGLJ0534.5+2201i	1	83.633	22.02	184.553	-5.78148		10000.		PWN		Crab Nebula		Crab

Unassociated Source #1 to PSR

Table Browser for 2: gl_l_psc_v03_1FGL.fit

Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
1523 1FGL J1214.7-3358	321.183	-33.9791	10.9255	-45.4378	PSR		PSR J1214-3358		26.0896	11.915	991.297
1248 1FGL J2015.7+3708	303.929	37.1399	74.8571	1.16046					26.1037	13.8629	1050.77
306 1FGL J0538.9-6914	84.7396	-69.2398	279.625	-31.6338	gal		LMC field		26.1676	14.6717	793.557
880 1FGL J1536.5-4949	234.146	-49.818	328.223	4.76784					26.3268	9.86395	952.782
452 1FGL J0825.8-2230	126.468	-22.5104	243.973	8.89926	bzb		PKS 0823-223		26.5001	13.3148	890.931
1064 1FGL J1751.5+0937	267.89	9.6275	34.8997	17.6326	bzb		4C +09.57		26.5094	15.0708	773.708
910 1FGL J1603.8-4903	240.972	-49.0662	332.158	2.57019	agu				26.8036	9.10865	1453.07
1354 1FGL J2203.5+1726	330.882	17.4387	75.6957	-29.6413	bzq				27.799	15.9251	574.137
1268 1FGL J2032.2+4127	308.054	41.4567	80.224	1.02802	PSR				27.134	7.49773	1696.68
1261 1FGL J2027.6+3335	306.915	33.592	73.3268	-2.86108					27.9593	18.1954	702.555
669 1FGL J1217.7+3007	184.446	30.1206	188.893	82.0351	bzb				27.3569	8.84911	1325.08
53 1FGL J0050.6-0928	12.6646	-9.4832	122.299	-72.3539	bzb		PKS 0048-09		27.4003	16.1701	764.423
74 1FGL J0112.0+2247	18.0179	22.7945	129.129	-39.8288	bzb		CGRaBS J0112+2...		27.5516	14.694	717.733
740 1FGL J1311.7-3429	197.947	-34.4944	307.689	28.1893					27.7331	14.9313	881.639
165 1FGL J0303.5-2406	45.889	-24.1124	214.619	-60.15	bzb		PKS 0301-243		27.8553	14.2392	1086.94

Unassociated
Class1 == null

Table Browser for 5: gl_l_psc_v09_2FGL.fit

Source_Name	RA	DEC	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
1724 2FGL J2139.3-4236	324.841	-42.608	358.291	-48.3171	bzb					39.1281	21.3043	805.105	2.09528
86 2FGL J0112.8+3208	18.2069	32.1387	128.188	-30.5146	bzq					39.4861	23.6512	561.489	2.31038
1130 2FGL J1536.4-4949	234.124	-49.8294	328.205	4.76692						39.3338	16.5487	865.906	1.94332
95 2FGL J0118.8-2142	19.7243	-21.7062	173.461	-81.7284	bzq					40.3492	26.6873	439.865	2.12153
1597 2FGL J2000.0+6509	300.02	65.1572	98.016	17.6657	bzb					40.4049	17.3951	1402.26	1.93676
873 2FGL J1221.4+2814	185.374	28.2391	201.691	83.2804	bzb					40.5612	19.8744	1106.2	2.01857
150 2FGL J0217.9+0143	34.4833	1.73026	162.199	-54.4104	bzq					42.2751	23.2729	724.696	2.15159
1387 2FGL J1800.5+7829	270.147	78.4834	110.057	29.0718	bzb					42.5332	22.2863	665.466	2.23857
764 2FGL J1058.6+5628	164.666	56.4861	109.575	54.4212	bzb					42.8403	19.4494	1236.75	1.92735
947 2FGL J1311.7-3429	197.947	-34.4944	307.689	28.1893						43.0656	26.4636	706.553	2.01592
1355 2FGL J1745.6-2858	266.423	-28.9786	359.973	-0.036044	spp					43.1212	23.2346	1618.9	2.33888
743 2FGL J1142.7+0127	174.044	1.3493	267.576	59.4367						43.2702	20.6608	7.59733	2.33888

Unassociated
Class1 == null

Associated to
PSR in 3FGL

Table Browser for 9: gl_l_psc_v16_3FGL.fit

Table Browser for 3FGL_PSR_V1200_01_Galaxy													
	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1
1372	3FGL J1136.1+7523	174.044	75.3885	129.104	40.8251	1514.61	9.41672E-14	4.3202	0.850643	0.726054	2.30112	PSR	PSR J1137+75
1385	3FGL J1142.9+0120	175.742	1.3493	267.576	59.4367	1830.62	3.34363E-13	13.7362	3.78123	3.98751	1.18626	PSR	PSR J1142+01
1496	3FGL J1231.2-1411	187.801	-14.1981	295.539	48.383	917.884	2.16312E-11	105.655	51.8699	23.9051	1.1804	PSR	PSR J1231-14
1565	3FGL J1301.6+0832	195.418	8.5398	310.829	71.2544	765.506	2.14075E-12	13.6457	8.33066	2.59564	2.37894	PSR	PSR J1301+08
1566	3FGL J1302.3-3259	195.595	-32.9879	305.577	29.8276	1440.63	1.12547E-12	18.7593	6.91077	6.14404	1.12307	PSR	PSR J1302-32
1595	3FGL J1311.8-3430	197.959	-34.5021	307.699	28.1807	658.035	2.41142E-11	62.3248	35.3774	9.05289	1.90862	PSR	LAT PSR J1311-34
1600	3FGL J1312.7+0051	198.195	0.861	314.85	63.2377	837.744	4.26514E-12	20.6608	7.59733	7.17686	1.52349	PSR	PSR J1312+00

Prior
associations

1FGL_Name	2FGL_Name	1FHL_Name	ASSOC_CAM1	ASSOC_CAM2	ASSOC_
1FGL J1142.7+0127	2FGL J1142.9+0121				
1FGL J1231.1-1410	2FGL J1231.2-1411	1FHL J1231.2-1414			EGR J1
1FGL J1301.8+0837	2FGL J1301.5+0835				
1FGL J1302.3-3255	2FGL J1302.4-3257				
1FGL J1311.7-3429	2FGL J1311.7-3429			3EG J1314-3431	
1FGL J1312.6+0048	2FGL J1312.7+0051				

Unassociated Source #2 confirmed PSR in 3FGL

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
91	1FGL J0136.5+3905	24.1305	39.0856	132.416	-22.9545	bzb		B3 0133+388		25.6661	6.22098	2336.53
781	1FGL J1344.2-1723	206.071	-17.3925	320.482	43.671	bzq				25.7139	13.8585	918.528
81	1FGL J0118.7-2137	19.6825	-21.6276	172.877	-81.7038	bzq				25.7165	16.3633	533.797
501	1FGL J0909.0+0126	137.273	1.4439	228.85	30.9519	bzq				25.7542	16.6642	385.549
217	1FGL J0403.9-3603	60.9774	-36.0636	237.713	-48.4798	bzq				25.8157	17.0821	452.558
1323	1FGL J2124.7-3358	321.183	-33.9791	10.9255	-45.4378	PSR		PSR J2124-3358		26.0896	11.915	991.297
1248	1FGL J2015.7+3708	303.929	37.1399	74.8571	1.16046					26.1037	13.8629	1050.77
306	1FGL J0538.9-6914	84.7396	-69.2398	279.625	-31.6338	gal		LMC field		26.1676	14.6717	793.557
880	1FGL J1536.5-4949	234.146	-49.818	328.223	4.76784					26.3268	9.86395	952.782
452	1FGL J0825.8-2230	126.468	-22.5104	243.973	8.89926	bzb		PKS 0823-223		26.5001	13.3148	890.931

Unassociated
Class1 == null

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
239	2FGL J0334.2-4008	53.5596	-40.1484	244.777	-54.0715	bzb				38.3389	23.6149	696.373	2.18878
566	2FGL J0831.9+0429	127.987	4.48609	220.719	24.3573	bzb				38.3503	22.1135	527.964	2.04695
1778	2FGL J2236.4+2828	339.102	28.4755	90.1205	-25.6591	bzb				38.7141	22.8297	460.793	2.05423
350	2FGL J0521.7+2113	80.449	21.2204	183.603	-8.69891	bzb		PSR J0521+211		38.9956	13.0665	1874.43	1.92896
1724	2FGL J2139.3-4236	324.841	-42.608	358.291	-48.3171	bzb		MH 2136-428		39.1281	21.3043	805.105	2.09528
86	2FGL J0112.8+3208	18.2069	32.1387	128.188	-30.5146	bzq		4C 31.03		39.4861	23.6512	561.489	2.31038
1130	2FGL J1536.4-4949	234.124	-49.8294	328.205	4.76692					40.3338	16.5487	865.906	1.94332
95	2FGL J0118.8-2142	19.7243	-21.7062	173.461	-81.7284	bzq		PKS 0116-219		40.3492	26.6873	439.865	2.12153
1597	2FGL J2000.0+6509	300.02	65.1572	98.016	17.6657	bzb		1ES 1959+650		40.4049	17.3951	1402.26	1.93676

Unassociated
Class1 == null

Unassociated Source #2 confirmed a PSR

Gets associated in 3FGL and appears in 4FGL also

Table Browser for 9: gll_psc_v16_3FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1
1349	3FGL J1125.6-6012	171.403	-60.2093	292.46	0.905667	2066.48	2.24512E-13	5.64072	3.52324	2.56519	psr	PSR J1125-6014
1427	3FGL J1203.9-6243	180.996	-62.7268	297.527	-0.354157	1619.7	2.34999E-12	14.8116	9.35934	5.75902	psr	PSR J1203-63
1434	3FGL J1207.2-5052	181.824	-50.87	295.854	11.3941	1525.98	3.26278E-13	7.73423	3.00148	3.92012	psr	PSR J1207-5050
1489	3FGL J1227.9-4854	186.976	-48.9012	298.954	13.7886	444.993	3.73728E-11	38.6305	22.9382	4.63259	psr	PSR J1227-4853
1704	3FGL J1400.5-1437	210.145	-14.617	326.929	45.0087	949.195	1.13245E-12	10.1603	5.40907	2.29915	psr	PSR J1400-1438
1844	3FGL J1513.4-2549	228.368	-25.8271	338.848	26.9653	1158.19	1.23075E-12	11.0081	5.65408	4.87454	psr	PSR J1513-25
1889	3FGL J1536.3-4949	234.099	-49.8225	328.196	4.7821	1043.57	1.21817E-11	60.0751	23.3951	8.53276	psr	PSR J1536-4948
1984	3FGL J1622.9-0312	245.741	-3.2039	10.758	30.7141	1228.08	6.82104E-13	9.47165	5.1476	2.29817	psr	PSR J1622-03
2002	3FGL J1628.0-3203	247.016	-32.0608	347.447	11.5151	1000.78	2.71948E-12	11.8772	6.40167	5.45218	psr	PSR J1628-3205

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC3	Pivot_Energy	Signif_Peak	LP_SigC
4483	4FGL J2032.2+4127	308.063	41.459	80.2294	1.02414	86.9727	PSR	PSR J2032-41			2235.47		22.
1474	4FGL J0809.8+5218	122.462	52.3143	166.248	32.9144	87.0045	BLL	1E5 0806+524			1297.	44.2755	3.
4695	4FGL J2143.5+1743	325.894	17.7306	72.1129	-26.0814	87.5121	fsrq	OX 169			394.828	36.8899	4.
2733	4FGL J1413.5-6205	213.382	-62.0903	312.378	-0.733427	87.7349	PSR	PSR J1413-6205			2051.79		22.
2021	4FGL J1058.4+0133	164.624	1.5641	251.514	52.7728	87.9966	BLL	4C +01.28			777.085	40.389	4.
329	4FGL J0144.6+2705	26.1502	27.0899	137.276	-34.2967	88.0051	bll	TXS 0141+268			767.909	40.1945	7.
2189	4FGL J1146.9+3958	176.74	39.9775	164.951	71.4682	91.0036	fsrq	S4 1144+40			540.053	57.648	5.
3038	4FGL J1536.4-4948	234.1	-49.8164	328.2	4.78665	91.2146	PSR	PSR J1536-49			1475.37		18.
3572	4FGL J1732.5-3131	263.145	-31.5174	356.315	1.00632	91.5022	PSR	PSR J1732-3131			2426.08		32.
403	4FGL J0210.7-5101	32.6946	-51.0218	276.106	-61.7737	91.9811	FSRQ	PKS 0208-512			458.644	52.4501	7.
4113	4FGL J1855.9+0121e	283.99	1.355	34.6534	-0.388004	92.3315	SNR	W 44		W 44	2448.21		28.

Associated to
PSR in 3FGL

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC3	ASSOC4	ASSOC5
5370	4FGL J2147.3-7536	1	326.827	-75.6024	315.801	-36.5338	97.4077	FSRQ		PKS 2142-75				
606	4FGL J0252.8-2219	1	43.2007	-22.3203	209.742	-62.0938	97.4665	fsrq		PKS 0250-225				
3080	4FGL J1413.5-6205	1	213.382	-62.0903	312.378	-0.733427	98.4685	PSR		PSR J1413-62...				
2467	4FGL J1146.9+3958	1	176.74	39.9775	164.951	71.4682	99.3588	fsrq		S4 1144+40				
761	4FGL J0339.5-0146	1	54.8771	-1.7769	188.	-42.4563	99.6133	fsrq		PKS 0336-01				
1825	4FGL J0854.8+2006	1	133.707	20.1159	206.805	35.8264	100.244	BLL		OJ 287				
5520	4FGL J2236.5-1433	1	339.144	-14.5557	47.8811	-56.2157	100.773	BLL		PKS 2233-148				
370	4FGL J0144.6+2705	1	26.1502	27.0899	137.276	-34.2967	100.824	bll		TXS 0141+268				
3421	4FGL J1536.4-4948	1	234.1	-49.8164	328.2	4.78665	101.719	PSR		PSR J1536-49				
1069	4FGL J0522.9-3628	1	80.737	-36.4686	240.618	-32.7216	101.878	AGN		PKS 0521-36				

Unassociated Source #2 changed class

spp in 1FGL to HMB in 2FGL

Table Browser for 2: gll_psc_v03_1FGL.fit

Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
975 1FGLJ1653.9+3945	253.49	39.7527	63.5928	38.8413	bzb		Mkn 501		40.2217	15.5982	1476.79
546 1FGLJ1015.1+4927	153.788	49.4526	165.496	52.7181	bzb		IES 1011+496		40.3215	15.9248	1122.12
31 1FGLJ0030.4+0451	7.6143	4.861	113.142	-57.6114	PSR		PSR J0030+0451		40.6079	23.8943	831.998
1217 1FGLJ1952.9+3252	298.242	32.8779	68.7655	2.82324	PSR		PSR J1952+3252		40.7442	22.6044	824.035
449 1FGLJ0818.2+4222	124.552	42.3816	178.224	33.392	bzb		B3 0814+425		40.8058	22.0282	758.552
1124 1FGLJ1833.6-2103	278.406	-21.0637	12.1595	-5.7045	bzq		PKS 1830-21		41.0232	27.4163	481.52
70 1FGLJ0108.6+0135	17.1741	1.5879	131.852	-60.9845	bzq		PKS 0108+013		42.1262	27.3067	434.662
1170 1FGLJ1907.9+0602	286.978	6.0378	40.1822	-0.894193	PSR		PSR J1907+0...		42.4456	24.3516	1008.68
1188 1FGLJ1923.5-2104	290.885	-21.0771	17.178	-16.2559	bzq		OV -235		42.5887	24.6697	751.494
549 1FGLJ1018.6-5856	154.673	-58.9388	284.32278	-1.70069	spp				42.9696	28.2287	766.352
1110 1FGLJ1826.1-1256	276.535	-12.9434	18.5552	-0.377947	PSR		LAT PSR J1826-1...		43.0842	20.5847	980.67

spp

Table Browser for 5: gll_psc_v09_2FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
992 2FGLJ1344.2-1723	206.068	-17.3956	320.477	43.6688	bzq		PMN J1344-1723		30.7749	16.3747	641.316	1.95405
1336 2FGLJ1740.2+5212	265.06	52.2012	79.5656	31.8057	bzq				30.7753	19.8098	481.282	2.50226
1622 2FGLJ2017.3+0603	304.35	6.05396	48.6261	-16.0298	PSR		PKS 2017+063		31.2384	8.00843	1608.34	1.04592
763 2FGLJ1058.4+0133	164.615	1.56646	251.501	52.7681	bzb				31.4827	16.1794	991.442	2.21744
1779 2FGLJ2236.5-1431	339.133	-14.5331	47.9086	-56.1957	BZB		PKS 2233-148		31.5358	18.6361	673.396	2.23955
1527 2FGLJ1911.1-2005	287.791	-20.097	16.8922	-13.2106	bzq		PKS B1908-201		31.851	19.5889	437.277	2.20548
708 2FGLJ1019.0-5856	154.759	-58.9416	284.362	-1.67841	HMB		1FGLJ1018.6-5856		31.8745	17.4002	1263.62	2.59492
1852 2FGLJ2339.6-0532	354.909	-5.54204	81.3579	-62.4673					32.2053	11.3601	973.939	1.66875
1580 2FGLJ1108.6-1845	200.554	-18.7633	65.0764	0.352095	PSR		LAT PSR J1108-1845		32.2511	16.717	2108.44	1.24055

HMB

Table Browser for 9: gll_psc_v16_3FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1
2913 3FGLJ2254.0+1608	343.502	16.1459	86.1214	-38.1918	412.747	1.33123E-9	480.741	308.06	18.2082	1.63463	FSRQ	3C 454.3
2940 3FGLJ2311.0+3425	347.764	34.4231	100.411	-24.021	368.599	7.98852E-11	66.8727	40.6246	5.56969	2.19157	FSRQ	
3003 3FGLJ2345.2-1554	356.307	-15.9131	65.6972	-70.9857	825.789	2.02701E-11	82.3238	40.1362	6.27866	2.01772	FSRQ	
123 3FGLJ0059.0-7242e	14.75	-72.7	302.145	-44.4167	665.532	1.05933E-11	27.9457	18.7552	2.50357	2.22098	GAL	SMC
603 3FGLJ0526.6-6825e	81.65	-68.42	278.843	-32.8502	396.322	1.83021E-10	83.8399	51.3288	6.56124	2.01708	GAL	MC
320 3FGLJ0240.5+6113	40.1379	61.2281	135.679	1.08628	547.579	2.79382E-10	196.606	127.731	16.7763	2.18968	HMB	LSI+61 303
1173 3FGLJ1018.9-5856	154.73	-58.9456	284.351	-1.69011	1350.21	2.15732E-11	52.2459	22.9137	21.1823	2.54647	HMB	1FGLJ1018.6-5856
2359 3FGLJ1826.2-1450	276.568	-14.8473	16.8856	-1.2938	280.328	7.86167E-10	35.0597	24.3102	5.33324	2.37947	HMB	LS 5039
994 3FGLJ1040.0-5108	162.487	-51.1445	167.603	20.1703	786.662	5.31324E-12	22.3503	18.0837	2.75833	2.28245	MCV1	CRC 0846+513

HMB

Unassociated Source #4 to Blazar

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
155	1FGL J0252.8-2219	43.2096	-22.323	209.753	-62.0864	bzq		PKS 0250-225		34.455	22.8489	513.47
675	1FGL J1221.5+2814	185.39	28.2467	201.611	83.2931	bzb		W Com		34.724	17.1719	988.293
547	1FGL J1016.1+0514	154.032	5.2351	236.514	47.0585	bzq		CRATES J1016+0...		34.7958	20.7301	692.145
962	1FGL J1642.5+3947	250.6398	39.7941	63.4273	41.0283					34.8074	21.9664	548.416
392	1FGL J0714.0+1935	108.515	19.5866	197.659	13.6204	bzq		CLASS J0713+19...		35.1604	22.1187	558.232

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
170	2FGL J0229.3-3644	43.2096	-36.7336	243.952	-67.1972	bzq				16.1813	10.452	475.342
1243	2FGL J1647.5+4950	251.881	49.84	76.6545	40.0848	agn				16.1842	9.63322	554.614
1515	2FGL J1901.1+0427	285.298	4.45755	38.0103	-0.132545					16.2615	12.2794	913.074
249	2FGL J0340.5+5307	55.1495	53.1314	146.755	-1.73135					16.2887	10.1242	250.669
1186	2FGL J1618.2-7718	244.562	-77.3007	313.438	-18.8793	bzq		PKS 1610-77		16.3579	10.4627	503.719
48	2FGL J0043.7+3426	10.9409	34.4394	121.134	-28.4055	bzq		GB6 J0043+3426		16.498	8.31669	1071.46
1237	2FGL J1642.9+3949	250.748	39.8487	63.4801	40.9473	BZQ		3C 345		16.5409	7.8223	972.728
1144	2FGL J1549.5+0237	237.388	2.62742	10.8693	40.8972	bzq		PKS 1546+027		16.6181	8.75505	572.201

Unassociated
Class1 == null

Table Browser for 9: gll_psc_v16_3FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	BZQ	Avg	Sqrt_TS300_1000	Signif_Curve	Spectral_Index	CLASS1	ASSOC1	
1804	3FGL J1451.4-5539	224.333	-55.6501	329.000	40.330	410.111		39.2333	39.2333	3.17304	2.10093	FSRQ	PKS 1451-5539	
1817	3FGL J1504.4+1029	226.105	10.4936	11.382	54.5796	345.173		189.471	118.838	9.19739	2.07509	FSRQ	PKS 1502+106	
1841	3FGL J1512.8-0906	228.213	-9.1032	351.289	40.1348	445.227		207.272	131.263	7.7404	2.30471	FSRQ	PKS 1510-08	
1995	3FGL J1626.0-2951	246.518	-29.8624	348.809	13.3169	938.06		5.04271E-12	20.9228	14.1516	4.38472	FSRQ	PKS 1622-29	
2021	3FGL J1635.2+3809	248.812	38.1525	61.1099	42.3395	701.191		4.47701E-11	86.9665	53.7671	9.3616	FSRQ	4C +38.41	
2036	3FGL J1640.6+3945	250.159	39.7553	63.3465	41.3952	1222.25		3.50900E-12	28.2796	13.7391	4.73747	FSRQ	NRAO 512	
2043	3FGL J1642.9+3950	250.748	39.8487	63.5063	40.9489	1156.74		2.54784E-12	20.5269	9.98583	0.273285	2.4474	FSRQ	3C 345
2179	3FGL J1733.0-1305	263.256	-13.0858	12.0253	10.8129	516.34		3.74577E-11	36.8071	21.9722	3.76453	2.24467	FSRQ	PKS 1730-13

BZQ

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_So...	Pivot_Energy	Signif_Peak	LP_Sig
3344	4FGL J1647.5+4950	251.892	49.8336	76.6451	40.0778	37.1897	bl						808.93	18.7059	1
2487	4FGL J1306.3-6043	196.587	-60.7199	304.755	2.09904	37.2479						FGES J1036...	2121.84		1
1947	4FGL J1036.3-5833e	159.094	-58.563	286.081	-0.181491	37.2569							5171.9		1
1494	4FGL J0816.3+5739	124.1	57.6616	159.851	33.8992	37.2642	bl						1483.76	15.9236	1
4278	4FGL J1932.3+1916	293.076	19.2685	54.65	0.083537	37.3203	FSRQ		PSR J1932+1916				1537.65		1
4705	4FGL J2147.1+0931	326.783	9.5181	65.808	-32.241	37.3583	FSRQ		PKS 2144+092				467.316	36.2753	1
2993	4FGL J1522.6-2730	230.664	-27.5059	339.588	24.4001	37.3679	bl		PKS 1519-273				1203.56	22.2033	1
3316	4FGL J1642.9+3948	250.734	39.8164	63.4624	40.9576	37.381	FSRQ		3C 345				1408.69	20.9635	1
2491	4FGL J1307.6-4259	196.91	-42.995	306.079	19.7777	37.7179	bl		1RXS J130737.8...				2525.42		1
2553	4FGL J1323.0+2941	200.762	29.6871	54.9945	82.5956	38.0299	bcu		4C +29.48				1142.81		1
4096	4FGL J1852.4+4856	283.116	48.935	78.5979	19.9487	38.0499	fsrq		S4 1851+48				766.87	33.2299	1

FSRQ

FSRQ

Unassociated Source #5 to Binary

Table Browser for 1: gll_psc3month_BSL_v2.fit

	Source_Name	RA	DEC	GLON	GLAT	Sqrt_TS	CLASS1	CLASS2	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
4	0FGLJ0030.3+0450	7.6	4.848	113.111	-57.622	18.7	PSR			0.138	8.		
14	0FGLJ0137.1+4751	24.286	47.854	130.818	-14.317	18.8	bzq	CGRAB5 J2147+0929	DA 55	0.12	1.		
135	0FGLJ1653.9+3946	253.492	39.767	63.612	38.841	19.	bzb	CGRAB5 J1653+3945	Mrk 501	0.068	2.		
22	0FGLJ0229.5-3640	37.375	-36.681	243.801	-67.189	19.2	bzq	CGRAB5 J0229-3643	PKS 0227-369	0.138	1.		
147	0FGLJ1805.3-2138	271.329	-21.649	8.536	-0.165	19.2				0.186	7.		
77	0FGLJ1045.6-5937	161.409	-59.631	287.637	-0.548	19.5				0.123	3.		
189	0FGLJ2147.1+0931	326.777	9.519	65.805	-32.236	19.9	bzq	CGRAB5 J2147+0929	PKS 2144+092	0.137	1.		

Unassociated
Class1 == null

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
547	1FGLJ1016.1+0514	154.032	5.2351	236.514	47.0585	bzq			34.7958	20.7301	692.145	
962	1FGLJ1642.5+3947	250.6398	39.7941	63.4273	41.0283				34.8074	21.9664	548.416	
392	1FGLJ0714.0+1935	108.515	19.5866	197.659	13.6204	bzq	CLASS J0713+19...		35.1604	22.1187	558.232	
1333	1FGLJ2139.3-4235	324.849	-42.588	358.319	-48.3256	bzb	CRATES J2139-4...		35.2554	18.8526	856.989	
585	1FGLJ1058.4+0134	164.62	1.5815	251.49	52.7827	bzq	PKS 1055+01		35.5381	20.491	660.283	
572	1FGLJ1045.2-5942	161.305	-59.7057	287.625	-0.63878				35.6078	19.7799	1191.51	
1071	1FGLJ1801.3-2322c	270.331	-23.383	6.57232	-0.21566	sdp			35.6355	18.8774	1052.3	

Unassociated
Class1 == null

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	ASSOC2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
1732	2FGLJ2147.3+0930	326.835	9.51476	65.8455	-32.2825	bzq			35.8819	23.1634	455.413	2.514
1812	2FGLJ2302.7+4443	345.698	44.7202	103.403	-13.9924	PSR	PSR J2302+4442		35.9022	14.5797	1156.86	1.02671
1784	2FGLJ2241.7-5236	340.448	-52.6161	337.436	-54.935	PSR	PSR J2241-5236		36.0254	15.4306	924.773	1.35051
744	2FGLJ1045.0-5941	161.25116	-59.692	287.595	-0.639366		Eta Carinae		36.0695	16.2337	1847.65	2.32184
1817	2FGLJ2311.0+3425	347.771	34.4298	100.42	-24.0173	bzq	B2 2308+34		36.0738	21.088	421.488	2.16433
907	2FGLJ1248.2+5820	192.063	58.3499	123.738	58.7735	bzb	PG 1246+586		36.1449	17.2007	1329.24	1.94943

Unassociated
Class1 == null

Unassociated Source #5 to Binary

Table Browser for 9: gll_psc_v16_3FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Signif_Curve	Signif_Index	CLASS1	ASSOC1
249	3FGLJ0210.7-5101	32.6972	-51.0282	276.112	-61.7677	385.06	5.92996E-11	62.9476	38.0937	4.74627	2.30171	BCU	PKS 0208-512
591	3FGLJ0522.9-3628	80.7416	-36.4715	240.623	-32.7183	321.474	1.14793E-10	70.4297	44.9991	2.77891	2.30171	BCU	PKS 0521-36
612	3FGLJ0532.0-4827	83.0004	-48.4585	255.024	-32.8055	791.888	1.58149E-11	73.8326	35.9432	4.12102	2.03153	BCU	PMN J0531-4827
707	3FGLJ0622.9+3326	95.7346	33.4357	179.941	9.15656	507.631	3.72538E-11	56.1443	30.2938	2.97232	2.09513	BCU	B2 0619+33
1737	3FGLJ1418.5+3543	214.65	35.7186	63.1902	69.5785	1220.98	1.77572E-12	36.3181	15.6653	2.26862	2.05842	BCU	NVSS J141828+354250
1243	3FGLJ1045.1-5941	161.277	-59.6916	287.606	-0.633075	3281.82	1.54551E-12	33.0856	8.16258	8.88966	2.3901	BIN	Eta Carinae
112	3FGLJ0050.6-0929	12.6744	-9.4849	122.331	-72.3557	1023.06	3.79958E-12	41.9779	21.0804	1.25283	2.09308	BLL	PKS 0048-09

Associated to
BIN

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_So...	Pivot_Energy	Signif_Peak
4905	4FGLJ2302.7+4443	345.692	44.7182	103.398	-13.9925	81.5316	PSR		PSR J2302+4442				1285.56	
979	4FGLJ0532.0-4827	83.0022	-48.4607	255.026	-32.8046		BLL		PMN J0531-4827				910.36	78.8813
4646	4FGLJ2124.7-3358	321.192	-33.9786	10.9268	-45.4451		PSR		PSR J2124-3358				1142.05	
4918	4FGLJ2311.0+3425	347.768	34.4223	100.414	-24.0231		FSRQ		B2 2308+34				499.986	66.5599
2973	4FGLJ1517.7-2422	229.425	-24.373	340.681	27.5766	83.55	bll		AP Librae		AP Lib		824.093	35.5893
308	4FGLJ0136.5+3906	24.1423	39.1	132.422	-22.9387	84.7523	bll		B3 0133+388		RGB J0136+391		1922.08	39.5364
4247	4FGLJ1923.2+1408e	290.818	14.145	49.1158	-0.461602	85.2124	SNR		W 51C		W 51	W 51C	2529.51	
1974	4FGLJ1045.1-5940	161.277	-59.6822	287.601	-0.624566	85.7256	BIN		Eta Carinae	FGES J1036.3-5833 field	Eta Carinae		2014.83	33.5116
4483	4FGLJ2032.2+4127	308.063	41.459	80.2294	1.02414	86.9727	PSR		PSR J2032+4127	Cygnus Cocoon field	PSR J2032+4127		2235.47	
1474	4FGLJ1040.8+5218	122.462	52.3143	166.248	32.9144	87.0045	RII		1FS 0806+524		1FS 0806+524		1297.	44.2755

BIN

BIN

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

	Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extended_Source_Name	Signif_Peak
4844	4FGLJ1923.2+1408e	1	290.818	14.145	49.1158	-0.461602	93.9032	2456.97	SNR		W 51C		W 51	W 51C	
2221	4FGLJ1045.1-5940	1	161.277	-59.6822	287.601	-0.624566	94.0432	2109.55	BIN		Eta Carinae	FGES J1036....	Eta Carinae		33.4744
1643	4FGLJ0808.2-0751	1	122.065	-7.8556	229.044	13.1616	94.4208	620.534	fsrq		PKS 0805-07				61.7413
2276	4FGLJ1058.4+0133	1	164.624	1.5641	251.514	52.7728	94.7605	802.762	BLL		4C +01.28				39.9912
3351	4FGLJ1517.7-2422	1	229.425	-24.373	340.681	27.5766	84.057	911.196	bll		AP Librae		AP Lib		35.7046

Unassociated Source #6 until 4FGL-DR1

Table Browser for 2: gll_psc_v03_1FGL.fit

Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Flux_Density	Spectral_Index
1339	1FGL J2146.6-1345	326.672	-13.7535	40.654	-44.982		9.79046	2.03775	3173.25	1.41837E-13	1.82421
982	1FGL J1702.4-4147c	255.604	-41.7859	344.45	0.003416		9.79595	5.71774	1192.53	7.48946E-12	2.39258
93	1FGL J0137.5-2428	24.3927	-24.4815	201.197	-79.2893	bzq	9.80455	6.51846	635.124	4.47998E-12	2.44899
1140	1FGL J1844.3-0309c	281.09	-3.157	29.3163	0.132846		9.8154	0.	2848.19	1.07250E-12	2.15436
565	1FGL J1036.2-6719	159.0655	-67.3229	290.432	-7.79229		9.81986	3.55857	914.016	3.80255E-12	2.34984
1334	1FGL J2139.9+4715	324.987	47.2579	92.627	-4.03097		9.82447	4.17288	789.853	6.14558E-12	2.36444

Unassociated
Class1 == null

Table Browser for 5: gll_psc_v09_2FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
1292	2FGL J1719.3+1744	259.83	17.7425	39.5258	28.0706	bzb	17.6009	4.67481	2048.13	1.84168
1657	2FGL J2039.8-5620	309.965	-56.3406	341.192	-37.1778		17.6212	9.81252	834.837	2.19114
390	2FGL J0543.9-5532	85.9921	-55.5337	263.513	-31.4719	bzb	17.7402	5.0146	3006.36	1.74014
957	2FGL J1315.9-3339	198.992	-33.6632	308.763	28.9306	bzq	17.7481	10.6962	905.7	2.30522
734	2FGL J1036.1-6722	159.039	-67.3716	290.448	-7.83956		17.7547	6.97096	1096.66	1.88395
1509	2FGL J1857.6+0211	284.419	2.18709	35.5898	-0.389722		17.7549	10.269	1475.28	2.56639

Unassociated
Class1 == null

PKS 1313-333

Table Browser for 9: gll_psc_v16_3FGL.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Flux_Density	Spectral_Index	CLASS1	CLASS2
1212	3FGL J1033.0-5945	158.266	-59.7513	286.308	-1.4227	1555.61	1.13956E-12	11.3052	5.49087	1.70744	2.7961		
1214	3FGL J1033.4-5035	158.374	-50.5946	281.724	6.50911	2603.06	1.33345E-13	10.4081	1.91365	0.694635	2.013		
1216	3FGL J1034.0-5731c	158.516	-57.5174	285.295	0.572408	1999.22	4.86822E-13	7.54238	5.12755	2.93315	2.28352		
1218	3FGL J1035.7-6720	158.926	-67.3335	290.391	-7.82838	778.904	8.65322E-12	30.4761	15.5561	8.39136	1.95648		
1220	3FGL J1036.4-4348	159.113	-43.812	278.678	12.6117	1252.55	2.79663E-13	4.61889	1.20748	1.22018	2.27681		

Unassociated
Class1 == null

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	ASSOC1	ASSOC2	ASSOC_TEV	Extend...	Pivot_Energy
4680	4FGL J2140.0+4716	325.001	47.2686	92.6408	-4.0291	43.5448	PSR J2139+4716				941.996
407	4FGL J0212.1+5321	33.0445	53.3507	134.926	-7.63041	43.5448					1033.57
1944	4FGL J1035.4-6720	158.865	-67.338	290.372	-7.84417	43.6775	PSR J1035-6720				1640.89

Associated to
PSR

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	Pivot_Energy	CLASS1	CLASS2	ASSOC1	ASSOC2
5163	4FGL J2045.2+5026e	1	311.32	50.44	88.7529	4.67043	48.3538	SNR		HB 21	
5538	4FGL J2241.7-5236	1	340.438	-52.6084	337.452	-54.9341	48.4299	PSR		PSR J2241-5236	
2185	4FGL J1035.4-6720	1	158.865	-67.338	290.372	-7.84417	48.495	PSR		PSR J1035-6720	
3959	4FGL J1719.2+1745	1	259.806	17.7533	39.5272	28.0959	48.5274	bll		PKS 1717+177	

Unassociated Source #7 thru 4FGL-DR4

Table Browser for 2: gll_psc_v03_1FGL.fit

	Source_Name	RA	DEC	GLON	GLAT	CLASS1	CLASS2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy
1090	1FGL J1813.4+3141	273.355	31.694	58.6128	21.383	bzb		12.683	4.94638	1209.52
959	1FGL J1639.5-5152	249.896	-51.8772	334.19	-3.42661			12.7029	8.11435	722.549
1385	1FGL J2243.1-2541	340.788	-25.685	28.4482	-61.3159	bzb	PKS 2240-260	12.7208	6.89736	736.055
1315	1FGL J2112.5-3044	318.137	-30.7357	14.8952	-42.4425			12.7843	5.55027	1107.29
54	1FGL J0051.1-0649	12.7826	-6.8184	122.713	-69.69	bzq	PKS 0048-071	12.821	8.70666	781.285

Unassociated
Class1 == null

Table Browser for 5: gll_psc_v09_2FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	CLASS1	CLASS2	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Spectral_Index
1330	2FGL J1917.0-1921	289.423	-19.3503	18.2433	-14.3047	DZD		19.5752	5.87047	2203.19	1.91033
797	2FGL J1118.8-6128	169.703	-61.4735	292.107	-0.563835	PSR	PSR J1119-6127	19.5814	11.2809	946.988	1.91202
970	2FGL J1326.7-5254	201.7	-52.9149	308.325	9.58611	bzb	PMN J1326-5256	19.6766	10.6327	997.444	2.22986
1700	2FGL J2112.5-3042	318.149	-30.7106	14.9311	-42.4479			19.6973	6.58634	1152.89	1.59954
409	2FGL J0608.0-1521	92.0034	-15.3667	222.112	-16.4071	bzq	PMN J0608-1520	19.7407	13.23	387.331	2.17306

Unassociated
Class1 == null

Table Browser for 9: gll_psc_v16_3FGL.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Pivot_Energy	Flux_Density	Signif_Avg	Sqrt_TS300_1000	Pivot_Energy	Signif_Peak
2735	3FGL J2109.4+1437	317.359	14.6197	63.6886	-21.881...	445.743	7.90816E-12	10.2063	7.78551	0.053823	2.947
2736	3FGL J2110.0+0442	317.513	4.7045	54.942	-27.836...	2533.46	5.72569E-14	5.13641	1.95131	0.585085	2.005
2740	3FGL J2110.4+5946	317.619	59.7793	98.2528	7.96758	840.834	1.53239E-12	7.38731	4.20488	1.27706	2.53383
2742	3FGL J2112.5-3044	318.145	-30.7344	14.8984	-42.448...	1133.51	2.86716E-12	30.1418	10.7088	7.62829	1.67709
2744	3FGL J2114.0+5239	318.524	52.6523	93.3661	2.73405	866.783	5.27712E-12	6.05006	5.31072	4.09598	2.38577

Unassociated
Class1 == null

Table Browser for 19: gll_psc_v22_4FGL_DR1.fit

	Source_Name	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC_TEV	Extend...	Pivot_Energy	Signif_Peak
472	4FGL J0228.3-5547	37.0854	-55.7931	278.263	-56.4474	50.2735	fsrq				1146.11	30.7008
4612	4FGL J2112.5-3043	318.14	-30.7293	14.9039	-42.4435	50.3083					1343.1	
1797	4FGL J0053.0-0840	14.263	-8.6605	246.186	33.0000	50.3263	hll	PMN J0053-0840			1046.86	20.4404

Unassociated
Class1 == null

Unassociated Source #7 thru 4FGL-DR4

Table Browser for 27: gll_psc_v27_4FGL_DR2.fit

	Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1
1806	4FGL J0850.0+4855	1	132.508	48.9217	170.465	39.2731	55.953	bll		GB6 J0850+4855
251	4FGL J0106.4+4855	1	16.6094	48.9269	125.469	-13.871	55.9827	PSR		PSR J0106+4855
2302	4FGL J1103.9-5357	1	165.976	-53.9647	287.428	5.63829	56.1967	bll		PKS 1101-536
5258	4FGL J2112.5-3043	1	318.14	-30.7293	14.9039	-42.4435	56.3453			

Unassociated
Class1 == null

Table Browser for 34: gll_psc_v31_4FGL_DR3.fit

	Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Signif_Avg	CLASS1	CLASS2	ASSOC1	ASSOC2
5727	4FGL J2001.2+4353	1	300.302	43.8862	79.0708	7.11348	2029.25	bll		MG4 J200112+...	
1197	4FGL J0510.0+1800	1	77.5181	18.0135	184.733	-12.7821	1006.97	fsrq		PKS 0507+17	
4620	4FGL J1728.3+5013	1	262.078	50.2267	77.0763	33.5371	2982.84	bll		I Zw 187	
6037	4FGL J2112.5-3043	1	318.14	-30.7293	14.9039	-42.4435	1431.25				
280	4FGL J0106.4+4855	1	16.6094	48.9269	125.469	-13.871	1577.5	PSR		PSR J0106+4855	

Unassociated
Class1 == null

Table Browser for 41: gll_psc_v33_4FGL_DR4.fit

	Source_Name	DataRelease	RAJ2000	DEJ2000	GLON	GLAT	Extended_Source_Name	Signif_Avg	CLASS1	CLASS2
3499	4FGL J1312.6+4828	1	198.169	48.4701	113.396	68.263		64.3245		
2942	4FGL J1120.0-2204	1	170.002	-22.0779	276.492	36.0559		64.3332		
4384	4FGL J1603.8-4903	1	240.967	-49.0617	332.158	2.57594		64.5449		
6518	4FGL J2112.5-3043	1	318.14	-30.7293	14.9039	-42.4435		64.5728		
54	4FGL J0009.3+5030	1	2.3466	50.5115	116.124	-11.8036		64.8969		

Unassociated
Class1 == null

Table Browser for 41: gll_psc_v33_4FGL_DR4.fit

	ASSOC_4FGL	ASSOC_FGL	ASSOC_FHL
3499	4FGL J1312.6+4828	3FGL J1312.7+4828	3FHL J1312.6+4828
2942	4FGL J1120.0-2204	3FGL J1119.9-2204	3FHL J1119.9-2204
4384	4FGL J1603.8-4903	3FGL J1603.9-4903	3FHL J1603.8-4903
6518	4FGL J2112.5-3043	3FGL J2112.5-3044	3FHL J2112.5-3044
54	4FGL J0009.3+5030	3FGL J0009.3+5030	3FHL J0009.4+5030

Prior
association